

Chapter I: General Provisions

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I. Introduction

Chapter I of the AI Act, comprising just four articles, lays the foundational groundwork for the regulation's subsequent substantive and procedural provisions. Article 1 delineates the Act's subject matter and overarching objectives, while Article 2 outlines its temporal, geographical, personal, and material scope. Article 3 provides a long list of definitions, crucial for interpreting the provisions that follow. Finally, Article 4 introduces an obligation focused on AI literacy.

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II. Article 1: Subject Matter

1. The purpose of this Regulation is to improve the functioning of the internal market and promote the uptake of human-centric and trustworthy artificial intelligence (AI), while ensuring a high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, the rule of law and environmental protection, against the harmful effects of AI systems in the Union and supporting innovation.
2. This Regulation lays down:
 - (a) harmonised rules for the placing on the market, the putting into service and the use of AI systems in the Union;
 - (b) prohibitions of certain AI practices;
 - (c) specific requirements for high-risk AI systems and obligations for operators of such systems;
 - (d) harmonised transparency rules for certain AI systems;
 - (e) harmonised rules for the placing on the market of general-purpose AI models;
 - (f) rules on market monitoring, market surveillance, governance and enforcement;
 - (g) measures to support innovation, with a particular focus on SMEs, including start-ups.

A. Overview

- 2.02** As is commonplace in European Union secondary legislation, the inaugural article defines the regulation's subject matter. Article 1(1) lays down the AI Act's (also referred to as 'the AIA' or 'the regulation') objectives and Article 1(2) outlines the various legal sub-regimes that are created by the regulation.
- 2.03** Article 1(1) defines the dual purpose of the AIA, which is to, first, improve the functioning of the internal market, inter alia, through the promotion of innovation, and the uptake of human-centric and trustworthy AI while, second, ensuring a high level of protection of health, safety and fundamental rights.¹ These elements are crucial as they will guide future judicial interpretations of other provisions of the AI Act. Given the open-ended nature and insufficient specificity of many of its operating provisions, practitioners and judges will need to rely on these provisions to interpret the regulation effectively. Article 1(2) lists the seven legal sub-regimes laid down by the regulation. First, harmonized rules for the placing on the market, the putting into service and the use of AI systems in the EU.² Second, absolute prohibitions and qualified prohibitions of some AI practices deemed contrary to EU values.³ Third, a *sui generis* legal regime for HRAIS.⁴ Fourth, transparency rules for particular kinds of AI systems.⁵ Fifth, rules concerning the placing on the market of GPAIM.⁶ Sixth, the creation of rules on market monitoring, surveillance, governance and enforcement.⁷ Finally, measures to support innovation, with a particular focus on small and medium sized enterprises ('SMEs'), including start-ups.⁸ These legislative objectives can be best conceptualized as various legal 'sub-regimes' created by the AIA as the purposes and methods of these regimes significantly diverge. Regarding legislative history, the list of sub-regimes enumerated by Article 1(1) evolved over time as various regimes were adjusted and expanded, including the regime for general-purpose AI models, which was not initially foreseen by the draft regulation.⁹

B. Analysis

i. Article 1(1): The objectives of the AIA

- 2.04** Article 1(1) articulates the **twin objectives** of the regulation, namely to improve the internal market and promote innovation, through the uptake of human-centric and trustworthy AI while also ensuring a high level of protection of health, safety, and fundamental rights.¹⁰ The legislative incentivization to further use AI in the internal market, coupled with a reminder

¹ The objectives of the AIA and their respected relation is formulated somewhat differently by Article 1(1) as well as Recitals 1 and 8 AIA. I use the above categorization as I believe that it best reflects the logic behind these formulations. See also line 102a of the Proposal for a Regulation of the European Parliament and of the Council laying down harmonized rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts (Four Columns AI Act) [2024] COM (2021) 206 final (Final draft as updated on 21 January 2024) <<https://artificialintelligenceact.eu/wp-content/uploads/2024/01/AIA-Final-Draft-21-January-2024.pdf>> accessed 17 March 2025.

² Chapter I AIA, General Provisions.

³ Chapter II AIA, Prohibited AI Practices.

⁴ Chapter III AIA, High-Risk AI Systems.

⁵ Chapter IV AIA, Transparency Obligations for Providers and Deployers of Certain AI Systems.

⁶ Chapter V AIA, General-Purpose AI Models.

⁷ Chapter IX AIA, Post-Market Monitoring, Information Sharing and Market Surveillance.

⁸ Chapter VI AIA, Measures in Support of Innovation.

⁹ See further analysis of Chapter V AIA, in particular the introduction. Four Columns AI Act (n 1) line 107.

¹⁰ See also Recitals 1 and 8 AIA.

that fundamental rights ought to be respected, reflects the multifacetedness of AI.¹¹ On one hand, AI is an important innovation that triggers positive change and it needs to be developed and used in the EU in order to maintain its competitiveness. This is echoed by the pre-emption of national competence and by legal provisions designed to stimulate innovation. On the other hand, AI is a potent tool that can be used in undesirable manners. This is reflected by rules applicable where AI is placed on the market, put into service, or used, as well as in prohibitions on some systems, rules applicable to high-risk AI, as well as transparency rules applicable to some AI systems. A provision to this effect was notably absent in the draft regulation.¹² The European Parliament, however, pushed for the introduction of a programmatic statement into the AIA. A revised version of the Parliament proposal was ultimately adopted in the trilogues.¹³ Each of the three elements of Article 1(1) is examined in detail, one after the other.

a. Improving the functioning of the internal market and promoting innovation

The main objective of the AIA is to improve the functioning of the EU internal market¹⁴, inter alia, through the uptake of ‘human-centric and trustworthy’ AI and the promotion of innovation.¹⁵ The **internal market** is the EU’s ‘area without internal frontiers in which the free movement of goods, persons, services and capital is ensured.’¹⁶ From its inception in the aftermath of World War II, the main objective of the EU has been to secure peace and prosperity in Europe through the approximation of national markets.¹⁷ According to the CJEU, the idea behind the internal market is that: ‘by establishing a common market and progressively approximating the economic policies of the Member States seeks to unite national markets in a single market having the characteristics of a domestic market.’¹⁸ It is well-known that divergent national regulations hinder the internal market by imposing additional costs on operators seeking to do business across multiple Member States. For this reason, EU law has long emphasized the importance of a free movement of goods, services, capital and workers across the internal market.¹⁹ Amid ongoing digitalization, these concerns have recently led the EU to implement various measures aimed at facilitating the free movement of personal and non-personal data across the Union.²⁰ The AI Act pursues these objectives

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¹¹ Recitals 3 and 4 AIA.

¹² Indeed, the dAIA only defined its purpose through Recital 1. On the legal status of recitals in EU law, see further related analysis in the Introduction.

¹³ Four Columns AI Act (n 1) line 102a.

¹⁴ See also Recitals 1, 3 and 4 AIA. This is also echoed by the fact that Article 114 TFEU is the primary legal basis of the AIA.

¹⁵ Recitals 1, 6 and 8 AIA contain explicit references to EU values. Recital 6 AIA specifies that these are the values enshrined in Article 2 TEU.

¹⁶ Article 26(2) TFEU.

¹⁷ On these beginnings, see ‘Schuman Declaration May 1950’ (European Union) <https://european-union.europa.eu/principles-countries-history/history-eu/1945-59/schuman-declaration-may-1950_en> accessed 18 February 2025.

¹⁸ Case C-207/83 *Commission v UKEU*:C:1985:161, [1985] ECRI-1201.

¹⁹ Article 28(1) TFEU, ‘The Union shall comprise a customs union which shall cover all trade in goods and which shall involve the prohibition between Member States of customs duties on imports and exports and of all charges having equivalent effect, and the adoption of a common customs tariff in their relations with third countries’; Article 45(1) TFEU, ‘Freedom of movement for workers shall be secured within the Union’; Article 56 TFEU, ‘Within the framework of the provisions set out below, restrictions on freedom to provide services within the Union shall be prohibited in respect of nationals of Member States who are established in a Member State other than that of the person for whom the services are intended’; Article 63(1) TFEU, ‘Within the framework of the provisions set out in this Chapter, all restrictions on the movement of capital between Member States and between Member States and third countries shall be prohibited’.

²⁰ DMA; DSA; DGA; Data Act and Free Flow of Non-Personal Data Regulation (Regulation (EU) 2018/1807 of the European Parliament and of the Council of 14 November 2018 on a framework for free flow of non-personal data in the European Union (2018) OJ L303/59).

by, first, pre-empting Member States' competence on AI and, second, by establishing supra-national norms, as outlined in Article 1(2), deemed to advance these objectives.

b. The promotion of innovation

- 2.06** The AI Act fosters innovation through two primary avenues: by establishing legal certainty, which is anticipated to stimulate novel developments, and by introducing targeted measures specifically designed to achieve this objective.
- 2.07** First, the AI seeks to create '**clear and robust**' rules that are expected to drive innovation, inter alia through the promotion of legal certainty.²¹ The Impact Assessment to the AIA highlighted that lacking legal certainty has economic consequences as without clear legal frameworks, start-ups and developers are unable to attract investments.²² In particular, 'companies must also calculate the risk that legislation or other requirements will be introduced, without knowing what this will imply for their business models. Such legal uncertainty is detrimental for investment and especially for innovation.'²³ As such, the AIA is intended to further innovation through harmonized rules. In addition, Chapter VI foresees **several mechanisms specifically designed to further innovation**, such as regulatory sandboxes and the testing of AI systems in real world environments. Other rules, such as lighter rules for start-ups and SMEs or the promotion of AI literacy can also be understood as innovation-facilitating mechanisms.²⁴
- 2.08** Moreover, the AIA also **shapes innovation** as it promotes a particular kind of innovation in the EU, namely innovation focused on human-centric and trustworthy AI.

c. Promoting the uptake of human-centric and trustworthy artificial intelligence

- 2.09** The AIA also seeks to promote innovation and the uptake of 'human centric and trustworthy AI'.²⁵ This objective is closely linked to the overarching goal of ensuring the competitiveness of the EU internal market, as the adoption of AI is expected to stimulate market growth.²⁶ However, it also serves as an independent regulatory goal, as the AIA aims not merely to promote the adoption of any AI, but specifically to encourage the use of 'human-centric and trustworthy AI' in line with EU values.
- 2.10** The AIA does not offer explicit **definitions** for either 'human-centric' or 'trustworthy AI'. There are some indications that these notions refer to AI that supports 'socially and environmentally beneficial outcomes'²⁷, AI that is developed 'in accordance with Union values'²⁸ as well as AI that 'meets a high level of protection of public interests'.²⁹ Human-centeredness is described as AI that serves 'as a tool for people, with the ultimate goal of increasing human well-being'.³⁰

²¹ Recital 8 AIA.

²² Commission, 'Commission staff working document - Impact Assessment - Accompanying the Proposal for a Regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain union legislative acts' SWD (2021) 84 final, 33.

²³ Ibid.

²⁴ See, inter alia, Articles 4 and 17(2) AIA.

²⁵ Recitals 1, 6 and 8 AIA contain explicit references to EU values. Recital 6 AIA specifies that these are the values enshrined in Article 2 TEU.

²⁶ According to Recital 4 AIA, the use of AI can provide 'key competitive advantages' to companies.

²⁷ Recital 4 AIA.

²⁸ Article 2 TEU and Recital 6 AIA.

²⁹ Recital 8 AIA.

³⁰ Recital 6 AIA.

The notion of **trustworthy AI** is shorthand for a form of AI the risks of which are managed to a degree that individuals and companies are willing to use it, which in turn relates to the market goals of the AIA. Trustworthiness is hence understood in terms of the acceptability of risk.³¹ This needs to be understood in the context of the AIA's underlying market objective as the legislative proposal was rooted in the assumption that market participants are more likely to use AI where they consider it to be trustworthy.³² The concept of 'trustworthy AI' has also been used in this manner in other jurisdictions. For instance, the United States NIST AI Risk Management Framework, a form of soft law, also announces that 'understanding and managing the risks of AI systems will help to enhance trustworthiness, and in turn, cultivate public trust'.³³ The Ethics Guidelines for Trustworthy AI issued by the HLEGAI in 2019 defined trustworthy AI as having three components, which should be met throughout its life cycle: '(1) it should be lawful, complying with all applicable laws and regulations; (2) it should be ethical, ensuring adherence to ethical principles and values; and (3) it should be robust, both from a technical and social perspective since, even with good intentions, AI systems can cause unintentional harm. Each component in itself is necessary but not sufficient for the achievement of Trustworthy AI'.³⁴ These guidelines have to be taken into account regarding the interpretation of Chapter III on HRAIS.³⁵

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d. Ensuring a high level of protection of health, safety, fundamental rights

The market integration, trustworthiness and human-centredness objectives of the regulation need to be pursued 'while ensuring a high level of protection of health, safety, fundamental rights'.³⁶ Article 1(1) underscores that the protection of health, safety, and fundamental rights should be achieved as **accessory goals** within the framework of the AIA's first two objectives, rather than as independent goals. Indeed, Article 1(1) refers to the 'purpose', not 'purposes' of the regulation³⁷ and the use of 'while' denotes that these aspects ought to be respected in the course of the pursuit of the economic aim, not independently thereof.³⁸ The inclusion of fundamental rights protection within the AI Act, a dimension not typically central to the EU's new legislative framework ('NLF'), which primarily focuses on health and safety risks, introduces significant conceptual and interpretational complexities. This expansion of regulatory scope, indeed, marks a novel approach that inherently generates numerous challenges in its application and understanding.

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Article 1(1) underlines that one of the objectives of the AIA is the protection of health and safety. The **protection of health** forms a classical objective of product safety law³⁹ and is,

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³¹ Johann Laux, Sandra Wachter and Brent Mittelstadt, 'Trustworthy Artificial Intelligence and the European Union AI Act: On the Conflation of Trustworthiness and Acceptability of Risk' (2024) 18 Regulation & Governance 3.

³² Commission Impact Assessment (n 22) 1, 24, 25, 29.

³³ NIST, 'Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile' (National Institute of Standards and Technology (US) 2024) NIST AI 600-1 <<https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>> accessed 16 January 2025.

³⁴ High-Level Expert Group on AI, 'Ethics Guidelines for Trustworthy AI' (European Commission 2019) <<https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>> accessed 24 March 2025.

³⁵ Recital 7 AIA reads as follows: 'In order to ensure a consistent and high level of protection of public interests as regards health, safety and fundamental rights, common rules for high-risk AI systems should be established. Those rules should be consistent with the Charter, non-discriminatory and in line with the Union's international trade commitments. They should also take into account the European Declaration on Digital Rights and Principles for the Digital Decade and the Ethics guidelines for trustworthy AI of the High-Level Expert Group on Artificial Intelligence ("AI HLEG").'

³⁶ Article 1(1) AIA.

³⁷ Compare, for instance, to Article 1 GDPR.

³⁸ See also Recital 1 AIA.

³⁹ See, by the way of example, Article 3(2) GPSR.

indeed, also an objective mentioned in primary law.⁴⁰ It is considered a ‘public interest’.⁴¹ Indeed, AI can pose risks to health such as in case of malfunctioning. It is important to note that the AIA protects the health of natural persons, not that of animals or of natural ecosystems.⁴² This echoes the fact that the Charter of Fundamental Rights (‘CFR’) also protects aspects of health of natural persons.⁴³ The **protection of safety** is another EU objective mentioned by primary law.⁴⁴ It also constitutes a public interest traditionally protected by product safety law.⁴⁵ Indeed, AI systems can pose risks to safety if not developed and deployed responsibly, particularly when functioning as safety components within products.

- 2.14** None of the 23 other secondary legislation instruments that have been adopted under the NLF so far explicitly aim to **protect fundamental rights**.⁴⁶ The inclusion of fundamental rights in Article 1(1) echoes broader concerns about the impact of AI on fundamental rights. The support study to the Impact Assessment of the AIA pointed out that there was ‘strong evidence that certain uses of AI systems can significantly impact all fundamental rights recognised in the Charter’.⁴⁷ In this context, the scalability of AI is especially troubling, as it amplifies the reach of harm, affecting a greater number of individuals. While a single prejudiced person might discriminate against a limited number of individuals, a biased algorithm has the capacity to perpetuate such discrimination on a vastly more extensive scale.
- 2.15** Numerous concerns have surfaced over the years about the **impact of AI on particular fundamental rights**. Bias and discrimination are probably the most well-documented fundamental rights risks to AI. Already in 2013, Latanya Sweeney highlighted racist discrimination in online ad delivery.⁴⁸ A 2018 study for the Council of Europe lists the uses and sectors in which bias and discrimination had emerged.⁴⁹ Algorithmic discrimination can have its source in various factors, such as in the definition of the target variable and class levels, in the collection and/or labelling of training data, feature selection and algorithmic design, as well as the use of proxies and redundant encodings.⁵⁰ The Council of Europe has also stressed concerns for the right to freedom of expression and information as the ‘enabling’ nature of AI, including search engines, provides new possibilities for freedom of expression.⁵¹ AI also raises concerns for human dignity as AI can be used to generate dark patterns, have a

⁴⁰ See Articles 114(3), 168 and 169(1) TFEU.

⁴¹ Recital 7 AIA.

⁴² See Recital 46 AIA.

⁴³ See Articles 2, 3 and 35 CFR (Charter of fundamental rights of the European Union [2012] C 326/02).

⁴⁴ See Articles 114(3) and 169(1) TFEU.

⁴⁵ See, by way of example, Article 1(1) GPSR.

⁴⁶ See also Marco Almada and Nicolas Petit, ‘The EU AI Act: A Medley of Product Safety and Fundamental Rights?’ (2023) Robert Schuman Centre for Advanced Studies Research Paper 2023/59 <<https://cadmus.eui.eu/handle/1814/75982>> accessed 14 March 2025.

⁴⁷ Commission, ‘Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe’ (Publications Office 2021) Final Report (D5), 7 <<https://data.europa.eu/doi/10.2759/523404>> accessed 4 February 2025.

⁴⁸ Latanya Sweeney, ‘Discrimination in Online Ad Delivery’ (2013) 56 Communications of the ACM 44 <<https://dl.acm.org/doi/10.1145/2447976.2447990>> accessed 21 March 2025.

⁴⁹ Frederike Zuiderveen Borgesius, ‘Discrimination, Artificial Intelligence, and Algorithmic Decision-Making’ (Council of Europe, Directorate General of Democracy 2018).

⁵⁰ Solon Barocas and Andrew D Selbst, ‘Big Data’s Disparate Impact’ (Social Science Research Network, 2016) <<https://papers.ssrn.com/abstract=2477899>> accessed 21 March 2025.

⁵¹ Council of Europe, ‘Recommendation of the Committee of Ministers to member States on the protection of human rights with regard to search engines (Adopted by the Committee of Ministers on 4 April 2012 at the 1139th meeting of the Ministers’ Deputies)’ CM/Rec(2012)3, para 1 <<https://search.coe.int/cm?i=09000016805caa87>> accessed 15 July 2025.

manipulative influence on the subconscious mind, and in particular anthropomorphic AI can be particularly efficient to lower human autonomy.

Existing fundamental right norms in EU and national law apply fully to AI where their scope of application is triggered.⁵² Fundamental rights are not only relevant between the state and private parties but also between private parties to the extent that the right in question has horizontal direct effect.⁵³ Yet, the Charter and the general principles of EU law merely announce broad principles, which leads to open questions as to how they should be applied in a given context, especially one as complex and recent as AI. During the legislative process, one of the most debated issues was to what extent, and in what form, fundamental rights should be explicitly safeguarded within this legislative instrument.

This controversy needs to be placed into the **broader context of EU law-making**. According to the principle of attributed competences in Article 5(2) of the Treaty on European Union ('TEU'), the EU can act only within the limits of the competences that Member States have attributed to it. Article 114 TFEU enables the EU to adopt measures for the approximation of law for the establishment and functioning of the internal market. Article 114 TFEU holds an 'immense functional breadth'⁵⁴ and can stretch to any sector. Indeed, whereas the other legal basis for EU legislative action are sector-specific, Article 114 is generally about legal harmonization for the benefit of the internal market at large.⁵⁵ Article 114 TFEU is the main legal basis of the AIA. Article 16(2) TFEU serves as an accessory legal basis insofar as it contains specific rules on data protection.⁵⁶ The EU, indeed, does not have a general competence to adopt secondary legislation on fundamental rights.⁵⁷ Indeed, Article 51(1) CFR merely obliges the EU to protect fundamental rights in the context of the competences that have been conferred upon it, such as its internal market competence. This implies that there was no competence to include provisions related to fundamental rights in the AIA that go beyond the right to data protection and do not improve the internal market. The boundary between the broad protection of fundamental rights and the protection of these rights in connection to the internal market is ambiguous, especially in the context of a versatile technology such as AI.

While the odd status of fundamental rights in the AIA can hence be explained constitutionally, this explanation does not resolve the **conceptual mismatch** that is caused by the embedding of fundamental rights into an instrument bedded in product safety law. The AIA introduces fundamental rights language and concerns into a legal framework based on product legislation, i.e. technical internal market law. Product safety law has been tailored

⁵² To the extent that they are protected by the EU Charter of Fundamental Rights, these rights have the status of EU primary law. See further Article 6 TEU.

⁵³ The CJEU has spelled out that some provisions in the EU Charter may produce horizontal direct effects, provided that the provision is sufficiently clear and precise, unconditional, and mandatory. See Case C-414/16 *Egenberger* EU:C:2018:257, para 57; Case C-68/17 *IR* EU:C:2018:696; Joined Cases C-569/16 and C-570/16 *Bauer* EU:C:2018:871; Case C-684/16 *Max-Planck* EU:C:2018:874 and Case C-193/17 *Cresco Investigation* EU:C:2019:43.

⁵⁴ Stephen Weatherill, *The Internal Market as a Legal Concept* (OUP 2017) 10.

⁵⁵ Examples include Article 169 TFEU on consumer protection or Article 168 TFEU in the field of public health. Article 352 TFEU also generally permits EU action to achieve Treaty objectives when the Treaties have not provided the necessary powers. However, unlike Article 114 TFEU, it cannot be used in sectors where the Treaties exclude harmonization and also required unanimity in the Council.

⁵⁶ Article 16(2) TFEU enables the EU to 'lay down the rules relating to the protection of individuals with regard to the processing of personal data by Union institutions, bodies, offices and agencies, and by the Member States when carrying out activities which fall within the scope of Union law, and the rules relating to the free movement of such data.'

⁵⁷ Article 19 TFEU provides that the EU may adopt legislation to combat **discrimination based on sex, racial or ethnic origin, religion or belief, disability, age or sexual orientation**.

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to products, such as elevators, which typically do not raise fundamental rights issues. Commentators have indeed warned that this mismatch will result in an inadequate protection of fundamental rights as product safety measures will be used to achieve fundamental rights protection.⁵⁸ Both areas of the law differ in substantial methodological ways as ‘the instruments used for protecting individuals from product risks rely on a formalized logic of evaluation, which differs from the logic of proportionality that guides fundamental rights assessments. Furthermore, product safety instruments aim to keep risk below a level deemed satisfactory rather than minimize it.’⁵⁹ The regulatory approach by the EU raises further challenges, such as how to incorporate fundamental rights risks into technical standards.⁶⁰ Risk regulation is prominent in sectors related to health and safety due to its dependence on scientific expertise and quantitative tools for risk analysis and management.⁶¹ The primary objective of this approach is to ascertain the tolerable levels of risk concerning health and safety. The increasing reliance on risk regulation has been linked to the increasing role played by science and technology in our age as this, indeed, allows many objects of regulatory intervention to be framed as scientific risks.⁶² In such contexts, qualitative concepts of risk assessment and management can be used.⁶³ Fundamental rights do not lend themselves easily to quantitative and scientific calculations.⁶⁴

2.19 Under the AI Act, risks to fundamental rights must be assessed independently, despite their inherent **resistance to quantification**. Even identifying which rights under the Charter warrant consideration in a provider’s risk assessment presents a significant challenge. Determining the relevance, weighing the severity of potential infringements, and striking a proportionate balance between risks and benefits are complex evaluative tasks—typically reserved for courts—and are further complicated by divergent constitutional traditions across Member States. These multifaceted issues demand thorough consideration and often require nuanced interpretations tailored to specific contexts within the European legal landscape. Yet, under the AIA they fall within the realm of private rule-making.

2.20 Despite the accessory role of fundamental rights in the AIA, the **narrative surrounding of the AI Act has been unduly narrow**, often fixating exclusively on fundamental rights risks. For instance, the Commission Communication on the Definition of the AI System in Article 3(1) states that the risk-based approach of the AIA ‘means that only those systems giving rise to the most significant risks to fundamental rights and freedoms will be subject to its prohibitions laid down in Article 5 AI Act, its regulatory regime for high-risk AI systems covered by Article 6 AI Act and its transparency requirements for a limited number of pre-defined AI systems laid down in Article 50 AI Act’, making no mention of risks to health and safety.⁶⁵

⁵⁸ Almada and Petit (n 46) 18.

⁵⁹ Ibid.

⁶⁰ See further below.

⁶¹ David Demortain, *Scientists and the Regulation of Risk: Standardising Control* (Edward Elgar 2011) 1.

⁶² Maria Weimer, *Risk Regulation in the Internal Market: Lessons from Agricultural Biotechnology* (1st edn, OUP 2019) 24.

⁶³ Julia Black, ‘The Role of Risk in Regulatory Processes’ in Robert Baldwin, Martin Cave and Martin Lodge (eds), Julia Black, *The Oxford Handbook of Regulation* (OUP 2010) 302.

⁶⁴ Raphaël Gellert, ‘The Role of the Risk-Based Approach in the General Data Protection Regulation and in the European Commission’s Proposed Artificial Intelligence Act: Business as Usual?’ (2021) 3 *Journal of Ethics and Legal Technologies* 15, 23; Elham Tabassi, ‘Artificial Intelligence Risk Management Framework (AI RMF 1.0)’ (National Institute of Standards and Technology (US) 2023) NIST AI 100-1, 5–6 <<http://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf>> accessed 18 June 2025.

⁶⁵ Commission, ‘Guidelines on the definition of an artificial intelligence system established by Regulation (EU) 2024/1689 (AI Act)’ C(2025) 924 final, 12.

This is misleading as, for instance, the risk classification rule for high-risk AI systems in Article 6(1) is based on risks to health and safety and not fundamental rights.⁶⁶

Regarding **legislative history**, the dAIA only defined its purpose through recitals and more succinctly in referring only to ‘health, safety and fundamental rights.’⁶⁷ The final version of Article 1(1), which specifies that its role is to ensure a ‘high level of protection of health, safety, fundamental rights enshrined in the Charter, including democracy, rule of law and environmental protection’, is in line with Article 114(3) TFEU, which obliges the EU Commission to strive for a high level of protection in legislative proposals based on this harmonizing clause that concerns ‘health, safety, environmental protection and consumer protection.’⁶⁸ The Parliament had suggested a reference to fundamental rights generally, whereas the final legislative text limits this to only those fundamental rights that are protected through the Charter (and hence does not include rights protected as general principles of EU law and potentially general principles to be distilled by the CJEU in the future). This legislative history confirms that, during the legislative process, there was debate over the significance that fundamental rights should hold in the final text.

The formulation ‘fundamental rights enshrined in the Charter, including **democracy, the rule of law and environmental protection**’ is inherently confusing as these are, in fact, not fundamental rights protected by the Charter. Rather, environmental protection is a principle.⁶⁹ The reference to democracy and the rule of law is even more puzzling as these are not even principles but merely mentioned in the preamble to the CFR.⁷⁰ In contrast, no actual fundamental rights are explicitly singled out by Article 1(1), such as consumer protection⁷¹ despite repeated calls that it should be explicitly mentioned.⁷² This raises the question of how the explicit reference to these principles ought to be interpreted.

The draft regulation also made no mention of **environmental protection**. Environmental protection gained more prominence during the legislative procedure to the point Recital 1 of the final text states that the purpose of the AIA consists in ‘ensuring a high level of protection of health, safety, fundamental rights [...] including [...] environmental protection.’⁷³ The formulation Article 1(1) denotes it as a fundamental right, whereas it in fact is only a principle according to the CFR.⁷⁴ This formulation is awkward, and constitutes a sign of, at

⁶⁶ See further analysis of Article 6 AIA.

⁶⁷ Recital 1 dAIA.

⁶⁸ Article 114(3) TFEU.

⁶⁹ On this distinction, see further Tobias Lock, ‘Rights and Principles in the EU Charter of Fundamental Rights’ (2019) 56 Common Market Law Review 1201.

⁷⁰ The second sentence of the preamble of CFR states that ‘conscious of its spiritual and moral heritage, the Union is founded on the indivisible, universal values of human dignity, freedom, equality and solidarity; it is based on the principles of democracy and the rule of law. It places the individual at the heart of its activities, by establishing the citizenship of the Union and by creating an area of freedom, security and justice’.

⁷¹ Contrast this with Article 1(1) DSA, which does explicitly mention consumer protection: ‘The aim of this Regulation is to contribute to the proper functioning of the internal market for intermediary services by setting out harmonised rules for a safe, predictable and trusted online environment that facilitates innovation and in which fundamental rights enshrined in the Charter, including the principle of consumer protection, are effectively protected.’

⁷² The European Consumer Organisation (BEUC), ‘Regulating AI to Protect the Consumer’ (BEUC 2021) Position Paper BEUC-X-2021-088 <<https://www.beuc.eu/position-papers/regulating-ai-protect-consumer>> accessed 9 December 2024.

⁷³ Recital 1 AIA.

⁷⁴ See Article 37 CFR: ‘A high level of environmental protection and the improvement of the quality of the environment must be integrated into the policies of the Union and ensured in accordance with the principle of sustainable development’.

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times, the suboptimal legislative quality of the legislative text. It does not, however, alter the fact that, under EU primary law, environmental protection is considered a principle rather than a fundamental right.⁷⁵ Whereas the co-legislators deemed environmental protection to be one of the particularly important public interests⁷⁶ protected by the CFR that, *inter alia*, should also receive pronounced attention in risks assessment. However, given the hierarchy of norms in EU law, the AIA, as secondary legislation, cannot alter the status of environmental protection under EU law, as primary law defines it as a principle.

ii. Article 1(2): New supranational legal regimes for artificial intelligence

2.24 The AIA seeks to improve the functioning of the internal market by laying down a uniform legal framework for AI systems.⁷⁷ A closer examination reveals that the AIA does not establish a single legal framework for all AI systems. Instead, it introduces **multiple legal sub-regimes** tailored to different types of AI systems, based on their perceived level of risk. The second paragraph of Article 1 lists these various legal sub-regimes, which have all been designed to contribute to the overarching goal of ensuring ‘the free movement, cross-border, of AI-based goods and services’.⁷⁸

2.25 Article 1(2) lists the seven distinct sub-regimes. First, harmonized rules for the placing on the market, the putting into service and the use of AI systems in the EU.⁷⁹ Second, absolute prohibitions and qualified prohibitions of some AI practices deemed contrary to EU values.⁸⁰ Third, a *sui generis* legal regime for HRAIS.⁸¹ Fourth, transparency rules for particular kinds of AI systems.⁸² Fifth, rules for the placing on the market of general-purpose AI models.⁸³ Sixth, the creation of rules on market monitoring, surveillance, governance and enforcement.⁸⁴ Finally, measures to support innovation, with a particular focus on small and medium sized enterprises, including start-ups.⁸⁵ These legislative objectives can be best conceptualized as various legal ‘sub-regimes’ created by the AIA. **The wording of Article 2(1) is somewhat inconsistent.** At times, the provision refers to the creation of ‘obligations for operators’ of certain AI systems, while at other times, it references rules imposed directly on the AI systems themselves.⁸⁶ In practice, however, operators, by-and-large providers, are responsible to ensure compliance with the obligations set out in the regulation.⁸⁷ The various legal regimes for different categories of artificial intelligence are explored in greater detail in the corresponding sections of this book.

2.26 Article 1(2) only implicitly formulates the maybe most important characteristic of the AIA, namely that it **pre-empts Member States’ competence** in relation to AI systems. This crucial constitutional dimension has thus far been largely overlooked by commentators. There can,

⁷⁵ See further Elisa Morgera and Gracia Marín Durán, ‘Article 37 – Environmental Protection’ in Steve Peers and others (eds), *The EU Charter of Fundamental Rights: A Commentary* (2nd edn, Hart Publishing 2021) 1041.

⁷⁶ See also Recital 8 AIA.

⁷⁷ Recital 1 AIA.

⁷⁸ *Ibid.*

⁷⁹ Chapter I AIA, General Provisions.

⁸⁰ Chapter II AIA, Prohibited AI Practices.

⁸¹ Chapter III AIA, High-Risk AI Systems.

⁸² Chapter IV AIA, Transparency Obligations for Providers and Deployers of Certain AI Systems.

⁸³ Chapter V AIA, General-Purpose AI Models.

⁸⁴ Chapter IX AIA, Post-Market Monitoring, Information Sharing and Market Surveillance.

⁸⁵ Chapter VI AIA, Measures in Support of Innovation.

⁸⁶ Contrast for instance Article 2(1)(c) with Article 2(1)(d) AIA.

⁸⁷ Article 3(8) AIA defines operators as ‘a provider, product manufacturer, deployer, authorised representative, importer or distributor’.

however, be no question that the regulation functions as a maximum harmonization instrument that almost entirely eliminates national competence regarding AI.

The primary and overarching objective of the AIA is to prevent ‘Member States from imposing restrictions on the development, marketing and use of AI systems.’⁸⁸ Indeed, certain Member States had already explored the adoption of national rules on AI.⁸⁹ This was considered to constitute a danger to the internal market as diverging national rules lead to fragmentation and decrease legal certainty for operators of AI systems.⁹⁰ The AIA hence serves to **prevent national regulatory divergences** that hamper the free circulation, innovation, deployment and the uptake of AI systems and related products and services in the EU by laying down uniform norms through the adoption of EU secondary legislation on the basis of Article 114 TFEU, the legal basis for the adoption of Union legislation that has as its objective the establishment and functioning of the internal market.⁹¹ The supranational regulation of AI is deemed particularly important considering that AI systems ‘can be easily deployed in a large variety of sectors of the economy and many parts of society, including across borders, and can easily circulate throughout the Union.’⁹² Software can, indeed, circulate much more easily across the EU in comparison to physical goods and services originally covered by EU law.⁹³

Pre-emption arises where the EU legislates in an area of shared competence,⁹⁴ such as the internal market, meaning that Member States may no longer adopt measures in that field. It must, however, be underlined that the pre-emptive scope of the AI Act presents a **constitutional oddity** within the framework of EU law.⁹⁵ The Act’s pre-emption of national rules indeed extends to a domain significantly broader than the specific areas it directly regulates.⁹⁶ This is due to the fact that the AI Act seeks to prevent Member State competence in relation to the development, marketing, and use of AI systems while only imposing supranational rules in relation to the seven sub-regimes listed in Article 1(2), the scope of which is much narrower.⁹⁷

Throughout the **legislative process**, it was stressed that there was a need to prevent national norms that risk fragmenting the market and ‘to impose even higher regulatory burdens on operators developing or using AI systems’⁹⁸, to avoid fragmentation and to ensure legal certainty.⁹⁹ As per the Commission’s Impact Assessment, the intention to create only minimal transparency rules was that all ‘non-high-risk AI systems would be shielded from potentially diverging national regulations which would stimulate the creation of a single market for

⁸⁸ Recital 1 AIA.

⁸⁹ Recital 3 AIA.

⁹⁰ Ibid.

⁹¹ Ibid.

⁹² Recital 3 AIA.

⁹³ Recital 8 AIA.

⁹⁴ Article 2(2) TFEU, ‘When the Treaties confer on the Union a competence shared with the Member States in a specific area, the Union and the Member States may legislate and adopt legally binding acts in that area. The Member States shall exercise their competence to the extent that the Union has not exercised its competence. The Member States shall again exercise their competence to the extent that the Union has decided to cease exercising its competence.’

⁹⁵ See further Michèle Finck, ‘The Constitutional Dimension of AI Regulation in the EU: The AI Act and Its Impact on Member State Competence’, forthcoming, *Common Market Law Review* 2026.

⁹⁶ See further *ibid.*

⁹⁷ See further Recital 1 and Article 1(2) AIA and *ibid.*

⁹⁸ Commission Impact Assessment (n 22) 54.

⁹⁹ Recital 3 AIA.

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trustworthy AI and prevent the market fragmentation for this substantial category of non-high-risk AI systems.¹⁰⁰ The Explanatory Memorandum expressed concern that national regulations would lead to two major issues. First, a fragmentation of the internal market on legal requirements for the AI products and services, their marketing, their use, the liability and the supervision by public authorities. Second, a ‘substantial diminishment of legal certainty for both providers and users of AI systems’ on applicable rules.¹⁰¹

- 2.30** The AIA maximally harmonizes the field of the development, marketing, and use of AI systems at EU level, meaning that Member States are pre-empted from further legislating in this domain.¹⁰² The AIA, indeed, creates ‘harmonized rules for the placing on the market, the putting into service and the use of AI systems in the Union.’¹⁰³ Crucially, however, the AIA only creates rules for a subset of AI systems, as is clear from Article 1(2). This reflects an **unusual misalignment** between its substantive obligations, which focus only on certain kinds of AI systems, and its broad scope, covering all AI systems. This wide scope also sets the AIA apart from other NLF frameworks, which neither create separate legal regimes based on product risk levels nor harmonize areas where they impose no requirements.¹⁰⁴ The AIA hence pre-empts national rules in a domain in which it does not itself create rules. Indeed, whereas legislative harmonization under Article 114 TFEU is usually de-regulatory and re-regulatory in nature—meaning that the instrument in question would pre-empt national regulation in a certain field while, at the same time, creating new supranational norms in the same field—the AIA only has the first effect for the majority of AI systems.¹⁰⁵
- 2.31** Determining the **effect of legislative harmonization on national legislative autonomy** requires two steps. First, identifying the material scope of the EU measure, the so-called ‘occupied field’, and second, the extent of pre-emption on that occupied field.¹⁰⁶ The EU has occupied the ‘field’¹⁰⁷ of ‘rules for the placing on the market, the putting into service and the use of AI systems in the Union.’¹⁰⁸ Member States may no longer regulate in this field, irrespective of whether these rules substantially enter into conflict with EU law.¹⁰⁹
- 2.32** **National competence** in relation to AI systems has thus been largely neutralized through the AI Act.¹¹⁰ This is why the specific subregimes of the regulation create a regulatory ceiling,

¹⁰⁰ Commission Impact Assessment (n 22) 54.

¹⁰¹ <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52022PC0496>.

¹⁰² This results from the combined reading of Article 1(2)(a) and Recital 1 AIA.

¹⁰³ This is confirmed by the title of the AIA, which is the regulation ‘laying down harmonised rules on artificial intelligence’.

¹⁰⁴ See Michael Veale and Frederik Zuiderveen Borgesius, ‘Demystifying the Draft EU Artificial Intelligence Act — Analysing the Good, the Bad, and the Unclear Elements of the Proposed Approach’ (2021) 22 Computer Law Review International 97, 109.

¹⁰⁵ Commission Guidelines on the definition of an artificial intelligence system (n 65) 12. (‘The vast majority of systems, even if they qualify as AI systems within the meaning of Article 3(1) AI Act, will not be subject to any regulatory requirements under the AI Act’).

¹⁰⁶ Stephen Weatherill, ‘Models of Harmonisation: Maximum or Minimum’, *Contract Law of the Internal Market* (Intersentia 2016); Stephen Weatherill, ‘The Fundamental Question of Minimum or Maximum Harmonisation’ in Sacha Garben and Inge Govaere (eds), *The Internal market 2.0* (1st edn, Hart Publishing, an imprint of Bloomsbury Publishing 2021).

¹⁰⁷ Robert Schütze, ‘EU Competences: Existence and Exercise’ in Damian Chalmers and Anthony Arnall (eds), *The Oxford Handbook of European Union Law* (OUP 2015).

¹⁰⁸ Article 1(2)(a) AIA.

¹⁰⁹ Case C-216/86 *Antonini v Prefetto di Milano* EU:C:1987:322, [1987] ECR I-2919.

¹¹⁰ For an overview of some detailed exceptions to this general rule, see further Michèle Finck, ‘The Constitutional Dimension of AI Regulation in the EU: The AI Act and Its Impact on Member State Competence’, forthcoming, *Common Market Law Review* 2026.

not a floor, which would leave space for Member States to go beyond the regulatory protections of the AIA.¹¹¹ Looking ahead, the EU's regulatory reach will likely expand as more products and services incorporate AI, bringing them within the scope of the AIA.

III. Article 2: Scope

1. This Regulation applies to:
 - (a) providers placing on the market or putting into service AI systems or placing on the market general-purpose AI models in the Union, irrespective of whether those providers are established or located within the Union or in a third country;
 - (b) deployers of AI systems that have their place of establishment or are located within the Union;
 - (c) providers and deployers of AI systems that have their place of establishment or are located in a third country, where the output produced by the AI system is used in the Union;
 - (d) importers and distributors of AI systems;
 - (e) product manufacturers placing on the market or putting into service an AI system together with their product and under their own name or trademark;
 - (f) authorised representatives of providers, which are not established in the Union;
 - (g) affected persons that are located in the Union.
2. For AI systems classified as high-risk AI systems in accordance with Article 6(1) related to products covered by the Union harmonisation legislation listed in Section B of Annex I, only Article 6(1), Articles 102 to 109 and Article 112 apply. Article 57 applies only in so far as the requirements for high-risk AI systems under this Regulation have been integrated in that Union harmonisation legislation.
3. This Regulation does not apply to areas outside the scope of Union law, and shall not, in any event, affect the competences of the Member States concerning national security, regardless of the type of entity entrusted by the Member States with carrying out tasks in relation to those competences. This Regulation does not apply to AI systems where and in so far they are placed on the market, put into service, or used with or without modification exclusively for military, defence or national security purposes, regardless of the type of entity carrying out those activities. This Regulation does not apply to AI systems which are not placed on the market or put into service in the Union, where the output is used in the Union exclusively for military, defence or national security purposes, regardless of the type of entity carrying out those activities.
4. This Regulation applies neither to public authorities in a third country nor to international organisations falling within the scope of this Regulation pursuant to paragraph 1, where those authorities or organisations use AI systems in the framework of international cooperation or agreements for law enforcement and judicial cooperation with the Union or with one or more Member States, provided that such a third country or international organisation provides adequate safeguards with respect to the protection of fundamental rights and freedoms of individuals.

¹¹¹ On this distinction, see further William W Buzbee, 'Asymmetrical Regulation: Risk, Preemption, and the Floor/Ceiling Distinction' (2007) 82 NYU Law Review 1547 <<https://nyulawreview.org/issues/volume-82-number-6/asymmetrical-regulation-risk-preemption-and-the-floor-ceiling-distinction/>> accessed 11 December 2024.

5. This Regulation shall not affect the application of the provisions on the liability of providers of intermediary services as set out in Chapter II of Regulation (EU) 2022/2065.
6. This Regulation does not apply to AI systems or AI models, including their output, specifically developed and put into service for the sole purpose of scientific research and development.
7. Union law on the protection of personal data, privacy and the confidentiality of communications applies to personal data processed in connection with the rights and obligations laid down in this Regulation. This Regulation shall not affect Regulation (EU) 2016/679 or (EU) 2018/1725, or Directive 2002/58/EC or (EU) 2016/680, without prejudice to Article 10(5) and Article 59 of this Regulation.
8. This Regulation does not apply to any research, testing or development activity regarding AI systems or AI models prior to their being placed on the market or put into service. Such activities shall be conducted in accordance with applicable Union law. Testing in real world conditions shall not be covered by that exclusion.
9. This Regulation is without prejudice to the rules laid down by other Union legal acts related to consumer protection and product safety.
10. This Regulation does not apply to obligations of employers who are natural persons using AI systems in the course of a purely personal non-professional activity.
11. This Regulation does not preclude the Union or Member States from maintaining or introducing laws, regulations or administrative provisions which are more favourable to workers in terms of protecting their rights in respect of the use of AI systems by employers, or from encouraging or allowing the application of collective agreements which are more favourable to workers.
12. This Regulation does not apply to AI systems released under free and open-source licences, unless they are placed on the market or put into service as high-risk AI systems or as an AI system that falls under Article 5 or 50.

A. Overview

- 2.33** Article 2 defines the material, personal, and territorial scope of the AIA. It also details various exemptions, sets out conflict rules with other areas of EU law, and includes opening clauses that further refine the legislation's scope.¹¹²
- 2.34** The AI Act applies to a broad range of entities—providers, deployers, importers, distributors, product manufacturers, authorized representatives, and affected persons—provided there is a sufficient nexus between their activities and the EU. Article 2, however, does not exhaustively list all entities covered; for example, it does not refer to national supervisory authorities or notifying authorities and notified bodies nor to third parties supplying instruments, services, or components.¹¹³ While the Act's personal scope is clarified by its specific provisions, the territorial scope detailed in Article 2(1) is not further elaborated in later sections, emphasizing the critical importance of this initial provision.

¹¹² The definition of scope *ratione temporis* can be found in Articles 111 and 113 AIA.

¹¹³ See further Chapter IX, Section 4 of Chapter III and Article 25(4) AIA.

B. Analysis

Article 2(1) defines the personal and territorial scope of the regulation by specifying the actors subject to its provisions and the circumstances under which their actions fall within the regulation's scope. It relies on the concepts of the AI system and the GPAIM, which are examined in further detail in relation to their respective definitions in Articles 3(1) and (63), respectively.

i. Article 2(1)(a): Providers established or located in the EU or a third country

Article 2(1)(a) establishes that the AI Act applies to **providers** placing on the market or putting into service an AI system or placing on the market a general-purpose AI model irrespective of where they are located. Article 2(1)(a) has been subject to modifications throughout the legislative process. The inclusion of providers 'placing on the market general-purpose AI models in the Union' is the result of the inclusion of the specific regime on GPAIMs in Chapter V, Articles 51 et seqq. during the last stages of the legislative process.¹¹⁴ Providers are one of the possible 'operators' of an AI system.¹¹⁵ They are defined based on their activities. The related **definition** can be found in Article 3(3), which defines a provider as the 'natural or legal person, public authority, agency or other body that develops an AI system or a general-purpose AI model or that has an AI system or a general-purpose AI model developed and places it on the market or puts the AI system into service under its own name or trademark, whether for payment or free of charge'. The regulation does not define the notion of development, which hence should be interpreted in line with commonly accepted definitions, e.g. 'the action or process of bringing something to a fuller or more advanced condition.'¹¹⁶

The concept of **placing on the market** refers to 'the first making available of an AI system or a general-purpose AI model on the Union market'.¹¹⁷ An AI system or general-purpose AI model can hence be deemed to have been placed on the market where it has been made available commercially within the internal market. In turn, the concept of **putting into service** refers to 'the supply of an AI system for first use directly to the deployer or for its own use in the Union for its intended purpose'.¹¹⁸ An AI system can hence be considered to have been put into service where it is used, with the exception of research settings, by the provider or a deployer.¹¹⁹ It is crucial to note that whereas the AI Act applies to providers that place on the market or put into service AI systems, it only applies to providers that place on the market **GPAIM**. Indeed, the obligations formulated in the corresponding legal regime in Articles 53 and 55 are focused on the phase before a model is placed on the market. These rules are motivated by general-purpose AI model providers' 'particular role and responsibility along the AI value chain, as the models they provide may form the basis for a range of downstream systems, often provided by downstream providers that necessitate a good understanding of the models and their capabilities, both to enable the integration of such models into their products, and to fulfil their obligations under this or other regulations'.¹²⁰

¹¹⁴ As is now the case under Article 2(1)(a) AIA.

¹¹⁵ Article 3 (8) AIA.

¹¹⁶ Oxford University Press (n.d.). Development. In *Oxford English Dictionary*. Accessed 02 October 2025, from <https://www.oed.com/dictionary/development_n>

¹¹⁷ Article 3(9) AIA.

¹¹⁸ Article 3(11) AIA.

¹¹⁹ Note, however, the discussion on the research exemptions in Articles 2(6) and (8) AIA.

¹²⁰ Recital 101 AIA.

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- 2.38** The definition of the provider also underlines that the AI Act applies to legal as well as natural persons. Indeed, at least in theory also natural persons developing AI systems during a purely personal non-professional activity could be caught as Article 2(10) only exempts natural persons using AI systems as deployers. This definition, moreover, underlines that **public authorities**, including EU institutions, bodies and offices can also be providers of AI systems or general-purpose AI models.¹²¹
- 2.39** Providers are the equivalent of **product manufacturers** under the NLF. Indeed, definitions of product manufacturers under EU product law resemble the definition of providers in Article 3(3). For instance, Article 2(3) of the European Accreditation Regulation defines product manufacturers as ‘any natural or legal person who manufactures a product or has a product designed or manufactured, and markets that product under his name or trademark’.¹²² The same definition is also found elsewhere in EU product safety law.¹²³ According to Article 2(30) of the Medical Device Regulation (‘MDR’), a product manufacturer is the ‘natural or legal person who manufactures or fully refurbishes a device or has a device designed, manufactured or fully refurbished, and markets that device under its name or trademark’.¹²⁴
- 2.40** Considering that the AI Act is largely a provider-centric legal framework, providers emerge as its **central figure**. This emphasis is evident throughout the legislation, as most obligations, particularly those related to the design, development, and placing on the market of AI systems and general-purpose AI models, are directly incumbent upon them.¹²⁵ Most legal obligations under the AI Act are assigned to providers, who play a central role in ensuring compliance with the regulation. Providers bear the primary responsibility for aligning their internal processes and the technology they develop with the AIA’s requirements. This includes, for example, implementing a risk management system for HRAIS or ensuring compliance with the rules for general-purpose AI models set out in Chapter V.¹²⁶ The Impact Assessment stressed the importance of placing compliance obligations on providers by noting that existing ‘secondary legislation on fundamental rights primarily places the burden for compliance on the user and often leaves the provider of the AI system outside its scope’.¹²⁷ This is, for instance, the case with data protection law where deployers, but not providers, will usually qualify as data controllers.¹²⁸ This was deemed problematic as ‘while users remain responsible for a possible breach of fundamental rights obligations, providers might be best placed to prevent and mitigate some of the risks already at an early development stage’.¹²⁹ This is due to the fact that providers have more control over the technical structures underlying AI systems compared to deployers.¹³⁰ Experts have warned, however,

¹²¹ Recital 23 AIA.

¹²² Article 2(3) of the European Accreditation Regulation (Regulation (EC) No 765/2008 of the European Parliament and of the Council of 9 July 2008 setting out the requirements for accreditation and market surveillance relating to the marketing of products and repealing Regulation (EEC) No 339/93 218 [2008] OJ L218/30).

¹²³ Article 3(8) MSR.

¹²⁴ Article 2(30) MDR.

¹²⁵ See, in particular, Chapters III and V AIA.

¹²⁶ See further Article 9 AIA and in particular Articles 53 and 55 AIA.

¹²⁷ Commission Impact Assessment (n 22) 22.

¹²⁸ EDPB-EDPS, ‘Joint Opinion 5/2021 on the Proposal for a Regulation of the European Parliament and of the Council Laying down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)’ (18 June 2021) 9 <https://www.edpb.europa.eu/our-work-tools/our-documents/edpb-edps-joint-opinion/edpb-edps-joint-opinion-52021-proposal_en> accessed 9 December 2024.

¹²⁹ Commission Impact Assessment (n 22) 22.

¹³⁰ Michèle Finck, ‘Cobwebs of Control: The Two Imaginations of the Data Controller in EU Law’ (2021) 11 International Data Privacy Law 333.

that there will be situations where entities will formally qualify as providers under the AIA yet are technically unable to comply with its requirements.¹³¹

If a provider meets the criteria outlined in Article 3(3), its **geographical location** is irrelevant. Article 2(1)(a) explicitly applies to providers established both within the EU and in third countries. Consequently, the AIA has an **extraterritorial scope**, meaning that it applies to facts occurring outside of the territory of the EU. This rule is rooted in a tradition within EU digital law, which ensures that altering the legal or factual establishment cannot be used to circumvent legal protections applicable within the EU.¹³² By subjecting both domestic and foreign providers to its scope, the Act aims to prevent any potential competitive advantage for non-EU providers who might otherwise operate without comparable regulatory burdens.¹³³ Such circumvention is of course much easier to achieve for software than it would be for goods, say trucks, which require physical border crossings.¹³⁴ Relatedly, providers only fall within the scope of the AIA *if* they place on the market or put into service an AI system or place on the market a general-purpose AI model **in the Union**. This rule implies that providers that develop AI in the EU to place it on the market in a third country are not caught by the AIA. This has been criticized because it allows providers to develop AI systems in the EU that are contrary to EU values and would be unlawful under the AIA if marketed in the EU, for third countries.¹³⁵ This rule reflects the AIA's market-oriented approach, allowing activities that may benefit the market, even if they do not fully align with the principles of 'human-centric' and 'trustworthy' AI as long as the implications thereof do not concern the EU directly. It also mirrors other areas of EU law, such as the fact that many pesticides are banned in the EU but can be produced in the Union for export to third countries.¹³⁶

The AIA's emphasis on the provider, which mirrors the structure of EU product law has been **criticized**. For an elevator, it is undoubtedly the manufacturer who knows the product as well as its risks and limitations best. However, many different sources, systems, and parties contribute to the creation of polycentric AI systems and GPAIM.¹³⁷ Indeed, it has been pointed out regarding AI that 'the way downstream deployers use it and adapt it, may be as significant as how it is originally built'.¹³⁸ The taxonomy of AI actors is not straightforward, as the roles of providers and users can easily overlap. This, for instance, happens when a user supplies training data to the developer.¹³⁹ This highlights that it is challenging to translate this complex network of actors, data, models, and services into a legal framework that assigns duties and rights to specific, identifiable parties.¹⁴⁰ Studies have for instance shown

¹³¹ Ibid; Lilian Edwards, 'Regulating AI in Europe: Four Problems and Four Solutions' (Ada Lovelace Institute 2022) Expert Opinion, 8 <<https://www.adalovelaceinstitute.org/report/regulating-ai-in-europe/>> accessed 9 December 2024.

¹³² See, e.g. Article 3 GDPR.

¹³³ Recital 21 AIA.

¹³⁴ Recital 7 AIA.

¹³⁵ 'EU: Lawmakers Reluctant to Stop EU Companies Profiting from Surveillance and Abuse through the AI Act' (*Amnesty International*, 5 December 2023) <<https://www.amnesty.org/en/latest/news/2023/12/eu-lawmakers-reluctant-to-stop-eu-companies-profiting-from-surveillance-and-abuse-through-the-ai-act/>> accessed 9 December 2024.

¹³⁶ Silke Bollmohr and Susan Haffmans, 'Imports and Exports: Banned but Sold Anyway' (Heinrich Böll Stiftung, 18 October 2022) <<https://eu.boell.org/en/PesticideAtlas-imports-exports>> accessed 9 December 2024.

¹³⁷ Please refer to Chapter 1 (Introduction) of this book for a discussion of the polycentricity of these systems.

¹³⁸ Edwards (n 131) 6.

¹³⁹ Norberto Nuno Gomes de Andrade and Antonella Zarra, 'Artificial Intelligence Act: A Policy Prototyping Experiment: Operationalizing the Requirements for AI Systems – Part I' (Open Loop 2022) <<https://openloop.org/reports/2022/10/ai-act-operationalizing-the-requirements-for-ai-systems.pdf>> accessed 9 December 2024.

¹⁴⁰ Edwards (n 131) 6.

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that almost half of AI-startups use GPAIM developed by other providers as the basis for their own AI product.¹⁴¹ Yet, only providers of GPAIM are subject to the AIA, not deployers. The European Data Protection Board ('EDPB') and the European Data Protection Supervisor ('EDPS') have warned that 'it will not always be possible for a provider to assess all uses for the AI system'.¹⁴²

- 2.43** The centrality of providers under the AIA has also been criticized for it leaves 'too much discretion to AI providers to decide on politically charged matters'¹⁴³, such as the definition of risk to fundamental rights in the context of the obligations applicable to providers of HRAIS.¹⁴⁴ Article 25 addresses these issues to some extent, as it anticipates that in certain conditions parties that do not initially qualify as providers can become providers, such as where deployers substantially modify a HRAIS. Notwithstanding, the complexity of AI supply chains is likely to raise numerous questions regarding the implementation of the AIA.
- 2.44** Whereas Article 2(1)(a) only refers to providers, Chapter VI also explicitly applies to **prospective providers** in allowing them to partake in AI regulatory sandboxes and testing in real world conditions.¹⁴⁵

ii. Article 2(1)(b): Deployers that have their place of establishment or are located in the European Union

- 2.45** Pursuant to Article 2(1)(b), the AI Act also applies to deployers of AI systems that have their place of establishment or are located within the EU.¹⁴⁶ Deployers are one of the possible 'operators' under the AI Act.¹⁴⁷ Although the AI Act is a provider-centric legal framework, it also explicitly imposes some obligations on deployers, such as the obligations for deployers of high-risk AI systems listed in Article 26 or the transparency obligations in Articles 50(3) and (4). Article 25, moreover, recognizes that in some circumstances the influence of a deployer of a high-risk AI system may change in such a way that they need to be requalified as providers.
- 2.46** Pursuant to Article 3(4), the **deployer** is the 'natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity'.¹⁴⁸ Deployers are hence natural or legal persons using AI systems (but not GPAIM). The notion of **using a system under one's authority** is not clearly defined and can, at least in most cases, be interpreted according to common sense with reference to the definition of an AI system in Article 3(1). Consequently, one could argue that 'use' refers to relying on an AI system to infer an output based on the input it receives. The draft regulation initially referred to 'users' instead of deployers but this term was later removed to avoid confusion as users in the conventional sense are typically

¹⁴¹ Christian Rammer, 'Das Ökosystem Für KI-Startups in Deutschland: Vermarktung, Finanzierung, Fachkräfte Und Vernetzung in Unternehmensgründungen Im Bereich Künstliche Intelligenz' (Bundesministerium für Wirtschaft und Klimaschutz (BMWK) 2023) <<https://www.de.digital/DIGITAL/Redaktion/DE/Digitalisierungsindex/Publikationen/publikation-download-ki-startups-2023.html>> accessed 9 December 2024.

¹⁴² EDPB-EDPS Joint Opinion on the proposal for a Regulation of the AIA (n 128) 9.

¹⁴³ Nathalie A Smuha and others, 'How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission's Proposal for an Artificial Intelligence Act' (*Social Science Research Network*, 5 August 2021) <<https://papers.ssrn.com/abstract=3899991>> accessed 18 March 2025.

¹⁴⁴ See in particular Section 2 of Chapter III AIA.

¹⁴⁵ Please refer to the analysis of Article 57(5) AIA for an analysis of this concept.

¹⁴⁶ This establishment criterion echoes that in Article 3(1) GDPR.

¹⁴⁷ Article 3(8) AIA.

¹⁴⁸ See also Recital 13 AIA.

not responsible for compliance.¹⁴⁹ Similarly, the notion of ‘under its authority’ is also not defined, which leaves open the crucial question of what degree of control over the system is required in order to qualify as a deployer.¹⁵⁰

The regulation explicitly recognizes that **public authorities**, including EU institutions, bodies and offices can also be deployers of AI systems and that they are accordingly subject to related obligations.¹⁵¹

The exemption for AI systems used exclusively in a **personal, non-professional capacity** is discussed further below under Article 2(10).

The imposition of legal obligations on deployers provides a **key distinction with traditional product safety law**, which only imposes obligations on participants in the production chain, excluding users. In contrast to static products like toys, AI systems, as dynamic and continuously evolving entities, present unique regulatory challenges. Deployers are hence **Janus-faced actors**. On one hand, the regulation aims to protect those using AI, yet on the other, those same persons may also be subject to the AIA’s legal obligations, to minimize the risks of using these systems. This duality underscores the complexity of regulating systems that interact with their environment. Indeed, ‘AI is not a dishwasher and the way downstream deployers use it and adapt it, may be as significant as how it is originally built’.¹⁵²

Importantly, **no obligations for deployers of GPAIM** arise under the related regime in Chapter V. The exclusion of deployers of GPAIM can be seen as ironic. This is because, unlike highly specific high-risk AI systems (such as those embedded in a medical device or a toy), general-purpose AI models are inherently adaptable. Their broad utility means that it is often the deployer, rather than the provider, who introduces the ultimate context, application, and fine-tuning that can generate unforeseen or amplified risks. The deployer’s choices regarding data input, application domain, and user interaction are crucial in determining the real-world impact and potential for harm. This dynamic highlights a potential regulatory gap, where the party most capable of introducing new risks through application might bear fewer direct responsibilities under certain aspects of the Act compared to the model’s original provider.¹⁵³

As per Article 2(1)(b), the AI Act only applies to deployers that have their place of establishment in the EU or are located in the EU. In EU law, establishment refers to ‘the actual pursuit of an economic activity through a fixed establishment in another Member State’ ‘on a stable and continuous basis’.¹⁵⁴ The reference to location highlights that a natural or legal person can be considered to be located in the EU if they operate from the EU in practice, even if they do not meet the formal legal criteria for establishment.

Whereas this provision only refers to deployers, the AI Act also applies to **prospective deployers** as it enables and governs their participation in real-world testing.¹⁵⁵ Whereas the AIA does not define the notion of the prospective deployer, it can be assumed that this is the

¹⁴⁹ Article 3(4) dAIA.

¹⁵⁰ Article 3(4) dAIA.

¹⁵¹ Recital 23 AIA.

¹⁵² Edwards (n 131) 6.

¹⁵³ Note, however, the provisions in Articles 50(3) and (4) AIA.

¹⁵⁴ Case C-221/89 *Factortame* EU:C:1991:320, [1991] ECR I-3905, para 20 and Case C-55/94 *Gebhard* EU:C:1995:411, [1995] ECR I-4165, para 25.

¹⁵⁵ See further analysis of Article 60(2) AIA.

natural or legal person, public authority, agency or other body that plans to use an AI system under its authority.

iii. Article 2(1)(c): Providers and deployers of artificial intelligence systems the output of which is used in the European Union

- 2.53** Pursuant to Article 2(1)(c), the regulation applies to deployers and providers that have their place of establishment or are located in a third country where an output of an AI system is used in the EU.¹⁵⁶ The provision only covers providers and deployers of AI systems and not those of general-purpose AI models.
- 2.54** Article 2(1)(c) covers providers and deployers¹⁵⁷ that are either **established or located in a third country**. Third countries are countries that are not EU Member States.¹⁵⁸ In EU law, establishment refers to ‘the actual pursuit of an economic activity through a fixed establishment in another Member State’ ‘on a stable and continuous basis.’¹⁵⁹ The reference to location highlights that a natural or legal person can be considered to be located in the EU if they operate from the EU in practice, even if they do not meet the formal legal criteria for establishment.
- 2.55** This definition of **territorial scope** was considered necessary alongside the market and establishment criteria because, given their digital nature, certain AI systems should fall within the scope of the AIA even if they are not placed on the market, put into service, or used within the Union. This is essential to prevent regulatory circumvention and to ensure the effective protection of natural persons located in the EU.¹⁶⁰ The physical location and legal incorporation of a provider or deployer abroad should not hinder the enforcement of AIA requirements for AI systems used within the EU. Indeed, considering that AI is software, it can be easily deployed across borders and circulate easily between jurisdictions.¹⁶¹ Given the critical role of cloud computing, many AI systems can be conveniently transferred to third countries. Indeed, the specific location of an AI system is difficult for outsiders to determine. Similarly, the outputs generated by these systems can be ‘imported’ into the EU without necessarily physically crossing a border, unlike traditional goods like a refrigerator. The agility and immaterial nature of software present challenges for the future enforcement of Article 2(1)(c), particularly given the practical difficulty in verifying whether encrypted signals contain the output of an AI system. In practice, distinguishing between general digital data and digital data that specifically originates from an AI system is quite challenging.
- 2.56** The notion of ‘**output**’ is not defined in respect of Article 2(1)(c). Yet, in respect of the definition of an AI system¹⁶², Recital 12 lists examples of outputs, namely predictions, content, recommendations or decisions. This indicates that the concept ought to be interpreted broadly. The Commission Guidelines on Article 3(1) provide an interpretation of these notions. Predictions are interpreted as ‘an estimate about an unknown value (the output) from known values supplied to the system (the input)’, whereas content is ‘the generation of new material by an AI system.’¹⁶³ Recommendations are ‘suggestions for specific actions,

¹⁵⁶ Article 2(1)(c) AIA.

¹⁵⁷ For a definition, refer to Articles 3(3) and (4) AIA respectively.

¹⁵⁸ See further Case T-370/19 *Spain v Commission* paras 31–38.

¹⁵⁹ Case C-221/89 *Factortame* para 20 and Case C-55/94 *Gebhard* para 25.

¹⁶⁰ Recital 22 AIA.

¹⁶¹ Recital 3 AIA.

¹⁶² See further Article 3(1) AIA.

¹⁶³ Commission Guidelines on the definition of an artificial intelligence system (n 65) 10–11.

products, or services to users based on their preferences, behaviours, or other data inputs' and decisions are interpreted as 'conclusions or choices made by a system'.¹⁶⁴ In any case, these four categories of output are merely illustrative, meaning that other types of outputs may also fall within the scope of the regulation. Furthermore, for the purposes of the definition of an AI system, the regulation foresees that outputs may influence physical or virtual environments.¹⁶⁵

The concept of 'use' is also not defined in relation to Article 2(1)(c). The Guidelines on the Interpretation of the Prohibited Practices in Article 5 however state that 'while the "use" of an AI system is not explicitly defined in the AI Act, it should be understood in a broad manner to cover the use or deployment of the system at any moment of its lifecycle after having been placed on the market or put into service'.¹⁶⁶ Whereas the notion of 'use' should be interpreted consistently across the regulation, it is questionable whether this interpretation can be relied on here considering that as per these non-binding guidelines, 'use' is not independent of the concepts of placing on the market and putting into service. Yet, a closer look at the regulation indicates that none of this may be needed to be caught by the AI Act. Remarkably, according to Recital 22 even outputs **intended to be used** are within the scope of the AIA, a formulation that is not found in Article 2(1)(c). This divergence can probably be explained by the controversial nature of this formulation as in EU law-making formulations that do not attract consensus but still have noticeable support are sometimes placed into the recitals instead of the text by way of compromise. The opening in Recital 22 recalls the familiar debate in EU law as to whether and when formulations in recitals that change the substantive meaning of an article are binding.¹⁶⁷ It also raises challenges in respect of **enforcement**. Whereas the detection of when an output generated outside of the EU enters the EU is difficult enough, a detection of an intention to use an output is even more difficult due to its subjective nature, and more intrusive into extraterritorial matters. The justification for the extraterritorial application of the AIA is also less clear as where there is merely an intention to use as opposed to an actual use, no risks materialize in the EU. Enforcement challenges can generally be anticipated in relation to Article 2(1)(c) as supervisory authorities will struggle to even identify when an output produced outside of the EU is used within the Union. Unlike classical goods, software can cross borders largely unnoticed.¹⁶⁸ Its digital nature allows it to move across jurisdictions without the physical checkpoints or regulatory scrutiny typically applied to tangible products, making it more challenging to monitor and control its circulation.¹⁶⁹ Article 2(1)(c) also potentially covers situations in which an output is used in the EU although the provider never intended for this to be the case. Imagine a South Korean company developing an AI system designed for the market in South Korea and the United States. The AI system is then acquired by a US company, which in turn uses it with one of its customers in the EU. In that case, the South Korean company could fall within the scope of Article 2(1)(c) although it never intended for its product to be used in the EU. This also raises important questions about the temporal scope of the regulation's application—specifically, whether provider obligations can arise

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¹⁶⁴ Ibid 11.

¹⁶⁵ Article 3(1) and Recital 12 AIA.

¹⁶⁶ Commission, 'Guidelines on prohibited artificial intelligence practices established by Regulation (EU) 2024/1689 (AI Act) C(2025) 884 final, 5.

¹⁶⁷ On the legal status of recitals in EU law see further related analysis in the Introduction.

¹⁶⁸ Recital 3 AIA.

¹⁶⁹ The definition of an AI system in Article 3(1) AIA does not presuppose the physicality of the system.

retrospectively if the provider falls within the scope of the regulation only after the system has been developed.¹⁷⁰

2.58 Article 2(1)(c) in any event underscores that the AIA has an **extraterritorial scope**, meaning that it applies to facts occurring outside of the territory of the EU. This is typical for EU digital regulation and rules regarding the extraterritorial application of EU law can also be found, e.g. in Article 3 GDPR, Article 1(3) of the Data Act or Article 2(1) of the Digital Services Act. This is also echoed by Article 2(1)(d), according to which importers and distributors of AI systems fall within the scope of the regulation and Article 2(1)(g), according to which affected persons that are located in the EU are also within its scope.

iv. Article 2(1)(d): Importers and distributors of AI systems

2.59 Article 2(1)(d) establishes that importers and distributors of AI systems, but not of GPAIM sometimes fall within the scope of the AI Act. Whereas this provision refers to importers of AI systems generically, the regulation only imposes obligations on providers of **high-risk AI systems**.¹⁷¹

2.60 The **importer** is a natural or legal person that is located or established in the Union that places on the market an AI system that bears the name or trademark of a natural or legal person established in a third country.¹⁷² Third countries are countries that are not EU Member States.¹⁷³ Importers are one of the possible ‘operators’ of an AI system.¹⁷⁴ Article 23 lists the various obligations applicable to importers. These include an obligation to ensure that a high-risk AI system is in conformity with the regulation before it can be placed on the market as well as other verification, risk identification, care, cooperation and record-keeping obligations.¹⁷⁵ Pursuant to Article 25(1), an importer can also be reclassified into a provider of a high-risk AI system in a number of circumstances, such as where they put their name or trade-mark on the system.¹⁷⁶

2.61 A **distributor** is a natural or legal person ‘in the supply chain, other than the provider or the importer, that makes an AI system available on the Union market’.¹⁷⁷ Distributors are one of the possible ‘operators’ of high-risk AI systems.¹⁷⁸ Article 24 is dedicated specifically to the responsibilities of distributors, which include, inter alia, an obligation to ensure that the HRAIS that they distribute bears the required CE marking, it is accompanied by a copy of the EU declaration of conformity and instructions for use, and that the provider and the importer of that system, as applicable, have complied with their respective obligations as laid down in Articles 16 and 23.¹⁷⁹ Distributors are also subject to other verification, risk identification, care, cooperation and record-keeping obligations.¹⁸⁰

¹⁷⁰ This discussion is also relevant in the context of Article 111 AIA, which contains transition rules for AI that pre-dates the dates on which the various regimes of the AI Act become binding.

¹⁷¹ Article 23 AIA. Whether an AI system qualifies as a high-risk AI system is determined by the classification rules in Article 6 AIA.

¹⁷² Article 3(6) AIA. See above for a discussion of the concepts of establishment and location.

¹⁷³ Case T-370/19 *Spain v Commission* paras 31–38.

¹⁷⁴ Article 3(8) AIA.

¹⁷⁵ Article 23(1) AIA.

¹⁷⁶ Article 25(1) AIA. See also Recitals 84 and 85.

¹⁷⁷ Article 3(7) AIA.

¹⁷⁸ Article 3(8) AIA.

¹⁷⁹ Article 24(1) AIA. See also Recital 83 AIA.

¹⁸⁰ See further Article 24 AIA.

v. Article 2(1)(e): Product manufacturers

Product manufacturers are also one of the possible ‘operators’ of HRAIS.¹⁸¹ The notion must be interpreted in accordance with the specific definition provided within the product law framework under which they qualify as manufacturers.¹⁸² In accordance with Article 25(3), for high-risk AI systems that are safety components of products covered by the Union harmonization legislation listed in **Section A of Annex I**, the product manufacturer is the provider of the high-risk AI system. Consequently, the product manufacturer is subject to related obligations, if either the system is placed on the market together with the product under the name or trademark of the product manufacturer, or the system is put into service under the name or trademark of the product manufacturer after the product has been placed on the market.¹⁸³ Assigning responsibility to manufacturers for AI systems integrated into their products under their name or trademark is intended to create unambiguous responsibility and enhance accountability within the regulatory framework. Considering that the regulation does not impose additional obligations or penalties on product manufacturers other than specifying that they are required to fulfil the provider obligations for HRAIS—it is questionable whether their explicit inclusion in Article 2 adds value.

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vi. Article 2(1)(f): Authorized representatives of providers

Prior to making their HRAIS or GPAIM available on the internal market, providers established in third countries have to, by written mandate, appoint an authorized representative which is established in the Union. Authorized representatives of HRAIS are governed by Article 22, whereas authorized representatives of general-purpose AI models are governed by Article 54. Third countries are countries that are not EU Member States.¹⁸⁴ The concept of the authorized representative serves to secure representation of a provider within the EU for providers established in a third country. Third countries are countries that are not EU Member States.¹⁸⁵ The concept of the permanent representative is well-established within the New Legislative Framework and can, for instance, also be found in Article 2(32) MDR. It is a well-established principle of EU product safety law that while manufacturers established in third countries are subject to an obligation to designate an authorized representative, those established in the EU may also do so.¹⁸⁶ Despite not being explicitly foreseen in the AIA, there is no reason why this principle should not also apply in respect of providers of HRAIS.

2.63

vii. Article 2(1)(g): Affected persons that are located in the EU

Article 2(1)(g) governs affected persons that are located in the EU. This is due to the fact that ‘depending on the type of AI system, the use of the system may affect persons other than the deployer.’¹⁸⁷ Unlike providers and deployers, affected persons can be subject to the AIA in

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¹⁸¹ Article 3(8) AIA.

¹⁸² Recital 87 AIA indeed provides that ‘in addition, where a high-risk AI system that is a safety component of a product which falls within the scope of Union harmonisation legislation based on the New Legislative Framework is not placed on the market or put into service independently from the product, the product manufacturer defined in that legislation should comply with the obligations of the provider established in this Regulation and should, in particular, ensure that the AI system embedded in the final product complies with the requirements of this Regulation’.

¹⁸³ Article 25(3) AIA.

¹⁸⁴ See further Case T-370/19 *Spain v Commission* paras 31–38.

¹⁸⁵ *Ibid.*

¹⁸⁶ Commission Notice ‘The “Blue Guide” on the implementation of EU products rules 2022’ [2022] OJ C247/1, point 3.2.

¹⁸⁷ Recital 13 AIA.

a passive capacity, without actively having decided to use or create an AI system. The provision does not define the notion of the ‘affected person’ nor ‘located’. Presumably, it only refers to natural persons and anyone located in the EU, even if just for a short period of time, without the need for, say, permanent residency. Its formulation leaves no doubt that EU citizenship is not a prerequisite.

- 2.65** The draft regulation did not refer to affected persons. This was criticized by the European Parliament as well as the EDPB and EDPS.¹⁸⁸ The omission of affected persons was also systemically incoherent, given that one of the primary objectives of the AIA is the mitigation of risks to fundamental rights. Moreover, their inclusion became necessary with the introduction of Articles 85 and 86, which respectively foresee a right to lodge a complaint with a Market Surveillance Authority and the right to an explanation of individual decision-making. Whereas these provisions allocate certain rights to affected persons, the latter have no obligations under the AI Act.

viii. Article 2(2): High-risk AI systems used in the area of mobility

- 2.66** Article 2(2) establishes how the regulation applies to certain HRAIS covered by existing sectoral EU harmonization legislation in Section B of Annex I. Whereas Section A of Annex I lists harmonizing legislation based on the NLF, Section B lists ‘other’ EU harmonizing legislation, which is not based on the NLF.

- 2.67** The draft regulation foresaw that for HRAIS that are safety components of products or systems, or which are themselves products or systems and which fall within the scope of Regulation 300/2008;¹⁸⁹ Regulation 167/2013;¹⁹⁰ Regulation 168/2013;¹⁹¹ Directive 2014/90;¹⁹² Directive 2016/797;¹⁹³ Regulation 2018/858;¹⁹⁴ Regulation 2018/1139;¹⁹⁵ Regulation 2019/2144,¹⁹⁶ only Article 84 dAIA would apply.¹⁹⁷ In accordance with the Parliament and Council mandates, the final legislative text no longer itself enumerates a closed list of

¹⁸⁸ EDPB-EDPS Joint Opinion on the proposal for a Regulation of the AIA (n 128) 8–9.

¹⁸⁹ Regulation (EC) No 300/2008 of 11 March 2008 on common rules in the field of civil aviation security and repealing Regulation (EC) 2320/2002 [2008] OJ L97/72.

¹⁹⁰ Regulation (EU) No 167/2013 of 5 February 2013 on the approval and market surveillance of agricultural and forestry vehicles [2013] OJ L60/1.

¹⁹¹ Regulation (EU) No 168/2013 of 15 January 2013 on the approval and market surveillance of two- or three-wheel vehicles and quadricycles [2013] OJ L60/52.

¹⁹² Directive 2014/90/EU of 23 July 2014 on marine equipment and repealing Council Directive 96/98/EC [2014] OJ L257/146.

¹⁹³ Directive (EU) 2016/797 of 11 May 2016 on the interoperability of the rail system within the European Union (recast) [2016] OJ L138/44.

¹⁹⁴ Regulation (EU) 2018/858 of 30 May 2018 on the approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles, amending Regulations (EC) No 715/2007 and (EC) No 595/2009 and repealing Directive 2007/46/EC [2018] OJ L151/1.

¹⁹⁵ Regulation (EU) 2018/1139 of 4 July 2018 on common rules in the field of civil aviation and establishing a European Union Aviation Safety Agency, and amending Regulations (EC) No 2111/2005, (EC) No 1008/2008, (EU) No 996/2010, (EU) No 376/2014 and Directives 2014/30/EU and 2014/53/EU of the European Parliament and of the Council, and repealing Regulations (EC) No 552/2004 and (EC) No 216/2008 of the European Parliament and of the Council and Council Regulation (EEC) No 3922/91 [2018] OJ L212/1.

¹⁹⁶ Regulation (EU) 2019/2144 of 27 November 2019 on type-approval requirements for motor vehicles and their trailers, and systems, components and separate technical units intended for such vehicles, as regards their general safety and the protection of vehicle occupants and vulnerable road users, amending Regulation (EU) 2018/858 of the European Parliament and of the Council and repealing Regulations (EC) No 78/2009, (EC) No 79/2009 and (EC) No 661/2009 of the European Parliament and of the Council and Commission Regulations (EC) No 631/2009, (EU) No 406/2010, (EU) No 672/2010, (EU) No 1003/2010, (EU) No 1005/2010, (EU) No 1008/2010, (EU) No 1009/2010, (EU) No 19/2011, (EU) No 109/2011, (EU) No 458/2011, (EU) No 65/2012, (EU) No 130/2012, (EU) No 347/2012, (EU) No 351/2012, (EU) No 1230/2012 and (EU) 2015/166 [2019] OJ L325/1.

¹⁹⁷ Article 2(2) dAIA.

legislation but a reference to harmonization legislation in Section B of Annex I, which currently lists Regulation 300/2008; Regulation 167/2013; Regulation 168/2013; Directive 2014/90; Directive 2016/797; Regulation 2018/858; Regulation 2018/1139; Regulation 2019/2144.¹⁹⁸ It also further provides that Article 57 also applies insofar that the requirements for HRAIS under the AIA have been integrated under EU harmonizing legislation.

The 'other' harmonizing legislation listed in Section B of Annex I all relate to the domain of **mobility**, namely the Regulation on common rules in the field of civil aviation security;¹⁹⁹ the Regulation on the approval and market surveillance of two- or three-wheel vehicles and quadricycles;²⁰⁰ the Regulation on the approval and market surveillance of agricultural and forestry vehicles;²⁰¹ the Directive on marine equipment;²⁰² the Directive on the interoperability of the rail system within the European Union;²⁰³ the Regulation on the approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles;²⁰⁴ the Regulation on type-approval requirements for motor vehicles and their trailers, and systems, components and separate technical units intended for such vehicles, as regards their general safety and the protection of vehicle occupants and vulnerable road users;²⁰⁵ and the Regulation on common rules in the field of civil aviation and establishing a European Union Aviation Safety Agency.²⁰⁶ Since these legal frameworks are not part of the NLF, aligning their mechanisms with Chapter III, which is grounded in the NLF, would have proven challenging.

Article 2(2) accordingly excludes HRAIS covered by section B of Annex I from all provisions of the AIA, with the exception of Article 6(1), which enables the classification of these systems; Articles 102 to 109, which govern updates to related secondary legislation; and Article 112, which governs the review of the AIA. Article 57 applies only insofar as the requirements for high-risk AI systems under the AIA have been integrated in that Union harmonization legislation.

ix. Article 2(3): Areas outside the scope of EU law, in particular military, defence and national security

Article 2(3) provides that the AIA does not apply to areas outside the scope of EU law, in particular in the domain of national security, defence or military purposes.²⁰⁷ The provision thus serves as a limitation on the regulation's pre-emptive effect over national competence. The scope of this exemption was one of the most **contentious** issues during the legislative procedure. Initially, the draft regulation was intended to 'not apply to AI systems developed or used exclusively for military purposes.'²⁰⁸ While the Parliament had no objections to this formulation, the Council, as the institution representing Member States, demanded a widening of the exemption, particularly concerning matters of national security. This emerged as one of the most controversial issues during the trilogue negotiations, as several Member States strongly advocated for the inclusion of a broad national security exemption.

¹⁹⁸ Section B of Annex I AIA.

¹⁹⁹ Regulation (EC) No 300/2008 (n 189).

²⁰⁰ Regulation (EU) No 168/2013 (n 191).

²⁰¹ Regulation (EU) No 167/2013 (n 190).

²⁰² Directive 2014/90/EU (n 192).

²⁰³ Directive (EU) 2016/797 (n 193).

²⁰⁴ Regulation (EU) 2018/858 (n 194).

²⁰⁵ Regulation (EU) 2019/2144 (n 196).

²⁰⁶ Regulation (EU) 2018/1139 (n 195).

²⁰⁷ This echoes other exemptions in EU digital regulation such as Article 2(2)(a) GDPR.

²⁰⁸ Article 2(3) dAIA.

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- 2.71** While Article 2(3) explicitly refers to AI systems, its first sentence also contains a broader statement that the regulation does not apply to areas **outside the scope of Union law** and shall not, under any circumstances, **affect the competences of the Member States with regard to national security**—regardless of the type of entity entrusted with related tasks.²⁰⁹ This raises the question of whether the exemption could also be interpreted as applying to general-purpose AI models. The wording of the provision appears to limit the exemption to AI systems, without extending it to general-purpose AI models. However, this narrow interpretation is open to debate, particularly considering the otherwise expansive nature of the national security exemption. The fact that both the provisions on general-purpose AI models and the exemption in Article 2(3) were finalized in the closing stages of the legislative negotiations further casts doubt on whether the exclusion of GPAIM a deliberate decision or was a drafting oversight.
- 2.72** **Military activities** are not automatically and entirely excluded from the scope of EU law but fall under the EU's Common Security and Defence Policy.²¹⁰ However, they are explicitly outside the scope of the AIA. The final text of Article 2(3) refers to systems that are placed on the market, put into service, or used 'exclusively for military, defence or national security purposes, regardless of the type of entity carrying out these activities'. This implies that this exemption also applies where Member States delegate tasks in the context of national security to **private actors**. Article 2(3) also provides that the regulation does not apply to AI systems which are not placed on the market or put into service in the EU 'where the output is used in the Union exclusively for military, defence or national security purposes, regardless of the type of entity carrying out these activities'.²¹¹ It follows that where an output of an AI system is used in the EU exclusively for military, defence or national security purposes, the AIA does not apply. This concerns situations where an AI system is simply used within the EU (but not placed on the market or put into service) or to AI system outside the EU, whose outputs are utilized for designated purposes within the EU. Pursuant to Recital 24, the exclusion of military, defence and national security is justified both by Article 4(2) TEU and by the specificities of the Member States and the common Union defence policy covered by Chapter 2 of Title V TEU that are subject to public international law, which is therefore the more appropriate legal framework for the regulation of AI systems in the context of the use of lethal force and other AI systems in the context of military and defence activities.²¹² It is worth noting that other product law instruments foresee similar exemptions as the Machinery Regulation ('MR') excludes 'machinery specially designed and constructed for military or police purposes'.²¹³

²⁰⁹ For a discussion of the notion 'outside the scope of EU law' see further Michèle Finck, 'The Constitutional Dimension of AI Regulation in the EU: The AI Act and Its Impact on Member State Competence', forthcoming, *Common Market Law Review* 2026.

²¹⁰ Vincenzo Randazzo, 'Article 346 and the Qualified Application of EU Law to Defence' (*European Union Institute for Security Studies (EUISS)*, 2014) <<https://www.iss.europa.eu/publications/briefs/article-346-and-qualified-application-eu-law-defence>> accessed 9 December 2024; Jelena von Achenbach, 'The EU's New Defence Policy – Beyond the Distinction Supranational / Intergovernmental' (*Verfassungsblog*, 28 March 2019) <<https://verfassungsblog.de/the-eus-new-defence-policy-beyond-the-distinction-of-supranational-and-intergovernmental/>> accessed 9 December 2024.

²¹¹ Article 2(3) AIA.

²¹² Recital 24 AIA.

²¹³ Article 2(2)(I) of the Machinery Regulation (Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC of the European Parliament and of the Council and Council Directive 73/361/EEC [2023] OJ L165/1).

There is no universally agreed upon definition of ‘military, defence or national security purposes’ in the AIA or, indeed, EU law more broadly, allowing Member States considerable discretion in determining what falls under the exemption. In *La Quadrature du Net* the Grand Chamber of the CJEU provided a definition of national security for the first time. It stated that ‘responsibility corresponds to the primary interest in protecting the essential functions of the State and the fundamental interests of society and encompasses the prevention and punishment of activities capable of seriously destabilising the fundamental constitutional, political, economic or social structures of a country and, in particular, of directly threatening society, the population or the State itself, such as terrorist activities.’²¹⁴ This is reminiscent of the derogation clause in Article 15 of the European Convention on Human Rights that refers to ‘time[s] of war or other public emergency threatening the life of the nation.’²¹⁵ Other case law has clarified that national security does not include activities related to road safety or the organization or administration of justice.²¹⁶

Importantly, the CJEU also ruled that ‘although it is for the Member States to define their essential security interests and to adopt appropriate measures to ensure their internal and external security, the mere fact that a national measure has been taken for the purpose of protecting national security cannot render EU law inapplicable and exempt the Member States from their obligation to comply with that law.’²¹⁷ It follows that if national security affects an area where the EU has competence and that is covered by EU law (such as where AI system is used in national security to generate insights that are then fed into systems outside the pure national security context), the exercise of that exclusive national competence may not impinge on the EU legal order or undermine the relevant EU legal rules. However, the definition of national security, particularly its distinction from the concept of public security, remains fraught with uncertainty, making it challenging to determine what falls within the AIA’s exemptions and what does not.²¹⁸

The scope of the exemptions in Article 2(3) hinges on the interpretation of ‘exclusively for military, defence or national security purposes.’²¹⁹ Recital 24 provides important clarification on the interpretation of the **condition of exclusivity**. It specifies that if an AI system, originally developed, placed on the market, put into service, or used for military, defence, or national security purposes, is temporarily or permanently utilized outside these domains, it falls within the scope of the regulation.²²⁰ This interpretation ensures that AI systems with **dual-use** potential do not automatically escape regulatory oversight under the AIA simply due to their initial military or security-related purpose.²²¹ The fact that an AI system may fall within the scope of the AI Act should not affect the ability of entities carrying out national

²¹⁴ Joined Cases C-511/18, C-512/18 and C-520/18 *La Quadrature du Net and Others* EU:C:2020:791, para 135.

²¹⁵ Article 15 ECHR.

²¹⁶ Case C-439/19 *Latvijas Republikas Saeima* EU:C:2021:504, para 68 and Case C-204/21 *Commission v Poland* EU:C:2023:442, para 319.

²¹⁷ Case C-623/17 *Privacy International* EU:C:2020:790, para 44.

²¹⁸ Plixavra Vogiatzoglou, ‘The AI Act National Security Exception’ (*Verfassungsblog*, 9 December 2024) <<https://verfassungsblog.de/the-ai-act-national-security-exception/>> accessed 19 February 2025.

²¹⁹ Article 2(3) AIA.

²²⁰ Recital 24 AIA, which provides that ‘Nonetheless, if an AI system developed, placed on the market, put into service or used for military, defence or national security purposes is used outside those temporarily or permanently for other purposes, for example, civilian or humanitarian purposes, law enforcement or public security purposes, such a system would fall within the scope of this Regulation. In that case, the entity using the AI system for other than military, defence or national security purposes should ensure the compliance of the AI system with this Regulation, unless the system is already compliant with this Regulation.’

²²¹ Recital 24 AIA.

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security, defence and military activities to use that system for national security, military and defence purposes, regardless of the type of entity carrying out those activities.²²²

2.76 AI systems used in military, defence, and national security contexts are among the most controversial²²³ and considering that they can intentionally lead to loss of life certainly also make them the riskiest. Yet, they fall outside the scope of the legal framework designed to manage AI risks in the EU. This has unsurprisingly **attracted controversy**. Experts have noted that these exemptions ‘make the entire military-industrial-political complex a largely digital rights-free zone’²²⁴ as the principled exclusion from the AI Act goes ‘well beyond what is required.’²²⁵ This is deemed ‘problematic considering that the European Union is itself actively involved in the development of military AI systems through the European Defence Fund.’²²⁶ Another important point to highlight is the challenging distinction between AI systems used in the context of national security—which are exempt under Article 2(3)—and those employed in law enforcement, which are classified as HRAIS.²²⁷ The distinction of these two areas is for instance challenging in relation to terrorist activity or large-scale organized crime.²²⁸

2.77 x. Article 2(4): Public authorities in a third country and international organizations Article 2(4) provides an **exemption to the otherwise broad territorial scope** of the regulation.²²⁹ It provides that the AIA does not apply to public authorities in a third country or international organizations that would otherwise fall within the scope of the AIA as per Article 2(1) where they use AI systems ‘in the framework of international cooperation or agreements for law enforcement and judicial cooperation’ with the EU or at least one Member State.²³⁰ Third countries are countries that are not EU Member States.²³¹

2.78 This exemption can, however, only be invoked provided that the country or organization ‘provides **adequate safeguards** with respect to the protection of fundamental rights and freedoms of individuals.’²³² The justification for this limitation is found in Recital 22, which considers it necessary ‘to take into account existing arrangements and special needs for future cooperation with foreign partners with whom information and evidence is exchanged.’²³³ The legislative formulation leaves open the question of whom ought to assess whether there are ‘adequate safeguards’ with respect to the protection of fundamental rights and freedoms of individuals. The combined reading of Article 74(8) and Recital 22, moreover, implies that this task falls on the data protection authorities acting as market surveillance authorities for the supervision of AI systems used in law enforcement and justice.

²²² Ibid. See also Commission Guidelines on prohibited artificial intelligence practices (n 166) 8.

²²³ Instead of many see Nathan Gabriel Wood, ‘The Problem with Killer Robots’ (2020) 19 *Journal of Military Ethics* 220; Douwe Korff, ‘Opinion on the Implications of the Exclusion from New Binding European Instruments on the Use of AI in Military, National Security and Transnational Law Enforcement Contexts’ (European Center for Non-for-Profit Law 2022) 8 <<https://ecnl.org/news/rights-free-zone-blanket-national-security-exemption-ai-legislation>> accessed 9 December 2024.

²²⁴ Korff (n 223) 2.

²²⁵ Ibid 8.

²²⁶ Smuha and others (n 143) 18.

²²⁷ Article 6(2) and Point 6 of Annex III AIA.

²²⁸ Articles 3(46) and 2(3) AIA respectively.

²²⁹ Article 2(1) AIA.

²³⁰ Third countries are countries that are not EU Member States. See further Case T-370/19 *Spain v Commission* paras 31–38.

²³¹ Ibid.

²³² Article 2(4) AIA.

²³³ Recital 22 AIA.

The **contours** of this exemption will undoubtedly be tested in the future, particularly in cases where the use of an AI system is initially covered by Article 2(4), but the system's output is subsequently used within the EU in scenarios that would otherwise fall under the AIA's purview, such as law enforcement.²³⁴ This raises important questions regarding the scope of the exemption and whether it remains applicable when the output of an AI system affects activities regulated under EU law. For instance, the results produced by AI systems operated by third countries or international organizations may be integrated into the systems of entities subject to EU law, thereby influencing or determining their decisions. This scenario is particularly relevant in the context of international agreements involving EU institutions, such as Europol and Frontex, EU missions, or national law enforcement agencies operating within the framework of EU law. It is hence unsurprising that the exemption in Article 2(4) has been **criticized**. The joint opinion of the EDPB and EDPS stressed 'serious concerns regarding the exclusion of international law enforcement cooperation' considering that it 'creates a significant risk of circumvention (e.g., third countries or international organisations operating high-risk applications relied on by public authorities in the EU)'.²³⁵ Douwe Koff has argued that rather than exempting them, 'no such "arrangements" and "agreements" should ever be entered into or adopted if they do not respect, or lead to violations of, the rights and freedoms of individuals under the Charter'.²³⁶

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xi. Article 2(5): Conflict rule for intermediary liability

Article 2(5), (7), (9) and (11) set out conflict rules for the relationship between the AIA and other instruments of EU law. While the AIA will intersect with many other areas of law beyond those listed here, the co-legislators considered the latter to be particularly relevant. Their reference leaves no doubt that there will be no *lex posterior derogat legi priori* effect in relation to these acts.²³⁷

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Pursuant to Article 2(5), the regulation does not affect the provisions on the liability of intermediary services as set out in Chapter III of the Digital Services Act ('DSA'). These rules establish the liability for such providers where they (a) merely transmit information; (b) cache information for the sole purpose of making the transmission of information more efficient or (c) only store information.²³⁸ This conflict rule is reminiscent of that found in other domains of EU digital regulation.²³⁹

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xii. Article 2(6): AI systems and models used for the sole purpose of scientific research and development

Article 2(6) is one of two **research exemptions** provided for in the regulation.²⁴⁰ It exempts any AI systems and models as well as their output from the scope of the AIA where it is 'specifically developed and put into service for the purpose of scientific research and

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²³⁴ See further Article 6(2) and Point 6 of Annex III AIA.

²³⁵ EDPB-EDPS Joint Opinion on the proposal for a Regulation of the AIA (n 128) 8.

²³⁶ Korff (n 223) 8.

²³⁷ On the status of these principles in EU law, see further Emily Hancox, 'Judicial Approaches to Norm Overlaps in EU Law: A Case Study on the Free Movement of Workers' (2021) 58 Common Market Law Review 1057.

²³⁸ See Chapter II DSA and Articles 12-15 of e-Commerce Directive (Directive 2000/31/EC of the European Parliament and of the Council of 8 June 2000 on certain legal aspects of information society services, in particular electronic commerce, in the Internal Market ('Directive on electronic commerce') [2000] OJ L178/1).

²³⁹ For an interpretation of these terms, see Articles 3(9) and (10) AIA. See further Recital 21 GDPR.

²⁴⁰ See also Article 2(8) AIA.

development.²⁴¹ The initial draft of the Commission did not include an explicit exemption for research, prompting calls for the introduction of a dedicated provision to safeguard academic freedom and promote innovation within the EU.²⁴²

- 2.83** Article 2(6) covers AI systems, models and outputs that are specifically developed and put into service for the purpose of scientific research and development. This presumably covers both fundamental research into various aspects of AI as well as AI models and systems used to do research in other disciplines, as AI is, indeed, gaining more importance in science.²⁴³
- 2.84** The interpretation of the AI Act's second research exemption is also far from straightforward. Article 2(6) specifically exempts from the scope of the regulation any AI systems or models that are developed and put into service (but not placed on the market!) for the exclusive purpose of scientific research and development. In medical research, for example, this could cover AI used to detect how the brain retains a sequence of events in memory. The scope of this provision mainly hinges on the interpretation of 'the sole purpose of scientific research'. The legislative text itself only provides little interpretational help. Recital 25 makes matters less clear in referring to 'scientific research and development'. This raises not only questions of legal interpretation, given that recitals are not strictly legally binding, but also about how the notion of 'the sole purpose' ought to be interpreted and whether there is a difference between 'scientific research' and 'scientific development'. Considering that the AI Act does not define these terms, reference must be made to other EU law frameworks. It is, indeed a well-established principle of EU law that notions within EU legislation are to be interpreted in the Union context.²⁴⁴
- 2.85** In **data protection law**, scientific research has to be interpreted 'in a broad manner including for example technological development and demonstration, fundamental research, applied research and privately funded research.[...] Scientific research purposes should also include studies conducted in the public interest in the area of public health.[...]'²⁴⁵ According to the EDPB the concept should not, however, 'be stretched beyond its common meaning and only include research projects set up in accordance with relevant sector-related methodologies and ethical standards.'²⁴⁶ There are two possible interpretations regarding the interplay between the AI Act and GDPR. One perspective suggests that the AIA's lack of a specific definition for scientific research implies that it should be interpreted consistently with the GDPR's established understanding. Conversely, the deliberate absence of a recital like the GDPR's and the additional research exemption in Article 2(8), which is not limited to scientific research, can be construed as an intentional divergence from such a broad interpretation.
- 2.86** In any event, Article 2(6) only applies where AI models and systems are specifically developed and put into service for the sole purpose of scientific research. While the intention behind this provision is clear, it overlooks the **intrinsic nature of innovation processes and**

²⁴¹ Article 2(6) AIA. Article 2(8) further exempts 'any research, testing and development activity regarding AI systems or models prior to them being placed on the market or put into service'.

²⁴² Four Columns AI Act (n 1) line 125a.

²⁴³ 'Science and the New Age of AI' (*Nature*, 6 December 2023) <<https://www.nature.com/immersive/d41586-023-03017-2/index.html>> accessed 19 December 2024; see further 'Machine Learning for Science' (*Eberhard Karls Universität Tübingen*) <<https://www.machinelearningforscience.de/>> accessed 30 June 2025.

²⁴⁴ Case C-11/76 *French v Commission* EU:C:1979:28, [1979] ECR I-245.

²⁴⁵ Recital 159 GDPR.

²⁴⁶ EDPB, 'Guidelines 03/2020 on the Processing of Data Concerning Health for the Purpose of Scientific Research in the Context of the COVID-19 Outbreak' (21 April 2020) 6 <https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines/guidelines-032020-processing-data-concerning-health-purpose_en> accessed 2 July 2025.

fundamental research. Specifically, it disregards the fact that researchers may not, at the outset, know the ultimate applications of their work. This further implies that research addressing significant, unsolved questions, which researchers should be actively encouraged to pursue from an innovation and societal welfare perspective (such as, for instance, AI explainability or AI able to reason causally), may not fall under Article 2(6). This is because foundational mechanisms developed in such research will undoubtedly find applications beyond the purely scientific domain. The specific implications of this horizontal research exemption for general-purpose AI models are examined further in the analysis of Article 53.

xiii. Article 2(7): Conflict rule for data protection and privacy law

Article 2(7) provides that the GDPR, the EU Data Protection Regulation ('EUDPR')²⁴⁷, the Law Enforcement Directive ('LED')²⁴⁸ and the e-Privacy Directive²⁴⁹ remain unaffected by the AI Act with the exceptions of Articles 10(5) and 59.

Given that AI is a technology with a voracious appetite for data, the **material overlap** between data (protection) law and the AIA is evident.²⁵⁰ As a result, the GDPR will often apply in conjunction with the AIA. The Impact Assessment stressed that 'the EU *acquis* on data protection is particularly relevant' for AI²⁵¹ and the joint opinion of the EDPB and EDPS stressed in 2021 that the regulation would have 'prominently important data protection implications.'²⁵²

Because of the **expansive interpretation of key data protection concepts**, data protection law often applies to AI. The concepts of 'personal data' and 'processing' must be interpreted expansively, underlying the very broad scope of application of data protection law.²⁵³ This implies that whenever AI uses or generates personal data, data protection law applies. The notion of the data controller, the legal or natural person liable to ensure compliance with data protection law, has also been judicially expanded.²⁵⁴ Sometimes, computer code also legally qualifies as personal data.²⁵⁵ This is, for instance, the case where large language models are subject to model inversion attacks.²⁵⁶ These attacks allow for the model's training data to be approximated with varying levels of precision.²⁵⁷

Article 2(7) clarifies that despite the AIA having been adopted later in time and arguably being the more specific norm it has **no effect of *lex specialis* or *lex posterior*** in relation to

²⁴⁷ EUDPR.

²⁴⁸ LED.

²⁴⁹ e-Privacy Directive (Directive 2002/58/EC of the European Parliament and of the Council of 12 July 2002 concerning the processing of personal data and the protection of privacy in the electronic communications sector (Directive on privacy and electronic communications) [2002] OJ L201/37).

²⁵⁰ Michèle Finck and Frank Pallas, 'They Who Must Not Be Identified—Distinguishing Personal from Non-Personal Data under the GDPR' (2020) 10 International Data Privacy Law 11.

²⁵¹ Commission Impact Assessment (n 22) 5.

²⁵² EDPB-EDPS Joint Opinion on the proposal for a Regulation of the AIA (n 128) 2.

²⁵³ Articles 4(1) and 4(2) GDPR. See further Finck and Pallas (n 250) 11; Nadezhda Purtova, 'The Law of Everything. Broad Concept of Personal Data and Future of EU Data Protection Law' (2018) 10 Law, Innovation and Technology 40.

²⁵⁴ Finck (n 130).

²⁵⁵ Nadezhda Purtova and Ronald Leenes, 'Code as Personal Data: Implications for Data Protection Law and Regulation of Algorithms' (2023) 13 International Data Privacy Law 245.

²⁵⁶ On model inversion attacks, see further Michael Veale, Reuben Binns and Lilian Edwards, 'Algorithms That Remember: Model Inversion Attacks and Data Protection Law' (2018) 376 Phil. Trans. R. Soc. A <<https://royalsocietypublishing.org/doi/10.1098/rsta.2018.0083>> accessed 9 December 2024; Claudio Novelli and others, 'Generative AI in EU Law: Liability, Privacy, Intellectual Property, and Cybersecurity' (2024) 55 Computer Law & Security Review 1, 14.

²⁵⁷ Veale, Binns and Edwards (n 256) 4.

data protection law.²⁵⁸ This explicit legislative clarification is valuable given the ambiguous status of these general principles in addressing instances of norm overlap in EU law.²⁵⁹ Formally, the AIA should hence not affect the application of the EU data protection instruments nor the status of data protection authorities, the independence of which is secured by EU primary law in the form of Article 8(3) CFR.²⁶⁰

2.91 Substantively, there are numerous intersections between the GDPR and the AIA, and many related questions will arise in the future.²⁶¹ A sign of this future dynamic was that the first legal headlines surrounding large language models mainly revolved around data protection law.²⁶² Additionally, the dual role of national Data Protection Authorities as market surveillance authorities raises intricate questions about institutional competence and functional coherence.²⁶³

2.92 Article 10(5), which enables the processing of special categories of personal data where this is strictly necessary for the purpose of ensuring bias detection and correction in relation to the high-risk AI systems, and Article 59, which governs the further processing of personal data in AI regulatory sandboxes, form **exceptions** to the rule announced by Article 2(7).

xiv. Article 2(8): Exemption for research, testing and development activities prior to placement on the market or putting into service

2.93 Articles 2(8) is the **second research exemption** foreseen in the AIA.²⁶⁴ It exempts ‘any research, testing and development activity regarding AI systems or models prior to them being placed on the market or put into service’. Importantly, there is no universally agreed definition of science or, indeed, scientific research, neither in general nor in EU law. It has been suggested that the absence of a clear definition, combined with uncertainties surrounding the interpretation of the research exemptions in the AIA, could create opportunities for regulatory arbitrage.²⁶⁵

2.94 The initial Commission draft contained no explicit research exemption, which led to calls to create a specific exemption on the matter to protect academic freedom and secure innovation in the EU.²⁶⁶ Article 2(8) exempts research, testing and development activity regarding AI systems or models prior to the placement on the market or putting into service. Unlike the scientific research exemption this provision covers AI systems and models that are later on placed on the market or put into service, whether they are related to science or not. Pursuant to Article 3(9), placing on the market ‘means the first making available of an

²⁵⁸ See further Recital 10 AIA.

²⁵⁹ Hancox (n 237).

²⁶⁰ Recital 10 AIA.

²⁶¹ See, by way of example, Guillermo Lazcoz and Paul De Hert, ‘Humans in the GDPR and AIA Governance of Automated and Algorithmic Systems. Essential Pre-Requisites against Abdicating Responsibilities’ (2023) 50 Computer Law & Security Review; and Lena Enqvist, ‘Rule-Based versus AI-Driven Benefits Allocation: GDPR and AIA Legal Implications and Challenges for Automation in Public Social Security Administration’ (2024) 33 Information & Communications Technology Law 222.

²⁶² Niko Härting, ‘Tutti Frutti guaranteed - Italy bans ChatGPT’ (*CR-online.de Blog*, 2 April 2023) <<https://www.cr-online.de/blog/2023/04/02/tutti-frutti-guaranteed-italy-bans-chatgpt/>> accessed 9 December 2024; ‘ChatGPT provides false information about people, and OpenAI can’t correct it’ (*Noyb*, 29 April 2024) <<https://noyb.eu/de/chatgpt-provides-false-information-about-people-and-openai-cant-correct-it>> accessed 9 December 2024.

²⁶³ See further Article 74(8) AIA.

²⁶⁴ See also Article 2(6) AIA.

²⁶⁵ Liane Colonna, ‘The AI Act’s Research Exemption: A Mechanism for Regulatory Arbitrage?’ in Eduardo Gill-Pedro and Andreas Moberg (eds), *YSEC Yearbook of Socio-Economic Constitutions 2023* (Springer 2023) 51.

²⁶⁶ Ibid 15.

AI system or a general-purpose AI model on the Union market.²⁶⁷ Pursuant to Article 3(11), putting into service means ‘the supply of an AI system for first use directly to the deployer or for own use in the Union for its intended purpose’.²⁶⁸

Multiple **interpretational questions**, indeed, arise in relation to Article 2(8).²⁶⁹ The first relates to the **notions of ‘placing on the market’ and ‘putting into service’**. As per Articles 3(9) and (10), an AI system or model is considered to have been placed on the market when it is first made available in the internal market. While a commercial purpose is a prerequisite for an AI model or system to be considered ‘placed on the market’, the assessment of this criterion—specifically, whether it should be objective or subjective—remains unclear. This ambiguity is particularly pertinent in computer science projects, where collaborations often involve numerous contributors, each potentially driven by diverse motivations, some commercial and others not. The commercial nature of research will be equally hard to assess in relation to corporate research departments and the commonplace research collaborations between these departments and academic institutions.²⁷⁰ Today, many research projects, including those specifically encouraged by the EU in order to promote innovation, involve close collaboration with industrial partners who may intend to commercialize the research once the exploratory phase is complete and a profit-oriented purpose is established.²⁷¹

According to Article 3(10), an AI system is deemed to have been put into service if it is supplied for first use directly to the deployer or for the provider’s own use in the EU for its intended purpose. Yet, the boundaries between the development of a system and the deployment for its own internal use are not always clear, such as where a system is tested internally. Computer science is a discipline that has embraced publicity of research, meaning that researchers are expected to publish research findings online.²⁷² Computer science conferences, moreover, require the publication of AI models and systems. Yet, there is a lack of clarity regarding whether uploading software to related platforms constitutes the act of placing it on the market or putting it into service of an AI system.²⁷³

The second interpretational question relates to the definition of **‘research, testing or development activity’**. Given the diverse forms that research collaborations can take, a fundamental question—not unique to this area of EU law²⁷⁴—arises as to how research should be defined. Particularly problematic, however, is the fact that the AI Act specifically imposes some requirements on the design and development phase²⁷⁵ of a high-risk AI system, respectively on the testing²⁷⁶ of these systems. This results in an interpretative conundrum.

²⁶⁷ See further analysis on Article 3(9) AIA below.

²⁶⁸ See further analysis on Article 3(11) AIA below.

²⁶⁹ This argument was first developed in Michèle Finck, ‘In Search of the Lost Research Exemption: Reflections on the AI Act’ [2025] GRUR International ikaf100.

²⁷⁰ Michèle Finck and Asia J Biega, ‘Reviving Purpose Limitation and Data Minimisation in Data-Driven Systems’ (2021) *Technology and Regulation* 44.

²⁷¹ ‘Horizon Europe’ (*European Commission*) <https://research-and-innovation.ec.europa.eu/funding/funding-opportunities/funding-programmes-and-open-calls/horizon-europe_en> accessed 2 July 2025.

²⁷² Alina Wernick and Kristof Meding, ‘Beware! The AI Act Can Also Apply to Your AI Research Practices’ (*arXiv*, 2025) <<https://arxiv.org/abs/2506.03218>> accessed 2 July 2025.

²⁷³ Katerina Yordanova, ‘The EU AI Act – Balancing Human Rights and Innovation Through Regulatory Sandboxes and Standardization’ (*PYMNTS.com*, 14 March 2022) <https://www.pymnts.com/cpi_posts/the-eu-ai-act-balancing-human-rights-and-innovation-through-regulatory-sandboxes-and-standardization/> accessed 2 July 2025.

²⁷⁴ João Pedro Quintais, ‘What Is a “Research Organisation” and Why It Matters: From Text and Data Mining to AI Research’ (2025) 74 *GRUR International* 397.

²⁷⁵ Article 13(1) AIA.

²⁷⁶ Article 9(6) AIA.

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The interplay between these requirements and the research exemption could be read in two ways. First, the research exemption does not apply in relation to requirements explicitly addressed at testing and development. This is, however, in tension with the broad formulation of Article 2(8). Second, the research exemption covers anything happening prior to the placing on the market or putting into service of an AI system and that requirements, such as those explicitly related to development and testing only need to be retroactively complied with once there is an intention to place on the market or put into service.²⁷⁷ In this case, the research exemption has no added value as the only criterion that matters is whether an AI system is placed on the market or put into service.

2.98 Third, the AI Act repeatedly refers to ‘the lifecycle’²⁷⁸ or ‘the lifetime’²⁷⁹ of an AI system without, however, defining this notion. According to the Organization for Economic Cooperation and Development (‘OECD’) conception of an AI system, which the AI Act generally is aligned with, the lifecycle of an AI system explicitly includes the ‘planning and design’ phase.²⁸⁰ The AIA does not clarify how ‘lifetime’ or ‘lifecycle’ should be interpreted or how they relate to the research exemption, again raising doubts about whether Article 2(8) adds any value, as these terms may simply imply that the key criterion is whether the AI system is placed on the market or put into service. The specific implications of this horizontal research exemption for general-purpose AI models are examined further in the analysis of Article 53.

xv. Article 2(9): Conflict rule for consumer protection and product safety law

2.99 Article 2(9) provides that the AIA is without prejudice to EU law on consumer protection, which protects the weaker party in business transactions, as well as EU product safety law, upon which the AIA has been modelled.

2.100 There are many intersections between the AI Act and consumer protection law. For instance, it has been argued that Article 5 is reminiscent of Article 5 of the Unfair Commercial Practices Directive (‘UCPD’)²⁸¹ which prohibits unfair commercial practices.²⁸² The UCPD is an unfair competition instrument adopted in 2005 in view of improving consumer confidence in cross-border trade within the internal market.²⁸³ Moreover, several provisions of the AIA overlap with protections in the UCPD targeting unfair commercial practices leading to economic or financial harm to consumers. Pursuant to Article 2(9), the AIA leaves these rules intact, meaning that they need to be applied alongside the rules in the AIA in case of substantive overlap.

2.101 Article 2(9) furthermore provides that the AI Act is **without prejudice to EU law on product safety**. AI systems that are placed on the market as a consumer good or component thereof

²⁷⁷ Article 10 AIA.

²⁷⁸ Recitals 65, 69, 73, 74, 110, 114 and 115, Articles 9(2), 15(1), 40(2) AIA and Point 6 of Annex IV AIA.

²⁷⁹ Recital 71 and Articles 12(1), 13(3)(e) and 72(2) AIA.

²⁸⁰ OECD, *Recommendation of the Council on Artificial Intelligence* (OECD/LEGAL/0449, 2019) <<https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449>> accessed 2 July 2025.

²⁸¹ UCPD. In 2019, the UCPD was updated through the Modernisation Directive (Directive (EU) 2019/2161 of the European Parliament and of the Council of 27 November 2019 amending Council Directive 93/13/EEC and Directives 98/6/EC, 2005/29/EC and 2011/83/EU of the European Parliament and of the Council as regards the better enforcement and modernization of Union consumer protection rules (Modernisation Directive) [2019] OJ L328/7).

²⁸² Catalina Goanta, ‘Regulatory Siblings: The Unfair Commercial Practices Directive Roots of the AI Act’ in Inge Graef and Bart Van Der Sloot (eds), *The Legal Consistency of Technology Regulation in Europe* (1st edn, Hart Publishing 2024).

²⁸³ For an overview see Stefan Wrška, ‘Review of European Law on Unfair Commercial Practices and Contract Law’ (2016) 79 *The Modern Law Review* 1141.

will be subject to the GPSR, as well as other applicable EU harmonization norms. Article 8(2) underlines that providers of HRAIS covered by Section A of Annex I remain responsible for ensuring that their product is fully compliant with all applicable requirements under applicable Union harmonization legislation.²⁸⁴ In this context, they have a choice of integrating the necessary testing and reporting processes, information and documentation they provide with regard to their product into documentation and procedures that already exist and are required under the Union harmonization legislation.²⁸⁵

xvi. Article 2(10): The household exemption

Pursuant to Article 2(10), the AIA does not apply to deployers who are natural persons that use an AI system in the course of a purely personal, non-professional activity.²⁸⁶ The rationale behind Article 2(10) is that it would be **disproportionate** to impose obligations incumbent on deployers under the AIA on natural persons, considering that the use of an AI system in the course of a purely personal, non-professional activity would generally only give rise to very limited risks to health, safety and fundamental rights, especially considering that the obligations of the provider to manage these risks still apply. This exemption is reminiscent of Article 2(2)(c) GDPR, which exempts a ‘purely personal or household activity’ from the scope of the GDPR. It has been interpreted restrictively by the CJEU.²⁸⁷

The Commission Guidelines on the Interpretation of the Prohibitions in Chapter II clarify that the derogation only applies as regards the **obligations of deployers** when using the system for purely personal non-professional activities. The AI system remains within the scope of the regulation as regards the obligations of providers placing the system on the market or putting it into service, other professional deployers, and other responsible actors, such as importers and distributors.²⁸⁸

It is unclear why there needed to be a specific provision to this effect as this exemption can also be deduced from the definition of deployer in Article 3(4).²⁸⁹ There is one notable distinction between the respective formulations in Articles 2(10) and 3(4), however, as the former refers to ‘purely personal non-professional activity’ whereas Article 3(4) omits the qualification of ‘**purely**’ and merely refers to personal non-professional activity. Recital 13 also merely refers to ‘personal non-professional activity’.²⁹⁰ The differing formulations in Articles 2(10) and 3(4) are likely to give rise to challenging questions about the scope of personal activity in future judicial disputes.

The AIA, moreover, does not provide a definition of what distinguishes ‘personal’ from ‘professional’. Determining what qualifies as ‘personal’ versus ‘non-professional’ can be complex, especially when it comes to transparency obligations. Consider the scenario where a professional politician, who, as is commonplace, maintains both official and personal social

²⁸⁴ See further analysis of Article 8(2) AIA.

²⁸⁵ Ibid.

²⁸⁶ See also Recital 13 AIA.

²⁸⁷ Case C-101/01 *Lindqvist* EU:C:2003:596, [2003] ECR I-12971.

²⁸⁸ Commission Guidelines on prohibited artificial intelligence practices (n 166) 11.

²⁸⁹ Article 3(4) AIA defines the deployer as ‘a natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity’.

²⁹⁰ Recital 13 AIA reads as follows: ‘The notion of “deployer” referred to in this Regulation should be interpreted as any natural or legal person, including a public authority, agency or other body, using an AI system under its authority, except where the AI system is used in the course of a personal non-professional activity. Depending on the type of AI system, the use of the system may affect persons other than the deployer’.

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media accounts, creates and shares a deepfake targeting a political opponent through their personal profile. Despite the use of a personal account, the content is clearly linked to their professional role.

- 2.106** While the interpretation of the household exemption in data protection law may offer some guidance, the term ‘personal’ has yet to be authoritatively defined. Advocate General Jääskinen opined that ‘personal’ ought to be interpreted as ‘activities which are closely and objectively linked to the private life of an individual and which do not significantly impinge upon the personal sphere of others.’²⁹¹ This test has, to date, however not been explicitly endorsed by the CJEU and it also is debatable whether a focus on the personal sphere of others should be relevant in the context of the AIA, as there is not the same focus on the reach of an activity as there is under the GDPR.

- 2.107** xvii. Article 2(11): No limitation of competences regarding worker protection Pursuant to Article 2(11), the regulation does not preclude the EU or Member States from maintaining or introducing measures that are more favourable to workers regarding the protection of their rights ‘in respect of the use of AI systems by employers’ or from encouraging or allowing the application of collective agreements that are more favourable to workers. This results from Article 114(2) TFEU according to which Article 114(1) TFEU cannot be used to harmonize areas ‘relating to the rights and interests of employed persons’. Indeed, the EU merely has supporting and complementing competences in relation to employment matters.²⁹² Regarding employment, the AIA serves as a minimum harmonization instrument, setting a floor of protection that Member States are free to exceed. Recital 9 underlines that the AIA shall moreover not affect the Platform Work Directive, which not only applies to workers, as the text of Article 2(11) indicates but also to independents, indicating that Article 2(11) should also be interpreted in that sense. Article 26(7) also creates an obligation for deployers who are employers to ‘inform workers representatives and the affected workers that they will be subject to the use of the high-risk AI system’ before putting into service or using a HRAIS.²⁹³

- 2.108** xviii. Article 2(12): A partial exemption for AI systems released under free and open-source licences Article 2(12) provides that the AI Act does not apply to AI systems that are released under free and open-source licences unless they are placed on the market or put into service as high-risk AI systems or as an AI system covered by the prohibitions in Article 5 or the transparency requirements in Article 50.
- 2.109** This exemption was not included in the draft regulation but emerged during the legislative process, where it was deemed essential to safeguard research and innovation within Europe.²⁹⁴ Free and open-source software has been a topic that remained contentious until the final stages of the legislative process. On one hand, the potentially innovation-enhancing

²⁹¹ Case C-212/13 *Ryneš* EU:C:2014:2072, Opinion of AG Jääskinen, para 51.

²⁹² Article 153(1) TFEU.

²⁹³ Article 26(7) AIA.

²⁹⁴ See Recital 102 AIA: ‘Software and data, including models, released under a free and open-source licence that allows them to be openly shared and where users can freely access, use, modify and redistribute them or modified versions thereof, can contribute to research and innovation in the market and can provide significant growth opportunities for the Union economy’. See also Recital 103 AIA. Note that these recitals relate to the open-source exemptions for GPAIM.

benefits of open-source software ('OSS') were highlighted²⁹⁵, while, on the other hand, it was also stressed that OSS can give rise to the same risks as proprietary software. Ultimately, the co-legislators opted to exempt AI systems that are released under free and open-source licences unless they are placed on the market or put into service as HRAIS or as an AI system covered by the prohibitions in Article 5 or the transparency requirements in Article 50 from the scope of the AI Act.

Article 2(12) only applies to AI systems, yet the AIA also foresees a partial exemption for **GPAIM** provided as open source in the form of Article 53(2), pursuant to which providers of GPAIM are exempted from some of their obligations where the models are released under a free and open-source licence that allows for the access, usage, modification, and distribution of the model, and whose parameters, including the weights, the information on the model architecture, and the information on model usage, are made publicly available.²⁹⁶ Importantly, this partial exemption does not apply to GPAIM with systemic risk.²⁹⁷

Article 2(12) does not specify the quantities of AI that must be open-sourced, such as the amount of data or code that must be made publicly available. Since open source exists on a spectrum, future scenarios are likely to raise questions about whether an open-source contract falls within the scope of this exemption.²⁹⁸

In relation to **GPAIM**, Recital 102 provides that 'the licence should be considered to be free and open-source also when it allows users to run, copy, distribute, study, change and improve software and data, including models under the condition that the original provider of the model is credited, the identical or comparable terms of distribution are respected'.²⁹⁹ This formulation leaves a lot of room for interpretation. According to Recital 103, which applies to GPAIM but is formulated in such a broad manner to also seem applicable to Article 2(12), provides that 'AI components that are provided against a price or otherwise monetised, including through the provision of technical support or other services, including through a software platform, related to the AI component, or the use of personal data for reasons other than exclusively for improving the security, compatibility or interoperability of the software, with the exception of transactions between microenterprises, should not benefit from the exceptions provided to free and open-source AI components'.³⁰⁰ Making AI components available through open repositories is not, in itself, considered to constitute a monetization.³⁰¹

Article 2(12) has a very limited scope considering that the rules in Chapters II and III as well as Article 50 make up the bulk of the norms in the AIA. Recital 89 encourages 'developers' of open-source software that is caught by the exemption to 'implement widely adopted documentation practices, such as model cards and data sheets, as a way to accelerate information

²⁹⁵ 'Open source in AI comes with a wide range of benefits: more innovation, more competition, more public-interest research. It can provide free-to-use building blocks to diffuse progress in AI and opens it up to public scrutiny'. See Maximilian Gahntz, 'The AI Act & Foundation Models: From Gridlock to Go-Ahead?' (*Mozilla Foundation*, 20 November 2023) <<https://foundation.mozilla.org/en/blog/the-ai-act-foundation-models-from-gridlock-to-go-ahead/>> accessed 9 December 2024.

²⁹⁶ See further analysis on Article 53(2) AIA.

²⁹⁷ Ibid.

²⁹⁸ Irene Solaiman, 'The Gradient of Generative AI Release: Methods and Considerations', *2023 ACM Conference on Fairness, Accountability, and Transparency* (ACM 2023) <<https://dl.acm.org/doi/10.1145/3593013.3593981>> accessed 4 January 2025.

²⁹⁹ Recital 102 AIA.

³⁰⁰ Recital 103 AIA.

³⁰¹ Ibid.

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sharing along the AI value chain, allowing the promotion of trustworthy AI systems in the Union.³⁰² It is worth noting that Recital 103 foresees that ‘free and open-source AI components covers the software and data, including models and general-purpose AI models, tools, services or processes of an AI system.’³⁰³

IV. Article 3: Definitions

For the purpose of this Regulation, the following definitions apply:

- (1) ‘AI system’ means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments;
- (2) ‘risk’ means the combination of the probability of an occurrence of harm and the severity of that harm;
- (3) ‘provider’ means a natural or legal person, public authority, agency or other body that develops an AI system or a general-purpose AI model or that has an AI system or a general-purpose AI model developed and places it on the market or puts the AI system into service under its own name or trademark, whether for payment or free of charge;
- (4) ‘deployer’ means a natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity;
- (5) ‘authorised representative’ means a natural or legal person located or established in the Union who has received and accepted a written mandate from a provider of an AI system or a general-purpose AI model to, respectively, perform and carry out on its behalf the obligations and procedures established by this Regulation;
- (6) ‘importer’ means a natural or legal person located or established in the Union that places on the market an AI system that bears the name or trademark of a natural or legal person established in a third country;
- (7) ‘distributor’ means a natural or legal person in the supply chain, other than the provider or the importer, that makes an AI system available on the Union market;
- (8) ‘operator’ means a provider, product manufacturer, deployer, authorised representative, importer or distributor;
- (9) ‘placing on the market’ means the first making available of an AI system or a general-purpose AI model on the Union market;
- (10) ‘making available on the market’ means the supply of an AI system or a general-purpose AI model for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge;
- (11) ‘putting into service’ means the supply of an AI system for first use directly to the deployer or for own use in the Union for its intended purpose;
- (12) ‘intended purpose’ means the use for which an AI system is intended by the provider, including the specific context and conditions of use, as specified in the information

³⁰² Recital 89 AIA.

³⁰³ Recital 103 AIA.

- supplied by the provider in the instructions for use, promotional or sales materials and statements, as well as in the technical documentation;
- (13) ‘reasonably foreseeable misuse’ means the use of an AI system in a way that is not in accordance with its intended purpose, but which may result from reasonably foreseeable human behaviour or interaction with other systems, including other AI systems;
 - (14) ‘safety component’ means a component of a product or of an AI system which fulfils a safety function for that product or AI system, or the failure or malfunctioning of which endangers the health and safety of persons or property;
 - (15) ‘instructions for use’ means the information provided by the provider to inform the deployer of, in particular, an AI system’s intended purpose and proper use;
 - (16) ‘recall of an AI system’ means any measure aiming to achieve the return to the provider or taking out of service or disabling the use of an AI system made available to deployers;
 - (17) ‘withdrawal of an AI system’ means any measure aiming to prevent an AI system in the supply chain being made available on the market;
 - (18) ‘performance of an AI system’ means the ability of an AI system to achieve its intended purpose;
 - (19) ‘notifying authority’ means the national authority responsible for setting up and carrying out the necessary procedures for the assessment, designation and notification of conformity assessment bodies and for their monitoring;
 - (20) ‘conformity assessment’ means the process of demonstrating whether the requirements set out in Chapter III, Section 2 relating to a high-risk AI system have been fulfilled;
 - (21) ‘conformity assessment body’ means a body that performs third-party conformity assessment activities, including testing, certification and inspection;
 - (22) ‘notified body’ means a conformity assessment body notified in accordance with this Regulation and other relevant Union harmonisation legislation;
 - (23) ‘substantial modification’ means a change to an AI system after its placing on the market or putting into service which is not foreseen or planned in the initial conformity assessment carried out by the provider and as a result of which the compliance of the AI system with the requirements set out in Chapter III, Section 2 is affected or results in a modification to the intended purpose for which the AI system has been assessed;
 - (24) ‘CE marking’ means a marking by which a provider indicates that an AI system is in conformity with the requirements set out in Chapter III, Section 2 and other applicable Union harmonisation legislation providing for its affixing;
 - (25) ‘post-market monitoring system’ means all activities carried out by providers of AI systems to collect and review experience gained from the use of AI systems they place on the market or put into service for the purpose of identifying any need to immediately apply any necessary corrective or preventive actions;
 - (26) ‘market surveillance authority’ means the national authority carrying out the activities and taking the measures pursuant to Regulation (EU) 2019/1020;
 - (27) ‘harmonised standard’ means a harmonised standard as defined in Article 2(1), point (c), of Regulation (EU) No 1025/2012;
 - (28) ‘common specification’ means a set of technical specifications as defined in Article 2, point (4) of Regulation (EU) No 1025/2012, providing means to comply with certain requirements established under this Regulation;

- (29) 'training data' means data used for training an AI system through fitting its learnable parameters;
- (30) 'validation data' means data used for providing an evaluation of the trained AI system and for tuning its non-learnable parameters and its learning process in order, inter alia, to prevent underfitting or overfitting;
- (31) 'validation data set' means a separate data set or part of the training data set, either as a fixed or variable split;
- (32) 'testing data' means data used for providing an independent evaluation of the AI system in order to confirm the expected performance of that system before its placing on the market or putting into service;
- (33) 'input data' means data provided to or directly acquired by an AI system on the basis of which the system produces an output;
- (34) 'biometric data' means personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, such as facial images or dactyloscopic data;
- (35) 'biometric identification' means the automated recognition of physical, physiological, behavioural, or psychological human features for the purpose of establishing the identity of a natural person by comparing biometric data of that individual to biometric data of individuals stored in a database;
- (36) 'biometric verification' means the automated, one-to-one verification, including authentication, of the identity of natural persons by comparing their biometric data to previously provided biometric data;
- (37) 'special categories of personal data' means the categories of personal data referred to in Article 9(1) of Regulation (EU) 2016/679, Article 10 of Directive (EU) 2016/680 and Article 10(1) of Regulation (EU) 2018/1725;
- (38) 'sensitive operational data' means operational data related to activities of prevention, detection, investigation or prosecution of criminal offences, the disclosure of which could jeopardise the integrity of criminal proceedings;
- (39) 'emotion recognition system' means an AI system for the purpose of identifying or inferring emotions or intentions of natural persons on the basis of their biometric data;
- (40) 'biometric categorisation system' means an AI system for the purpose of assigning natural persons to specific categories on the basis of their biometric data, unless it is ancillary to another commercial service and strictly necessary for objective technical reasons;
- (41) 'remote biometric identification system' means an AI system for the purpose of identifying natural persons, without their active involvement, typically at a distance through the comparison of a person's biometric data with the biometric data contained in a reference database;
- (42) 'real-time remote biometric identification system' means a remote biometric identification system, whereby the capturing of biometric data, the comparison and the identification all occur without a significant delay, comprising not only instant identification, but also limited short delays in order to avoid circumvention;
- (43) 'post remote biometric identification system' means a remote biometric identification system other than a real-time remote biometric identification system;
- (44) 'publicly accessible space' means any publicly or privately owned physical place accessible to an undetermined number of natural persons, regardless of whether

certain conditions for access may apply, and regardless of the potential capacity restrictions;

- (45) 'law enforcement authority' means:
 - (a) any public authority competent for the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security; or
 - (b) any other body or entity entrusted by Member State law to exercise public authority and public powers for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security;
- (46) 'law enforcement' means activities carried out by law enforcement authorities or on their behalf for the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including safeguarding against and preventing threats to public security;
- (47) 'AI Office' means the Commission's function of contributing to the implementation, monitoring and supervision of AI systems and general-purpose AI models, and AI governance, provided for in Commission Decision of 24 January 2024; references in this Regulation to the AI Office shall be construed as references to the Commission;
- (48) 'national competent authority' means a notifying authority or a market surveillance authority; as regards AI systems put into service or used by Union institutions, agencies, offices and bodies, references to national competent authorities or market surveillance authorities in this Regulation shall be construed as references to the European Data Protection Supervisor;
- (49) 'serious incident' means an incident or malfunctioning of an AI system that directly or indirectly leads to any of the following:
 - (a) the death of a person, or serious harm to a person's health;
 - (b) a serious and irreversible disruption of the management or operation of critical infrastructure.
 - (c) the infringement of obligations under Union law intended to protect fundamental rights;
 - (d) serious harm to property or the environment;
- (50) 'personal data' means personal data as defined in Article 4, point (1), of Regulation (EU) 2016/679;
- (51) 'non-personal data' means data other than personal data as defined in Article 4, point (1), of Regulation (EU) 2016/679;
- (52) 'profiling' means profiling as defined in Article 4, point (4), of Regulation (EU) 2016/679;
- (53) 'real-world testing plan' means a document that describes the objectives, methodology, geographical, population and temporal scope, monitoring, organisation and conduct of testing in real-world conditions;
- (54) 'sandbox plan' means a document agreed between the participating provider and the competent authority describing the objectives, conditions, timeframe, methodology and requirements for the activities carried out within the sandbox;
- (55) 'AI regulatory sandbox' means a controlled framework set up by a competent authority which offers providers or prospective providers of AI systems the possibility

- to develop, train, validate and test, where appropriate in real-world conditions, an innovative AI system, pursuant to a sandbox plan for a limited time under regulatory supervision;
- (56) ‘AI literacy’ means skills, knowledge and understanding that allow providers, deployers and affected persons, taking into account their respective rights and obligations in the context of this Regulation, to make an informed deployment of AI systems, as well as to gain awareness about the opportunities and risks of AI and possible harm it can cause;
 - (57) ‘testing in real-world conditions’ means the temporary testing of an AI system for its intended purpose in real-world conditions outside a laboratory or otherwise simulated environment, with a view to gathering reliable and robust data and to assessing and verifying the conformity of the AI system with the requirements of this Regulation and it does not qualify as placing the AI system on the market or putting it into service within the meaning of this Regulation, provided that all the conditions laid down in Article 57 or 60 are fulfilled;
 - (58) ‘subject’, for the purpose of real-world testing, means a natural person who participates in testing in real-world conditions;
 - (59) ‘informed consent’ means a subject’s freely given, specific, unambiguous and voluntary expression of his or her willingness to participate in a particular testing in real-world conditions, after having been informed of all aspects of the testing that are relevant to the subject’s decision to participate;
 - (60) ‘deep fake’ means AI-generated or manipulated image, audio or video content that resembles existing persons, objects, places, entities or events and would falsely appear to a person to be authentic or truthful;
 - (61) ‘widespread infringement’ means any act or omission contrary to Union law protecting the interest of individuals, which:
 - (a) has harmed or is likely to harm the collective interests of individuals residing in at least two Member States other than the Member State in which:
 - (i) the act or omission originated or took place;
 - (ii) the provider concerned, or, where applicable, its authorised representative is located or established; or
 - (iii) the deployer is established, when the infringement is committed by the deployer;
 - (b) has caused, causes or is likely to cause harm to the collective interests of individuals and has common features, including the same unlawful practice or the same interest being infringed, and is occurring concurrently, committed by the same operator, in at least three Member States;
 - (62) ‘critical infrastructure’ means critical infrastructure as defined in Article 2, point (4), of Directive (EU) 2022/2557;
 - (63) ‘general-purpose AI model’ means an AI model, including where such an AI model is trained with a large amount of data using self-supervision at scale, that displays significant generality and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market and that can be integrated into a variety of downstream systems or applications, except AI models that are used for research, development or prototyping activities before they are placed on the market;
 - (64) ‘high-impact capabilities’ means capabilities that match or exceed the capabilities recorded in the most advanced general-purpose AI models;

- (65) ‘systemic risk’ means a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain;
- (66) ‘general-purpose AI system’ means an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems;
- (67) ‘floating-point operation’ means any mathematical operation or assignment involving floating-point numbers, which are a subset of the real numbers typically represented on computers by an integer of fixed precision scaled by an integer exponent of a fixed base;
- (68) ‘downstream provider’ means a provider of an AI system, including a general-purpose AI system, which integrates an AI model, regardless of whether the AI model is provided by themselves and vertically integrated or provided by another entity based on contractual relations.

A. Overview

Article 3 sets out no fewer than 68 definitions, playing a pivotal role in delineating the scope of the AI Act and structuring its substantive provisions. Furthermore, many of these definitions represent the first legal articulation of technical and legal concepts at the core of AI governance, with an impact that will extend far beyond the boundaries of this specific regulation. As Oskar Grstein notes, ‘for the better or worse, many of these definitions will become crucial reference points for AI governance going forward. They have been established through a democratically legitimised legislative procedure following a multi-stakeholder consultation process, in contrast to definitions purely focusing on technological developments, or economic opportunities.’³⁰⁴ It is furthermore likely that the definitions espoused by the AIA will not only be replicated by future EU legislation but also form the starting point of legislative discussions in other jurisdictions. The following part of the chapter provides an analysis of the 68 definitions set out in the AI Act. It not only offers a textual examination of each definition but also contextualizes them within the broader framework of the regulation, highlighting their significance for the interpretation and application of the Act and, where relevant, beyond.

B. Analysis

i. Article 3(1): AI system

By defining what constitutes an AI system, the regulation delineates its material scope. Notably, despite its title, the Artificial Intelligence Act does not regulate artificial intelligence generally but specifically targets AI systems and GPAIM. The related definition was a key point of contention during the drafting process, highlighting the broader challenge of

³⁰⁴ Oskar J Gstrein, Noman Haleem, and Andrej Zwitter, ‘General-Purpose AI Regulation and the European Union AI Act’ (2024) 13 Internet Policy Review <<https://policyreview.info/articles/analysis/general-purpose-ai-regulation-and-ai-act>> accessed 2 May 2025.

defining AI considering that this concept denotes a ‘fast evolving family of technologies.’³⁰⁵ Article 96(1)(f) empowers the Commission to adopt guidelines on the interpretation of Article 3(1).³⁰⁶ The Commission has adopted Guidelines on the Definition of an Artificial Intelligence System of 6 February 2025.³⁰⁷

- 2.116** Regarding **legislative history**, the Impact Assessment stressed that AI is a general-purpose technology³⁰⁸, that is to say ‘a very powerful family of computer programming techniques that can be deployed for desirable uses, as well as for more harmful ones.’³⁰⁹ While acknowledging that the definition of AI is highly contested, the Impact Assessment also stressed the importance of a single definition of AI that builds on the OECD definition as this was deemed ‘technology neutral and future proof.’³¹⁰ The draft regulation initially defined an AI system as ‘software that is developed with one or more of the techniques and approaches listed in Annex I and can, for a given set of human-defined objectives, generate outputs such as content, predictions, recommendations, or decisions influencing the environments they interact with.’³¹¹ Whereas the Commission defined the system as the regulatory object subject to the AI Act, it also recognizes that the ‘key power of an AI system resides in its operational logic.’³¹² An important characteristic of the definition espoused by the draft regulation was that an annex was intended to specify which techniques and approaches would classify as AI systems.³¹³ This approach was criticized for being overly broad.³¹⁴ The annex listed virtually all computational techniques, including statistical methods not conventionally understood to be AI.³¹⁵ This would have made it difficult to determine with legal certainty when an AI system falls within the scope of the regulation and when it does not. The rationale behind the initial annex-based approach was to ensure legislative agility, enabling the AI Act to evolve through amendments to the annexes in response to technological developments and shifting risk perceptions. This mechanism would have conferred significant regulatory discretion on the Commission. However, such flexibility came at the expense of legal certainty, leaving market participants uncertain as to whether software not currently within the scope might later fall under the regulation. This reflects a common regulatory trade-off in the governance of emerging technologies. Given the political sensitivity of the legislative negotiations, it is unsurprising that the Parliament and the Council ultimately chose to anchor the definition of an AI system—and thus the scope of the Act—within the main text, rather than relying on a more fluid, annex-driven model. As a result,

³⁰⁵ Recital 4 AIA.

³⁰⁶ See further, analysis of Article 96 AIA.

³⁰⁷ Commission Guidelines on the definition of an artificial intelligence system (n 65).

³⁰⁸ On this, see further Ajay Agrawal, Joshua Gans and Avi Goldfarb, ‘Economic Policy for Artificial Intelligence’ (2019) 19 *Innovation Policy and the Economy* 139; Erik Brynjolfsson, Daniel Rock and Chad Syverson, ‘Artificial Intelligence and the Modern Productivity Paradox: A Clash of Expectations and Statistics’ (National Bureau of Economic Research 2017) w24001 <<http://www.nber.org/papers/w24001.pdf>> accessed 9 December 2024.

³⁰⁹ Commission Impact Assessment (n 22) 2.

³¹⁰ Ibid 49.

³¹¹ Article 3(1) dAIA.

³¹² Commission Impact Assessment (n 22) 2.

³¹³ A list of such techniques can now be found in Commission Guidelines on the definition of an artificial intelligence system (n 65) 8. Unlike an Annex, these guidelines are not binding and can at any point be overridden by a contrary judicial interpretation through the CJEU.

³¹⁴ Smuha and others (n 143) 14.

³¹⁵ Patrick Glauner, ‘An Assessment of the AI Regulation Proposed by the European Commission’ in Sepehr Ehsani and others (eds), *The Future Circle of Healthcare: AI, 3D Printing, Longevity, Ethics, and Uncertainty Mitigation* (Springer International Publishing 2022); Leonardo CT Bezerra and others, ‘How VADER Is Your AI? Towards a Definition of Artificial Intelligence Systems Appropriate for Regulation’ (*arXiv*, 2024) <<https://arxiv.org/abs/2402.05048>> accessed 4 January 2025.

the final text resulting from the trilogues is noticeably distinct from what the Commission initially suggested.

The definition of AI system in Article 3(1) pursues **three objectives**.³¹⁶ First, it is intended to provide legal certainty and to ensure ‘consistency across the various legislative frameworks applicable to AI, thus enhancing legal certainty for operators and reducing the risk of sectoral market fragmentation.’³¹⁷ Second, the definition is explicitly aligned with international efforts coordinated by the OECD to promote uniformity in the legal definition of AI across jurisdictions.³¹⁸ This approach seeks to facilitate international convergence.³¹⁹ Third, the broad definition is intended to ensure flexibility in the face of rapid technological advancements, as a more rigid definition could risk becoming outdated as innovations evolve. The Commission Staff Working Document to the AI Act, moreover, notes that the definition is intended as a reference point for future sectoral legislation at supranational level.³²⁰

The definition in Article 3(1) comprises **four distinct elements**: an AI system is a (i) machine-based system; which (ii) is designed to operate with varying levels of autonomy and potentially also adaptiveness; and which (iii) pursues an objective that (iv) infers, from the input it receives, outputs that influence physical or virtual environments.³²¹ These elements are examined in turn below.

First, an AI system is a **machine-based system**. The Commission draft used the terminology of ‘software’ as opposed to machine-based system.³²² The other two co-legislators were in agreement that this term should be omitted.³²³ The Parliament suggested ‘a machine-based system’ as an alternative, whereas the Council opted for ‘a system.’³²⁴ The notion of ‘machine-based’ refers to the fact that AI systems ‘run on machines.’³²⁵ Recital 12 offers a tautological definition of ‘machine-based’, simply stating it as ‘the fact that AI systems run on machines.’³²⁶ This highlights that the reference to ‘machine-based’ in Article 3(1) merely states the obvious, as AI inherently operates on a machine.³²⁷ A ‘machine’ has been defined as ‘a piece of equipment with several moving parts that uses power to do a particular type of work’ and is often used as a synonym for a computer, understood in a wider sense, thus

³¹⁶ Recital 12 AIA reads as follows: ‘the notion of “AI system” in this Regulation should be clearly defined and should be closely aligned with the work of international organisations working on AI to ensure legal certainty, facilitate international convergence and wide acceptance, while providing the flexibility to accommodate the rapid technological developments in this field’.

³¹⁷ Commission Impact Assessment (n 22) 49.

³¹⁸ The final version of Article 3(1) AIA indeed corresponds to the revised OECD definition of an artificial intelligence system. See further OECD, ‘Explanatory Memorandum on the Updated OECD Definition of an AI System’ (2024) OECD Artificial Intelligence Papers 8, 1 <https://www.oecd.org/en/publications/explanatory-memorandum-on-the-updated-oecd-definition-of-an-ai-system_623da898-en.html> accessed 23 December 2024. For a comment, see Marko Grobelnik, Karine Perset and Stuart Russell, ‘What Is AI? Can You Make a Clear Distinction between AI and Non-AI Systems?’ (*OECD.AI*, 6 March 2024) <<https://oecd.ai/en/wonk/definition>> accessed 9 December 2024.

³¹⁹ ‘EU AI Act to Serve as Blueprint for Global Rules, Benifei Says’ *Reuters* (9 November 2023) <<https://www.reuters.com/technology/eu-ai-act-serve-blueprint-global-rules-benifei-says-2023-11-08/>> accessed 19 February 2025.

³²⁰ Commission Impact Assessment (n 22) 49.

³²¹ For an interpretation of these terms, see further Commission Guidelines on the definition of an artificial intelligence system (n 65). This Communication is not legally binding but will likely factor into any judicial interpretations of the legislative definition formulated by Article 3(1) AIA.

³²² Article 3(1) dAIA.

³²³ Four Columns AI Act (n 1) line 128.

³²⁴ *Ibid.*

³²⁵ Recital 12 AIA.

³²⁶ *Ibid.*

³²⁷ Commission Guidelines on the definition of an artificial intelligence system (n 65) 2–3.

also including smartphones and many other electronic devices.³²⁸ Furthermore, AI systems ‘can be used on a stand-alone basis or as a component of a product, irrespective of whether the system is physically integrated into the product (embedded) or serves the functionality of the product without being integrated therein (non-embedded).’³²⁹ Legal frameworks in other jurisdictions also focus on AI systems, such as the proposed Canadian Artificial Intelligence and Data Act³³⁰ or the Chilean Draft Law Regulating AI Systems, Robotics and Related Technologies.³³¹

- 2.120** This formulation also recalls that the AIA applies to AI systems, not AI models except for GPAIM.³³² This approach denotes that the AIA does not regulate AI per se but rather its use. AI systems are often combinations of different AI models.³³³ Take for instance automated driving, which is an ensemble of different models such as sign recognition systems and lane holder systems.
- 2.121** The emphasis on the ‘system’, moreover, allows for the classification of AI systems that utilize hybrid approaches, including conventional software. In the widely discussed Dutch welfare benefits scandal, for example, data was collected using machine learning, but the actual analysis was conducted through Microsoft Excel.³³⁴
- 2.122** Second, AI systems are **designed to operate with varying levels of autonomy and may exhibit adaptiveness after deployment**. This requirement was absent from the initial draft, leading to recurring criticisms that the original definition was overly broad.³³⁵ In the legislative procedure, the European Parliament suggested that systems ‘designed to operate with varying levels of autonomy and that can, for explicit or implicit objectives, generate outputs’ should be within the scope of the AIA, whereas the Council wanted a reference to systems ‘designed to operate with elements of autonomy and that, based on machine and/or human-provided data and inputs, infer how to achieve a given set of objectives.’³³⁶
- 2.123** **Autonomy** distinguishes AI systems from products ordinarily governed by the New Legislative Framework, such as dishwashers or elevators. The formulation ‘varying levels of autonomy’ implies that AI systems at various points of a broader spectrum are caught by Article 3(1).³³⁷ Pursuant to Recital 12, ‘varying levels of autonomy’ ought to be interpreted

³²⁸ ‘machine, n’ (*Cambridge Dictionary Online*) <<https://dictionary.cambridge.org/dictionary/english/machine>> accessed 16 December 2024.

³²⁹ Recital 12 AIA.

³³⁰ Parliament of Canada [2022] Bill C-27.

³³¹ Cámara de Diputados y Diputados, Chile [2023] Regula los sistemas de inteligencia artificial, la robótica y las tecnologías conexas, en sus distintos ámbitos de aplicación.

³³² See further Article 1(1) AIA and Recital 12 AIA.

³³³ Models have been defined as a ‘physical, mathematical or otherwise logical representation of a system, entity, phenomenon, process or data’ by ‘ISO/IEC 22989:2022 - Artificial intelligence concepts and terminology’ (1st edn, ISO 2022) point 3.1.23 <<https://www.iso.org/standard/74296.html>> accessed 17 March 2025.

³³⁴ Autoriteit Persoonsgegevens, ‘De verwerking van de nationaliteit van aanvragers van kinderopvangtoeslag’ (Onderzoekrapport z2018-22445, 2020) <<https://www.rijksoverheid.nl/documenten/rapporten/2020/07/17/de-verwerking-van-de-nationaliteit-van-aanvragers-van-kinderopvangtoeslag>> 6, 21–24; Björn ten Seldam and Alex Brenninkmeijer, ‘The Dutch Benefits Scandal: A Cautionary Tale for Algorithmic Enforcement’ (*EU Law Enforcement*, 30 April 2021) <<https://eulawenforcement.com/?p=7941>> accessed 10 December 2024.

³³⁵ Emre Kazim and others, ‘Proposed EU AI Act—Presidency compromise text: select overview and comment on the changes to the proposed regulation’ (2023) 3 AI and Ethics 381, 383; David Bomhard and Marieke Merkle, ‘Regulation of Artificial Intelligence: The EU Commission’s Proposal of an AI Act’ (2021) 6 EuCML 257, 258; Hannah Ruschemeier, ‘AI as a Challenge for Legal Regulation – the Scope of Application of the Artificial Intelligence Act Proposal’ (2023) 23 ERA Forum 361, 368.

³³⁶ Four Columns AI Act (n 1) line 128.

³³⁷ Pursuant to Recital 4 AIA, AI is a ‘fast evolving family of technologies’.

as ‘some degree of independence of actions from human involvement and of capabilities to operate without human intervention.’³³⁸ The Commission Guidelines on the Definition of an Artificial Intelligence System of 6 February 2025 clarify that autonomy is a spectrum.³³⁹

Article 3(1) accordingly requires that there be *some* degree of independence from human control.³⁴⁰ Yet, the AIA does not determine the dividing line between rules defined ‘solely’ by natural persons and those that dispose of ‘some degree of independence of actions from human involvement.’³⁴¹ Article 3(1) establishes that AI systems are those ‘designed’ to operate with a degree of autonomy, implying that the mere intention to develop autonomous functionality—regardless of whether it is fully realized—may suffice to bring a system within the scope of the definition. This would be paradoxical for a legal framework concerned with the actual capabilities, and resulting risks, of AI. In any event, rule-based systems that exhibit no autonomy are outside the scope of Article 3(1) given that AI systems should be distinguished from ‘simpler traditional software systems or programming approaches and should not cover systems that are based on the rules defined solely by natural persons to automatically execute operations.’³⁴²

The advantage of a reference to autonomy is that it is in line with the OECD’s recently changed definition of an AI system, which acknowledges that different AI systems may exhibit different levels of adaptiveness after deployment.³⁴³ The OECD definition has, for instance, also been endorsed by the United States National Institute of Standards and Technology (‘NIST’).³⁴⁴ The explanatory memorandum to the revised OECD definition provides that autonomy ‘means the degree to which a system can learn or act without human involvement following the delegation of autonomy and process automation by humans. Human supervision can occur at any stage of the AI system lifecycle, such as during AI system design, data collection and processing, development, verification, validation, deployment, or operation and monitoring. Some AI systems can generate outputs without these outputs being explicitly described in the AI system’s objective and without specific instructions from a human.’³⁴⁵ The Commission Guidelines on the Definition of an Artificial Intelligence System

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³³⁸ Recital 12 AIA.

³³⁹ Commission Guidelines on the definition of an artificial intelligence system (n 65) 3. ‘At one extreme of possible human-machine interaction are systems which are designed to perform all tasks though manually operated functions. At the other extreme are systems that are capable to operate without any human involvement or intervention, i.e. fully autonomously.’

³⁴⁰ This can be derived from a document prepared by the Belgian presidency of the Council, which was shared with Member States just days before the final COREPER vote on the AIA, and which was intended to explain the changes operated in the trilogue in view of gathering agreement on the AIA. This document states that ‘the definition is not intended to cover simpler traditional software systems or programming approaches, which are based on the rules defined solely by natural persons to automatically execute operations’. See European Commission, ‘Proposal for a Regulation of the European Parliament and of the Council laying down harmonized rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts: Analysis of the final compromise text with a view to agreement’ [2024] 5662/24, 16 | Parlament Österreich’ <<https://www.parlament.gv.at/gegenstand/XXVII/EU/170765>> accessed 13 December 2024.

³⁴¹ Recital 12 AIA.

³⁴² Ibid.

³⁴³ This definition reads as follows: ‘An AI system is a machine-based system that can, for a given set of human-defined explicit or implicit objectives, infer, from the input it receives, how to generate outputs such as makes predictions, content, recommendations, or decisions that can influencing physical real or virtual environments. Different AI systems are designed to operate with varying in their levels of autonomy and adaptiveness after deployment.’ Stuart Russell, Karine Perset and Marko Grobelnik, ‘Updates to the OECD’s Definition of an AI System Explained’ (OECD.AI, 29 November 2023) <<https://oecd.ai/en/wonk/ai-system-definition-update>> accessed 10 December 2024.

³⁴⁴ Tabassi (n 64).

³⁴⁵ OECD Explanatory Memorandum (n 318) 5.

further clarify that the notions of autonomy and inference go hand in hand as ‘the inference capacity of an AI system (i.e., its capacity to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments) is key to bring about its autonomy’.³⁴⁶

- 2.126** Although Article 3(1) references **adaptiveness**, that is to say the capacity of AI to ‘learn’³⁴⁷, there is no strict requirement for AI systems to be adaptive—that is, capable of learning—to fall within the scope of the AIA. According to Article 3(1), an AI system ‘may’ be adaptive. Recital 12 similarly refers to the adaptiveness that an AIS ‘could exhibit after deployment’. The use of these verbs underlines that adaptiveness is not a necessary condition. Adaptiveness ‘refers to self-learning capabilities, allowing the system to change while in use’ and thus after its initial development.³⁴⁸ This occurs where ‘the system modifies its behaviour through direct interaction with input and data before or after deployment’.³⁴⁹
- 2.127** The concepts of autonomy and adaptiveness are closely related.³⁵⁰ They collectively denote the self-learning capacities of AI, which allow the system to change while in use.³⁵¹
- 2.128** The formulation of Article 3(1) indicates that these ‘criteria’ (which, it has been seen, are not hard criteria) ought to be assessed at the level of the system, not an individual model. This implies that even where an individual model is deemed autonomous or adaptive, the overall system may not be in cases where the model plays an only marginal role in a much more complex system.
- 2.129** Third, an AI system pursues **explicit or implicit objectives**. The draft regulation already included a reference to the fact that AI systems operate to achieve ‘human-defined objectives’.³⁵² As AI can also pursue objectives defined by software, there was soon consensus that the reference to humans ought to be removed. The Parliament mandate suggested referring to ‘explicit or implicit objectives’ whereas the Council mandate put forward the formulation of ‘objectives, using machine learning and/or logic- and knowledge-based approaches’.³⁵³
- 2.130** Article 3(1) foresees that an AI system infers outputs from inputs it receives to pursue explicit or implicit objectives. The reference to explicit or implicit objectives is intended to underscore ‘that AI systems can operate according to explicit defined objectives or to implicit objectives. The objectives of the AI system may be different from the intended purpose of the AI system in a specific context’.³⁵⁴ For instance, the broad (human-defined) objective may be an embedded AI that tidies up my living room every evening. In this very appealing scenario, the corresponding AI system would have more specific objectives, such as to pick up each item that meets predetermined criteria. Article 3(1) also indicates that objectives can be explicit, such as for reinforcement learning systems, or implicit, such as where an automated vehicle has a rule to stop at a red light, which is explicit, but implicit is the objective to

³⁴⁶ Commission Guidelines on the definition of an artificial intelligence system (n 65) 3.

³⁴⁷ See further Recital 12 AIA.

³⁴⁸ Ibid.

³⁴⁹ OECD Explanatory Memorandum (n 318) 6. ‘Examples include a speech recognition system that adapts to an individual’s voice or a personalised music recommender system. AI systems can be trained once, periodically, or continually and operate by inferring patterns and relationships in data. Through such training, some AI systems may develop the ability to perform new forms of inference not initially envisioned by their programmers’.

³⁵⁰ Commission Guidelines on the definition of an artificial intelligence system (n 65) 4.

³⁵¹ Recital 12 AIA and Commission Guidelines on the definition of an artificial intelligence system (n 65) 4.

³⁵² Four Columns AI Act (n 1) line 128.

³⁵³ Ibid.

³⁵⁴ Recital 12 AIA. On the notion of the intended purpose, see further 3(12) and Article 8(1) AIA.

not breach traffic regulations.³⁵⁵ There are also systems where the ultimate objective is not explicitly programmed but implicit in the training data, such as the reward of large language models for generating plausible responses.

Fourth, an AI system ‘**infers, from the input it receives, how to generate outputs** such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.’³⁵⁶ The Commission Guidelines on Article 3(1) provide an interpretation of the concept of the **output**. Predictions are interpreted as ‘an estimate about an unknown value (the output) from known values supplied to the system (the input)’ whereas content is ‘the generation of new material by an AI system.’³⁵⁷ Recommendations are ‘suggestions for specific actions, products, or services to users based on their preferences, behaviours, or other data inputs’ and decisions are interpreted as ‘conclusions or choices made by a system.’³⁵⁸ Yet, these four categories of output are merely illustrative, meaning that other types of outputs may also fall within the scope of the regulation. The draft regulation already envisaged a definition of an AI system, which accounts for the fact that these systems ‘generate outputs such as content, predictions, recommendations, or decisions, influencing the environments they interact with.’³⁵⁹ The Parliament suggested changing this to ‘outputs such as predictions, recommendations or decisions, that influence physical or virtual environments’, thus leaving out ‘content’, presumably because this is a very wide notion that, *inter alia*, includes predictions, recommendations or decisions and specifying that the influenced environments can be of a physical or virtual nature.³⁶⁰ The Council in turn suggested keeping ‘content’ and including a bracket that referred to generative artificial intelligence, presumably as a way to somehow try to house GPAIM in the AI Act prior to agreement on the related regime now found in Chapter V, and ‘environments with which the AI system interacts’, which would have supposed a direct link between the AI system and the affected environments, which is absent from the final text.³⁶¹ The generation of outputs is also a crucial element of the ISO definition of AI system, which defines those as ‘engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives.’³⁶² This focus on outputs as the AIA seeks to mitigate the risks.

According to Article 3(1), these outputs can, but do not have to, **influence physical or virtual environments**, which should be understood as ‘the contexts in which the AI systems operate.’³⁶³ This criterion is not truly a criterion, as anything can arguably influence its environment, and the use of the modal verb ‘may’ further indicates that it is not a binding requirement.³⁶⁴

The central concept in Article 3(1) is the verb ‘**to infer**.’ While many software systems are designed to pursue objectives, process inputs, and produce outputs, not all of them engage in inference. This distinction may prove pivotal in delineating which systems fall within the scope of the AI Act, as inference implies a level of complexity and functional autonomy that

³⁵⁵ See further Commission Guidelines on the definition of an artificial intelligence system (n 65) 4.

³⁵⁶ Article 3(1) AIA.

³⁵⁷ Commission Guidelines on the definition of an artificial intelligence system (n 65) 10–11.

³⁵⁸ *Ibid* 11.

³⁵⁹ Article 3(1) dAIA.

³⁶⁰ Four Columns AI Act (n 1) line 128.

³⁶¹ *Ibid*.

³⁶² ISO/IEC 22989:2022 (n 333).

³⁶³ Recital 12 AIA.

³⁶⁴ Commission Guidelines on the definition of an artificial intelligence system (n 65) 4.

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goes beyond conventional software. Inference ‘refers to the process of obtaining the outputs [...] and to a capability of AI systems to derive models or algorithms, or both, from inputs or data.’³⁶⁵ The techniques that enable inference include ‘machine learning approaches that learn from data how to achieve certain objectives, and logic- and knowledge-based approaches that infer from encoded knowledge or symbolic representation of the task to be solved.’³⁶⁶ The Commission Guidelines on Article 3(1), moreover, draw on the definition in the ISO/IEC 22989 standard, which defines inference ‘as reasoning by which conclusions are derived from known premises’ and which note that ‘[i]n AI, a premise is either a fact, a rule, a model, a feature or raw data.’³⁶⁷ The Guidelines further clarify that the notions of autonomy and inference go hand in hand as ‘the inference capacity of an AI system (i.e., its capacity to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments) is key to bring about its autonomy.’³⁶⁸ Stated differently, the capacity of an AI system to infer ‘transcends basic data processing by enabling learning, reasoning or modelling.’³⁶⁹

2.134 Inference is also a criterion of other definitions of AI. The OECD Explanatory Memorandum states that inference ‘generally refers to the step in which a system generates an output from its inputs, typically after deployment.’³⁷⁰ Inference was also a key element of the now rescinded October 2023 US Executive Order on AI.³⁷¹ Similarly, Brazil is contemplating to define an AI system as a ‘computational system, with different degrees of autonomy, designed to infer how to achieve a given set of objectives, using approaches based on machine learning and/or logic and knowledge representation, through input data from machines or humans, with the aim of producing predictions, recommendations or decisions that may influence the virtual or real environment.’³⁷² A similar definition has been espoused by Article 2 of the Council of Europe’s Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law.³⁷³ This broader alignment can be interpreted as a signal of an emerging international consensus on the definition of AI systems.³⁷⁴

2.135 The Commission Guidelines on the definition of an artificial intelligence system list various AI techniques that enable inference, namely supervised learning, unsupervised learning, self-supervised learning, reinforcement learning, deep learning, and logic and knowledge-based approaches such as knowledge representation, inductive (logic) programming, knowledge bases, inference and deductive engines, (symbolic) reasoning, expert systems and search and optimization methods.³⁷⁵ Recital 12 explicitly excludes simpler traditional

³⁶⁵ Recital 12 AIA.

³⁶⁶ Ibid.

³⁶⁷ Commission Guidelines on the definition of an artificial intelligence system (n 65) 5 stating that Article 3(1) ‘does not contradict’ these guidelines. See further ISO/IEC 22989:2022 (n 333).

³⁶⁸ Commission Guidelines on the definition of an artificial intelligence system (n 65) 3.

³⁶⁹ Recital 12 AIA.

³⁷⁰ OECD Explanatory Memorandum (n 318) 9.

³⁷¹ Executive Order No 14110, Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, 88 Federal Register 75193 (2023). This Executive Order defined the AI system as follows: ‘a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments. Artificial intelligence systems use machine- and human-based inputs to perceive real and virtual environments; abstract such perceptions into models through analysis in an automated manner; and use model inference to formulate options for information or action.’

³⁷² Article 4(1) of Brazil AI Bill 2338/2023.

³⁷³ Article 2 of Council of Europe, ‘Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law’, No. 225 (2024).

³⁷⁴ Gstrein, Haleem, and Zwitter (n 304).

³⁷⁵ Commission Guidelines on the definition of an artificial intelligence system (n 65) 6–8.

software systems or programming approaches that are based on the rules defined solely by natural persons to automatically execute operations.³⁷⁶ The non-binding Commission Guidelines on the definition of an artificial intelligence system similarly identify four classes of systems that should not be qualified as AI systems under Article 3(1). First, systems for improving mathematical optimization. These techniques should be interpreted as ‘including systems used to improve mathematical optimisation or to accelerate and approximate traditional, well established optimisation methods, such as linear or logistic regression methods.’³⁷⁷ This exclusion is justified by the fact that ‘while those models have the capacity to infer, they do not transcend ‘basic data processing’. An indication that a system does not transcend basic data processing could be that it has been used in consolidated manner for many years.’³⁷⁸ Also systems that solely intended for descriptive analysis, hypothesis testing, and visualization, fall outside the definition of an AI system.³⁷⁹ Second, the Guidelines exclude basic data processing methods from the scope of Article 3(1). These are predefined, explicit instructions or operations that do not involve ‘learning, reasoning or modelling’ at any stage of the system lifecycle.³⁸⁰ Third, the Guidelines exclude systems based on classical heuristics, defined as problem-solving techniques that rely on experience-based methods to find approximate solutions efficiently to apply predefined rules or algorithms to derive solutions.³⁸¹ These methods do not adjust their models based on input-output relationships, in contrast to machine-learning techniques, meaning that they learn from experience.³⁸² Fourth, the Guidelines exclude simple prediction systems whose performance can be achieved via a basic statistical learning rule.³⁸³ It must, however, be borne in mind that ‘no automatic determination or exhaustive lists of systems that either fall within or outside the definition of an AI system are possible.’³⁸⁴ Furthermore, the guidelines adopted by the Commission are not legally binding, meaning that future judicial interpretations of Article 3(1) might differ from those now espoused by the Commission.

The remaining ambiguities in Article 3(1) are likely to become the subject of future judicial interpretation. However, somewhat counterintuitively, companies may have limited incentive to challenge the breadth of this provision in court, as it not only delineates the scope of the AI Act but also defines the extent of Union competence in the field of AI, thereby constraining the regulatory space left to Member States. It can be resumed that precisely for this reason, there was strong industry pressure to secure the adoption of the AIA in the final days leading up to the Committee of Permanent Representatives (‘COREPER’) vote, at a time when its passage appeared uncertain.

ii. Article 3(2): Risk

The AI Act relies on the assumption that while AI can lead to beneficial outcomes, it may also ‘generate risks and cause harm to public interests and fundamental rights that are protected by Union law.’³⁸⁵ It is hence unsurprising that risk is a central concept in the

³⁷⁶ Recital 12 AIA.

³⁷⁷ Commission Guidelines on the definition of an artificial intelligence system (n 65) 8.

³⁷⁸ *Ibid.*

³⁷⁹ *Ibid.* 9.

³⁸⁰ *Ibid.*

³⁸¹ *Ibid.*

³⁸² *Ibid.*

³⁸³ *Ibid.* 10.

³⁸⁴ *Ibid.* 12.

³⁸⁵ Recitals 4 and 5 AIA.

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regulation.³⁸⁶ This centrality calls for a definition, which we find in Article 3(2), which defines risk as ‘the combination of the probability of an occurrence of harm and the severity of that harm’.

2.138 Risk has long been **defined** along these lines. Frederick Warner wrote in 1992 that ‘a commonly accepted definition of risk is that of a combination of the probability or likelihood of an event (i.e., the risk) and the severity or magnitude of its consequences (i.e., the harms it leads to).’³⁸⁷ The definition of risk has also been embraced by technical standards such as Clause 3.9 of ISO/IEC Guide 51:2014, which defines risk as the ‘combination of the probability of occurrence of harm and the severity of that harm.’³⁸⁸ Similar definitions have also been adopted elsewhere in EU law. The General Food Law Regulation, which establishes the principle of risk analysis in relation to food and feed, defines risk as ‘a function of the probability of an adverse health effect and the severity of that effect, consequential to a hazard.’³⁸⁹

2.139 Despite the established nature of this definition, the AIA does not define its key terms—specifically, ‘harm’, as well as the concepts of the **probability of harm and the severity of harm**. Harm was described as follows by the High-Level Expert Group on AI: ‘Harms can be individual or collective, and can include intangible harm to social, cultural and political environment.’³⁹⁰ These Ethics Guidelines formulate a principle of prevention of harm, which ‘entails the protection of human dignity as well as mental and physical integrity. AI systems and the environments in which they operate must be safe and secure. They must be technically robust and it should be ensured that they are not open to malicious use. Vulnerable persons should receive greater attention and be included in the development, deployment and use of AI systems. Particular attention must also be paid to situations where AI systems can cause or exacerbate adverse impacts due to asymmetries of power or information, such as between employers and employees, businesses and consumers or governments and citizens. Preventing harm also entails consideration of the natural environment and all living beings.’³⁹¹ Interestingly, while this description makes explicit reference to risks to fundamental rights and safety, risks to human health are not mentioned. According to Recitals 7 and 27, the Ethics Guidelines devised by the High-Level Expert Group ‘should be translated, when possible, in the design and use of AI models.’³⁹² Yet, the vague wording of these recitals, coupled with their lack of legally binding force, raises questions about the extent to which they can or should be relied upon in the context of Article 3(2).³⁹³ Regarding **probability**, the ISO Standard on Safety Aspects, provides that this is ‘a function of the exposure to a hazardous situation, the occurrence of a hazardous event, and the possibilities of avoiding or limiting the harm.’³⁹⁴

³⁸⁶ See e.g. the risk management system that providers of high-risk AI systems are required to implement pursuant to Article 9 AIA.

³⁸⁷ Frederick Warner, ‘Introduction’ in Royal Society (ed), *Risk: analysis, perception and management* (Royal Society 1992) 4.

³⁸⁸ ‘ISO/IEC Guide 51:2014 - Guidelines for Their Inclusion in Standards’ (3rd edn, ISO 2014) <<https://www.iso.org/standard/53940.html>> accessed 16 January 2025.

³⁸⁹ Article 3(9) of the General Food Law Regulation (Regulation (EC) No 178/2002 of the European Parliament and of the Council of 28 January 2002 laying down the general principles and requirements of food law, establishing the European Food Safety Authority and laying down procedures in matters of food safety [2002] OJ L31/1).

³⁹⁰ HLEGAI Ethics Guidelines for Trustworthy AI (n 34) 12.

³⁹¹ Ibid.

³⁹² Recital 27 AIA.

³⁹³ On the legal status of recitals in EU law, see further related analysis in the Introduction.

³⁹⁴ ISO/IEC Guide 51:2014 (n 388) Clause 5. The terms ‘hazard’, ‘hazardous event’ and ‘hazardous situation’ are defined in ISO/IEC Guide 51:2014 (n 388) Clause 3.2–3.4.

Notably, the AIA goes beyond the typical focus of EU product law on health and safety risks, extending its scope to include ‘risks’ to fundamental rights.³⁹⁵ This introduces a unique interpretational challenge, as ‘risk’ is not a well-established concept in EU fundamental rights law, which traditionally focuses on interferences and violations of rights.³⁹⁶ Notably, this raises questions regarding the evaluation of ‘risk’ in terms of ‘harm’, given that, under EU law, a violation of fundamental rights can occur without any actual harm being present.³⁹⁷ This introduces complex conceptual and interpretational challenges, as Article 3(2), being part of secondary legislation, must be interpreted in accordance with EU primary law, including the Charter of Fundamental Rights.

The definition of risk in Article 3(2) also forms the basis of another definition, namely that of systemic risk under Article 3(65). This is relevant in the context of Article 55, which creates additional legal obligations for providers of those general-purpose AI models that are deemed to present systemic risk.³⁹⁸

iii. Article 3(3): Provider

The AI Act is **provider-centric** as it assigns the bulk of legal obligations to providers, making the related definition pivotal for determining the regulation’s scope and ensuring accountability. In this context, Article 3(3) plays a crucial role in clarifying both the personal and territorial scope set out in Article 2(1).

Article 3(3) **defines** the provider as a natural or legal person, public authority, agency or other body that develops an AI system or a general-purpose AI model or that has an AI system or a general-purpose AI model developed and places it on the market or puts the AI system into service under its own name or trademark, whether for payment or free of charge. The final formulation of Article 3(3) largely aligns with the one originally proposed by the draft regulation, with only minor modifications, such as the reference to general-purpose AI models, which were not within the scope of the draft Act.³⁹⁹

The concept of the provider echoes that of the **manufacturer** in EU product safety law, which refers to ‘any natural or legal person who manufactures a product or has a product designed or manufactured, and markets that product under its name or trademark.’⁴⁰⁰ Yet, since ‘putting into service’ serves as an alternative criterion alongside ‘placing on the market’ for defining a provider, the concept of ‘provider’ under the AI Act extends well beyond that of ‘manufacturer’ in EU product law.⁴⁰¹

Article 3(3) is broad in its formulation, allowing for a wide interpretation of its provisions. Considering the heavily provider-centric nature of the AIA, the qualification as provider is something that economic operators will try to avoid. The provider can be a **natural or legal**

³⁹⁵ See further Article 1(1) AIA.

³⁹⁶ On EU fundamental rights law, see further Steve Peers and others (eds), *The EU Charter of Fundamental Rights: A Commentary* (2nd edn, Hart 2021).

³⁹⁷ Joined Cases C-293/12 and C-594/12 *Digital Rights Ireland* EU:C:2014:238, paras 33 and 37; Joined Cases C-465/00 C-138/01 and C-139/01 *Rundfunk* EU:C:2003:294, [2003] ECR I-04989, para 75; *Amann v Switzerland* App No 27798/95 (ECtHR, 16 February 2000), para 70.

³⁹⁸ According to Article 3(65) AIA, systemic risk is ‘a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain.’

³⁹⁹ See further Four Columns AI Act (n 1) line 129.

⁴⁰⁰ Article 3(8) MSR.

⁴⁰¹ *Ibid.*

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person, public authority, agency or other body. This broad formulation underscores the regulation's expansive scope and its intention to encompass any actor exercising decisive influence over an AI system or general-purpose AI model—particularly those capable of identifying and managing associated risks, as required under the relevant chapters of the regulation. In fact, public authorities, including EU institutions, bodies, offices and agencies, can also be explicitly included as providers.⁴⁰² The CJEU recently held in a case concerning the interpretation of Article 4(7) GDPR, which similarly refers to 'natural or legal person, public authority, agency or other body' that there is no need for public agencies or bodies to have legal personality by virtue of national law for them to fall within the scope of this provision as long as the scope of its responsibility is generally determined by law.⁴⁰³

- 2.146** With respect to natural persons, the AI Act does not provide a household exemption for providers. However, a **commercial context** is nonetheless presumed—evidenced by the requirement that an AI system or general-purpose AI model must be placed on the market or put into service.⁴⁰⁴
- 2.147** Article 3(3) applies to those **directing the development of AI systems and general-purpose AI models**, whether they develop the system themselves or delegate the task to a third party before bringing it to market or putting it into service. The concept of 'development' is not explicitly defined in the legislation. However, its use underscores that providers are subject to the majority—and, in the case of Chapter V, all—of the legal obligations imposed by the AI Act. This reflects the assumption that providers possess the greatest knowledge, expertise, and access to the AI system, positioning them as best placed to exercise *ex ante* risk control.
- 2.148** Providers are the actors that **place an AI system or general-purpose AI model on the market or put it into service under their own name or trademark**. Placing on the market 'means the first making available of an AI system or a general-purpose AI model on the Union market'.⁴⁰⁵ **Placing on the market** can occur for payment or free of charge. This formulation is designed to catch certain code released as open-source software. A commercial context is, however, required.⁴⁰⁶ **Putting into service** refers to 'the supply of an AI system for first use directly to the deployer or for own use in the Union for its intended purpose'.⁴⁰⁷
- 2.149** Pursuant to Article 2(1)(a), providers placing on the market or putting into service AI systems or placing on the market general-purpose AI models in the Union, are subject to the regulation irrespective of their geographical location. Providers established or located in a third country are also subject to the AIA where the output of the AI system (but not of a GPAIM) is used in the EU.⁴⁰⁸ Third countries are countries that are not EU Member States.⁴⁰⁹

⁴⁰² Recital 23 AIA reads as follows: 'This Regulation should also apply to Union institutions, bodies, offices and agencies when acting as a provider or deployer of an AI system'.

⁴⁰³ Case C-638/23 *Amt der Tiroler Landesregierung* EU:C:2025:127.

⁴⁰⁴ Contrast with Article 2(10) AIA, which provides that the regulation 'does not apply to obligations of deployers who are natural persons using AI systems in the course of a purely personal non-professional activity'. While one can imagine scenarios in which an AI system is put into service in non-commercial contexts most of these contexts will have a commercial nature.

⁴⁰⁵ See further analysis on Article 3(9) AIA below.

⁴⁰⁶ Article 3(10) AIA.

⁴⁰⁷ See further analysis on Article 3(11) AIA below.

⁴⁰⁸ Article 2 (1)(c) AIA.

⁴⁰⁹ See further Case T-370/19 *Spain v Commission* paras 31–38.

In addition, there are **other circumstances** through which a natural or legal person may come to qualify as a provider under the AI Act. First, where an entity rebrands a high-risk AI system that has already been placed on the market or put into service by affixing its own name, trademark, or logo.⁴¹⁰ Second, where the intended purpose of an AI system that has been placed on the market or put into service is changed from not qualifying as high-risk to high-risk.⁴¹¹ Third, where there is a substantial modification of a high-risk AI system that has already been placed on the market or put into service.⁴¹² Fourth, where a GPAIM is customized and put into use through fine-tuning or modification.⁴¹³ Fifth, where a GPAIM is integrated into an AI system, which may be a GPAIS, in which case the ‘downstream provider’ becomes a system, but not a model provider.⁴¹⁴

The definition of the provider is further broadened by Article 25(3) pursuant to which for HRAIS that are safety components of products covered by EU harmonizing legislation, the **product manufacturer** of the product subject to EU harmonization legislation is considered to be the provider where (a) the HRAIS is placed on the market together with the product under the name or trademark of the product manufacturer, or (b) where the HRAIS is put into service under the name or trademark of the product manufacturer after the product has been placed on the market.

Whereas Article 3(3) only refers to providers, Chapter VI also explicitly applies to **prospective providers** in allowing them to partake in AI regulatory sandboxes and testing in real world conditions.⁴¹⁵

iv. Article 3(4): Deployer

Article 3(4) defines the deployer as the natural or legal person, public authority, agency or other body **using an AI system under its authority** except where it is used during a personal non-professional activity.⁴¹⁶ Beyond providers, deployers hold a crucial role within AI ecosystems. A fundamental distinction from ordinary products regulated under the New Legislative Framework is that AI systems can be significantly modified or influenced by their users.⁴¹⁷

Article 3(4) helps define the **personal and territorial scope** under Article 2(1). By virtue of Article 2(1)(b), it specifically applies to deployers that (a) have their place of establishment or are located in the EU are subject to the AIA (Article 2(1)(b)), as well as (b) those that have their place of establishment or are located in a third country to the extent that the output of the AI system is used in the EU (Article 2(1)(c)). Third countries are countries that are not EU Member States.⁴¹⁸

Deployers of AI systems, moreover, have several **obligations** under the AIA—chiefly the general obligations for deployers under Article 26; the obligation for some deployers to

⁴¹⁰ Article 25(1)(a) AIA.

⁴¹¹ Article 25(1)(c) AIA.

⁴¹² Article 25(1)(b) AIA.

⁴¹³ Recital 109 AIA suggests that when such fine-tuning and customization occur, the entity undertaking these tasks becomes a provider. If not, they are a deployer.

⁴¹⁴ Article 3(68) AIA.

⁴¹⁵ Please refer to the analysis of Article 57(5) AIA for an analysis of this concept.

⁴¹⁶ See also Article 2(10) and Recital 13 AIA.

⁴¹⁷ This is acknowledged in Article 25(1) AIA, which recognizes that a deployer’s influence over a high-risk AI system can be so substantial that, in certain circumstances, they must be legally reclassified as providers.

⁴¹⁸ See further Case T-370/19 *Spain v Commission* paras 31–38.

carry out fundamental rights impact assessments under Article 27; the transparency obligations for deployers governed by Article 50(3) and (4); as well as the duty for some deployers to register high-risk AI systems in the EU database pursuant to Article 71(3). The AI Act does not impose obligations on deployers of general-purpose AI models.⁴¹⁹

- 2.156** It must also be stressed that the roles of providers and deployers are **not mutually exclusive**, meaning an entity can fulfil both under the AIA. This occurs, for example, when a provider develops an AI system for its own use.⁴²⁰ This underscores the dynamic relationship between the roles of provider and deployer. While Article 25 explicitly addresses scenarios in which a deployer's influence over a high-risk AI system is so significant that it should be reclassified as a provider, a provider simultaneously assumes the role of deployer when using an AI system beyond its development phase.
- 2.157** Regarding **legislative history**, the draft regulation referred to 'users' instead of 'deployers', but the term was later removed to avoid confusion, as 'users' in the conventional sense are typically not held responsible for compliance.⁴²¹ Indeed, deployers cannot be equated to end users in other EU product law instruments where the end user refers to the person that ought to be protected by the supranational norms.⁴²² In contrast, AI deployers are not the entities the regulation seeks to protect; rather, they are participants within a broader AI ecosystem. Their obligations focus on safeguarding others from the potential risks that AI systems, and particularly the use thereof, may pose to health, safety, and fundamental rights. In contrast to the definition of the provider, the AIA only applies to deployers of AI systems, not of GPAIM.⁴²³
- 2.158** The inclusion of **natural or legal persons, public authorities, agencies and bodies** highlights the broad scope of the AIA. In line with the functional definition of a deployer, anyone using an AI system is subject to the AIA's obligations, ensuring effective risk management. This is in line with the regulation's overarching goal of ensuring trustworthy AI. Recital 23 explicitly recognizes that EU institutions, bodies, offices and agencies can be providers subject to the AIA.⁴²⁴ The only exception to this broad definition is the household exemption for natural persons, which is examined further below. The CJEU recently held in a case concerning the interpretation of Article 4(7) GDPR—which similarly refers to a 'natural or legal person, public authority, agency or other body'—that there is no need for public agencies or bodies to have legal personality by virtue of national law for them to fall within the scope of this provision, as long as the scope of its responsibility is generally determined by law.⁴²⁵
- 2.159** Pursuant to Article 3(4), the deployer is the entity that **uses an AI system under its authority** except where such usage occurs in the context of a personal, non-professional activity.⁴²⁶ The central criterion to define a deployer hence is whether it 'uses' an AI system

⁴¹⁹ See further Chapter V AIA, which only creates obligations for providers of general-purpose AI models.

⁴²⁰ Veale and Borgesius (n 104) 112.

⁴²¹ Muhammed Demircan, 'Deployers of High-Risk AI Systems: What Will Be Your Obligations Under the EU AI Act?' (*Kluwer Competition Law Blog*, 2 June 2023) <<https://competitionlawblog.kluwercompetitionlaw.com/2023/06/02/deployers-of-high-risk-ai-systems-what-will-be-your-obligations-under-the-eu-ai-act/>> accessed 9 December 2024.

⁴²² Article 3(21) MSR.

⁴²³ See Articles 50(3) and 50(4) AIA.

⁴²⁴ Recital 23 AIA reads as follows: 'This Regulation should also apply to Union institutions, bodies, offices and agencies when acting as a provider or deployer of an AI system'.

⁴²⁵ Case C-638/23 *Amt der Tiroler Landesregierung*.

⁴²⁶ Article 3(4) AIA.

that is ‘under its authority’. The AIA does not provide a clear definition of what constitutes the ‘use’ of an AI system, leading to speculation about whether it, for instance, also extends to situations where a party utilizes the output of a system that was originally operated by another party. Articles 2(1)(b) and (c) could, indeed, be understood to indicate that only using an output does not amount to the use of an AI system. Usage can occur in many different constellations. For instance, a financial institution developing a fraud detection system or procuring such a system in a software-as-a-service (‘SaaS’) format will be a deployer of this system under both circumstances (whilst additionally being a provider in the first scenario). What matters is whether the financial institution determines the input and makes use of the output of the AI system. The concept of ‘under its authority’ goes similarly undefined. It can be interpreted to mean, for example, that an employee of a company that uses an AI system in the context of their professional activity is not a deployer as the AI system is used under the authority of the employer, not the employee.⁴²⁷

An entity does not qualify as a deployer when an AI system is used during a **personal, non-professional activity**.⁴²⁸ While Article 3(4) does not formally limit the availability of this exemption, only natural persons can act in purely personal, non-professional activities.⁴²⁹ This principle echoes the household exemption formulated in Article 2(10), according to which the AIA does not apply to deployers who are natural persons using AI systems in the course of a purely personal, non-professional activity.⁴³⁰ In contrast to Article 3(4), which simply refers to ‘personal and non-professional’ activity, Article 2(10) explicitly requires a ‘purely’ personal and non-professional activity.

The AIA does not provide a definition of **what distinguishes ‘personal’ from ‘non-professional’**. Determining what qualifies as ‘personal’ versus ‘non-professional’ can be complex, especially when it comes to transparency obligations. Consider the scenario where a professional politician, who, as is commonplace, maintains both official and personal social media accounts, creates and shares a deepfake targeting a political opponent through their personal profile.⁴³¹ Despite the use of a personal account, the content is clearly linked to their professional role.

While interpretations of the household exemption in data protection law may offer some guidance, the term ‘**personal**’ remains without a definitive legal definition in that context. In 2014, Advocate General Jääskinen opined that ‘personal’ activities ought to be interpreted as ‘activities which are closely and objectively linked to the private life of an individual and which do not significantly impinge upon the personal sphere of others.’⁴³² However, this test has, to date, not been explicitly endorsed by the CJEU and it also is debatable whether a focus on the personal sphere of others should be relevant in the context of the AIA as there is not the same focus on the reach of an activity than there is under the GDPR.

⁴²⁷ Paul Voigt and Nils Hallen, *Handbuch KI-Verordnung: FAQ zum EU AI Act* (1st edn, Springer, 2024) ch. 1, 14; Martin Ebers and Chiara Streitböcher, ‘Die Regulierung von Hochrisiko-KI-Systemen in Der KI-Verordnung’ (2024) 5 Recht Digital 393, 399.

⁴²⁸ Article 3(4) AIA.

⁴²⁹ This is also made explicit by the wording of Article 2(10) AIA.

⁴³⁰ Article 2(10) AIA.

⁴³¹ This question is relevant in the context of Article 50(4) AIA, pursuant to which ‘deployers of an AI system that generates or manipulates image, audio or video content constituting a deep fake, shall disclose that the content has been artificially generated or manipulated’.

⁴³² Opinion of AG Jääskinen in Case C-212/13 *Ryneš*, para 51.

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v. Article 3(5): Authorized representative

2.163 Article 3(5) defines the authorized representative as the natural or legal person located or established in the Union who has received and accepted a written mandate from a provider of an AI system or a GPAIM to, respectively, perform and carry out on its behalf the obligations and procedures established by the AIA.

2.164 The authorized representative ensures that providers established in a third country have **official representation within the EU**. Third countries are countries that are not EU Member States.⁴³³ The concept is well-established within the New Legislative Framework and can, for instance, also be found in Article 2(32) MDR.⁴³⁴ As the Blue Guide clarifies, manufacturers of products can appoint authorized representatives in the EU to act on their behalf in carrying out certain tasks.⁴³⁵ The authorized representative must be located or established in the EU and have received and accepted a written mandate from the provider.⁴³⁶

2.165 Article 3(5) builds on this **established concept** in the context of AI systems and general-purpose AI models. Article 22(1) obliges providers established outside the EU to designate an authorized representative by written mandate prior to making a high-risk AI system available in the internal market.⁴³⁷ This is designed to ensure that supervisory authorities as well as affected persons have a contact point within the Union. Article 54 creates the same obligation for providers of GPAIM established outside the EU.⁴³⁸ Pursuant to Article 22(3) and 54(2), the written mandate must be drafted so that the authorized representative is empowered to carry out several tasks foreseen by the AIA. Article 13(3)(a) requires that the name and contact details of the authorized representative must be mentioned in the instructions for use of a HRAIS.

2.166 It is worth noting that it is a well-established principle of EU product safety law that while manufacturers established in third countries are subject to an obligation to designate an authorized representative, those established in the EU may also do so.⁴³⁹

2.167 Whereas this provision only refers to deployers, the AI Act also applies to **prospective deployers** as it enables and governs their participation in real-world testing.⁴⁴⁰ Whereas the AIA does not define the notion of the prospective deployer, it can be assumed that this is the natural or legal person, public authority, agency or other body that plans to use an AI system under its authority.

vi. Article 3(6): Importer

2.168 The importer is the natural or legal person located or established in the EU that places on the market an AI system, which bears the name or trademark of a natural or legal person established in a third country.⁴⁴¹ Importers are one of the possible operators of HRAIS.⁴⁴²

⁴³³ See further Case T-370/19 *Spain v Commission* paras 31–38.

⁴³⁴ Article 2(32) MDR defines authorized representative as follows: ‘any natural or legal person established within the Union who has received and accepted a written mandate from a manufacturer, located outside the Union, to act on the manufacturer’s behalf in relation to specified tasks with regard to the latter’s obligations under this Regulation’.

⁴³⁵ Blue Guide 2022 (n 186), point 3.2.

⁴³⁶ Article 3(5) AIA.

⁴³⁷ Article 22(1) AIA.

⁴³⁸ Article 54 AIA.

⁴³⁹ Blue Guide 2022 (n 186).

⁴⁴⁰ See further analysis of Article 60(2) AIA.

⁴⁴¹ Third countries are countries that are not EU Member States See further Case T-370/19 *Spain v Commission* paras 31–38.

⁴⁴² Article 3(8) AIA.

The importer holds a special responsibility in place of the provider when the latter is located in a **third country**. Like the arrangements for permanent representatives, this ensures that accountability under the AIA can be upheld within the EU, even if the provider is based outside its borders. The notion of the importer is also a well-established concept under the New Legislative Framework.⁴⁴³

Any **natural or legal person located or established in the EU** can be an importer. This is in line with the general definition of importers under the NLF.⁴⁴⁴ It is worth noting that factually there is no necessity for an ‘importer’ importing AI system to be located or established in the EU. Rather, there can be business models where an entity based outside of the EU puts into service an AI system within the Union. The 2024 Product Liability Directive recognizes this and defines the importer as any natural or legal person who places a product from a third country on the Union market.⁴⁴⁵ Under the AIA, such natural or legal persons qualify as distributors and the definition of the importer adheres to a required nexus between the activity of an importer and its geographical location.

To determine the importer, the concept of **placing on the market** is crucial. Article 3(9) defines placing on the market as the first making available of an AI system or a general-purpose AI model on the Union market. The importer is hence the natural or legal person located or established in the EU who makes available an HRAIS on the Union market for the first time, bearing the name or trademark of entities established in a third country. Indeed, the limitation to HRAIS is in order as the AIA only creates obligations for importers of high-risk AI systems.⁴⁴⁶

A natural or legal person located or established in the EU who, for the first time, makes available an high-risk AI system on the Union market, ordinarily qualifies as a provider of an AI system.⁴⁴⁷ Importers are only those entities that, for the first time, place on the market an AI system bearing **the name or trademark of a natural or legal person established in a third country**. There, this applies to AI systems for which another provider can be identified but that entity does not have the required geographical nexus with the EU.

vii. Article 3(7): Distributor

Article 3(7) defines the distributor as the natural or legal person in the supply chain, other than the provider or importer, who makes an AI system available in the internal market. Making available in the internal market refers to the supply of an AI system or a general-purpose AI model for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge.⁴⁴⁸ Natural or legal persons in the supply chain that make available AI systems or GPAIM in the EU, and who are not caught by the definitions of the provider or the importer are defined residually as distributors.

Distributors are one of the possible operators of HRAIS.⁴⁴⁹ The figure of the distributor forms part of the New Legislative Framework and the same definition can be found, inter

⁴⁴³ See in lieu of many legislative provisions Blue Guide 2022 (n 186) point 3.3.

⁴⁴⁴ Blue Guide 2022 (n 186) point 3.3.

⁴⁴⁵ Article 4(12) of the Product Liability Directive (Directive (EU) 2024/2853 of the European Parliament and of the Council of 23 October 2024 on liability for defective products and repealing Council Directive 85/374/EEC [2024] OJ L2853).

⁴⁴⁶ See Articles 20, 23, 24 and 25 AIA.

⁴⁴⁷ Article 3(3) AIA.

⁴⁴⁸ Article 3(10) AIA.

⁴⁴⁹ Article 3(8) AIA.

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alia, in the MR or the Product Liability Directive.⁴⁵⁰ Article 24 details the duties of distributors of HRAIS, which must, for instance, verify whether the system bears the required CE marking⁴⁵¹, that it is accompanied by a copy of the EU declaration of conformity under Article 47 and instructions for use, as well as that both providers and importers have complied with their respective obligations under Articles 16(b) and (c), and also Article 23(3).⁴⁵² Distributors are also under an obligation to cooperate with supervisory authorities and to provide them with information and documentation.⁴⁵³ Pursuant to Article 7(3) of the Product Liability Directive, distributors can under certain circumstances be liable for damage caused by providers and importers.

viii. Article 3(8): Operator

- 2.175** Article 3(8) defines the operator as a provider, product manufacturer, deployer, authorized representative, importer, or distributor. This is thus the catch-all category for all actors involved in AI supply chains that are subject to legal obligations under the AIA. Many of these obligations are specific to a certain category of actors, such as the obligations resting on providers of HRAIS.⁴⁵⁴ Some provisions of the regulation impose obligations on operators more broadly, such as the derogations outlined in Article 63, according to which any operator that qualifies as a microenterprise under EU law can comply with the obligation for a quality management system under Article 17 in a simplified manner.

ix. Article 3(9): Placing on the market

- 2.176** In accordance with Article 3(9), ‘placing on the market’ means the first making available of an AI system or a GPAIM on the Union market. The importance of this concept can hardly be overstated for it determines the applicability of the AI Act to an AI system or GPAIM. Indeed, a provider must comply with the requirements arising under the regulation if they place an AI system or GPAIM on the internal market.⁴⁵⁵
- 2.177** The concept of ‘placing on the market’ is **key in EU product safety law** and can, inter alia, also be found in Article 3(2) of the Market Surveillance Regulation (‘MSR’) and Article 3(7) GPSR.⁴⁵⁶ ‘Placing on the market’ is also a key concept within the AIA and forms part of the **central definitions** such as that of the provider or the importer.⁴⁵⁷ Furthermore, the concept is crucial to clearly define what activity falls outside the scope of the regulation, as research, testing, or development activities related to AI systems or models before their market entry or deployment are shielded from the AIA’s application under the research exemption in Article 2(8).⁴⁵⁸ Crucially, AI systems must comply with the AIA at the time they are placed on the market.⁴⁵⁹ Article 3(57) clarifies that testing under real-world conditions does not amount to ‘placing on the market’ provided that the conditions under Article 57,

⁴⁵⁰ See Blue Guide 2022 (n 186) 39; Machinery Regulation (n 213); Article 3(21) AIA and Article 4(14) of the Product Liability Directive (n 445).

⁴⁵¹ Article 24(1) AIA.

⁴⁵² Ibid.

⁴⁵³ Article 24(5) and (6) AIA.

⁴⁵⁴ See further Article 16 AIA.

⁴⁵⁵ Articles 2(1) and 2(8) AIA.

⁴⁵⁶ MSR and Article 3(7) GPSR.

⁴⁵⁷ Articles 3(3) and (6) AIA.

⁴⁵⁸ Article 2(8) AIA.

⁴⁵⁹ Blue Guide 2022 (n 186) 20. ‘The placing on the market is the most decisive point in time concerning the application of the Union harmonised legislation. When made available on the Union market, products must be in compliance with the Union harmonisation legislation applicable at the time of placing on the market.’

which enables real world testing in AI regulatory sandboxes, or Article 60, which relates to real-world testing outside of AI regulatory sandboxes, are met. The concept ought to be interpreted with reference to each individual AI system.⁴⁶⁰

The definition in Article 3(9) contains **three important elements**: (i) the presence of an AI system and general-purpose AI model, which are being (ii) made available for the first time, (iii) in the internal market. The notions of the AI system and general-purpose AI model need to be interpreted in the respective definitions.⁴⁶¹ The notion of the Union market refers to the sum of all Member States' markets. It can be taken as a synonym for the internal market, which is defined in Article 26(2) TFEU as an 'area without internal frontiers in which the free movement of goods, persons, services and capital is ensured'.⁴⁶² This leaves us with the need to define when an AI system or general-purpose AI model is first made available on the Union market. This concept ought to be interpreted in line with the corresponding definition in Article 3(10), which is examined just below.

x. Article 3 (10): Making available on the market

In accordance with Article 3(10), 'making available on the market' means the supply of an AI system or a GPAIM for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge. Although Article 3(10) refers to 'the market' generically, it follows from Article 3(9) that this is shorthand for the internal market.

This formulation is borrowed from other **product safety law** norms such as, for instance, Article 3(11) of the MR, and is of general importance in the context of the NLF.⁴⁶³ It ought to be interpreted with reference to each individual AI system.⁴⁶⁴

The AI system needs to be **made available to other operators**, irrespective of whether a product is actually physically handed over or software is made available.⁴⁶⁵ The Blue Guide explains that the central role of the concept 'is related to the fact that all economic operators in the supply chain have traceability obligations and need to have an active role in ensuring that only compliant products circulate on the Union market'.⁴⁶⁶

An AI system or GPAIM can be considered to have been made available on the market if it is **supplied for distribution or use**.⁴⁶⁷ Supplying a product can only be considered synonymous with making available 'when the product is intended for end use on the Union market. The supply of products, whether for further distribution, for incorporation into a final product or for further processing or refinement, with the aim to export the final product outside the Union market is not considered as making available'.⁴⁶⁸ This is in line with the definition of territorial scope embraced by Article 2(1), pursuant to which AI systems or

⁴⁶⁰ Ibid 19.

⁴⁶¹ Articles 3(1) and 3(63) AIA.

⁴⁶² Article 26(2) TFEU.

⁴⁶³ Article 3(10) AIA.

⁴⁶⁴ Ibid. Blue Guide 2022 (n 186) point 2.2. stating that making available 'refers to each individual product, not to a type of product, and whether it was manufactured as an individual unit or in series'.

⁴⁶⁵ Case C-691/21 *Cafpi and Aviva assurances* EU:C:2022:926.

⁴⁶⁶ Blue Guide 2022 (n 186) point 2.2.

⁴⁶⁷ The Blue Guide embraces a broader definition of 'made available on the market when supplied for distribution, consumption or use on the Union market'. Blue Guide 2022 (n 186) point 2.2. Presumably, the word 'consumption' has been omitted as it is not fitting for AI systems or general-purpose AI models, which are rather 'used'.

⁴⁶⁸ Blue Guide 2022 (n 186) point 2.2.

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GPAIM created in the EU but not placed on the market or put into service are outside the scope of the regulation.

- 2.183** The Commission Guidelines on the definition of an AI system explain that placing on the market, *inter alia*, occurs when an AI system developed outside the Union by a third-country provider is placed on the Union market for the first time—whether it is offered in return for payment or free of charge—in one or more Member States. Such placing on the market may occur by providing access to the system online through an API or other user interface.⁴⁶⁹
- 2.184** The concept of ‘use’ refers to the intended purpose of the product as defined by the manufacturer under conditions which can be reasonably foreseen. Usually, this is the end use of the product.⁴⁷⁰ The notion of the **intended purpose** needs to be defined in line with the definition provided under Article 3(12).⁴⁷¹ The concept of ‘making available on the market’ only refers to the supply of an AI system for use by the provider. The entity using the AI system under its authority is a deployer.⁴⁷²
- 2.185** According to the Blue Guide, **making available a product** supposes ‘an offer or an agreement (written or verbal) between two or more legal or natural persons for the transfer of ownership, possession or any other right concerning the product in question after the stage of manufacture has taken place. The transfer does not necessarily require the physical handover of the product. This transfer can be for payment or free of charge, and it can be based on any type of legal instrument.’⁴⁷³ However, the requirement that the manufacturing stage has been completed maps more easily to physical products compared to AI system. Indeed, AI systems may, as per the definition in Article 3(1), ‘exhibit adaptiveness after deployment’. Unlike a dishwasher, an AI system may hence not be conclusively produced by the time it is placed on the market. Yet, as per the Blue Guide, each product can only be placed on the market once.⁴⁷⁴ This underlines the importance of the concept of ‘**substantial modification**’ in Article 3(23) as well as the related legal regime in Article 25(1)(b) pursuant to which any importer, deployer or third party may become a provider of a high-risk AI system and subject to the obligations under Article 16 if they make a substantial modification to a high-risk AI system that was already placed on the market or has already been put into service in such a way that it remains a high-risk AI system.⁴⁷⁵
- 2.186** Placing on the market **does not require the physical handover of the product**.⁴⁷⁶ Indeed, AI systems can be unembodied software.⁴⁷⁷ This poses a challenge in determining when an AI system is placed on the market, unlike more tangible products, such as medical devices or elevators, where market entry is easier to define. Presumably Article 3(10) should be interpreted to mean that an AI system or general-purpose AI model is made available on the market whenever it is put into circulation with the goal of allowing deployers to use it. In *Weltimmo*, the CJEU held in a case concerning the interpretation of the territorial application of data protection law that a data controller can be deemed to exercise economic activity

⁴⁶⁹ Commission Guidelines on the definition of an artificial intelligence system (n 65) 4.

⁴⁷⁰ Blue Guide 2022 (n 186) point 2.2.

⁴⁷¹ Article 3(12) AIA.

⁴⁷² Article 3(10) AIA.

⁴⁷³ Blue Guide 2022 (n 186) 20.

⁴⁷⁴ *Ibid* 39.

⁴⁷⁵ Article 25(1)(b) AIA.

⁴⁷⁶ Blue Guide 2022 (n 186) point 2.2.

⁴⁷⁷ See further Article 3(10) and related analysis.

in a Member State other than the one in which it is established to the extent that it ‘exercises, through stable arrangements in the territory of that Member State, a real and effective activity—even a minimal—in the context of which that processing is carried out’.⁴⁷⁸ This was deemed to be the case where a data controller offered real estate in a Member State and in the language of that Member State rather than the language of the Member State in which it was formally established.⁴⁷⁹ Whereas the establishment is of course easier to pinpoint where there is actual real estate on the soil of a Member State, the language criterion seems particularly pertinent regarding the interpretation of Article 3(10), as could be a domain name ending in the abbreviation of the Member State.⁴⁸⁰

Article 3(10) underlines that making available on the market always involves a **commercial activity**, irrespective of whether it is for payment or free of charge. Commercial activity has been defined ‘as providing goods in a business-related context. Non-profit organizations may be considered as carrying out commercial activities if they operate in such a context. This can only be appreciated on a case-by-case basis considering the regularity of the supplies, the characteristics of the product, the intentions of the supplier, etc. In principle, occasional supplies by charities or hobbyists should not be considered as taking place in a business-related context’.⁴⁸¹

xi. Article 3(11): Putting into service

Article 3(11) defines ‘putting into service’ as the supply of an AI system for first use directly to the deployer or for own use in the Union for its intended purpose. The importance of this concept can hardly be overstated for it determines the applicability of the AI Act to an AI system. Indeed, a provider must comply with the requirements arising under the regulation if they put an AI system into service.⁴⁸²

Putting into service an AI system refers to the **supply of that system for first use**. Putting into service takes place at the moment of first use within the Union by the end user for the purposes for which it was intended.⁴⁸³ The notion of ‘use’ is not defined in relation to Article 3(11), but the Blue Guide provides a definition thereof in the context of ‘making available on the market’ and according to which the concept of ‘use’ refers to the intended purpose of the product as defined by the manufacturer under conditions which can be reasonably foreseen. Usually, this is the end use of the product.⁴⁸⁴ This definition can also be embraced here.

It is unclear whether the supply needs to have a **commercial nature**. Interestingly, and in contrast to the activities previously defined, Article 3(11) does not require that the putting into service of an AI system has a commercial character. It is questionable whether this omission by the EU legislator was intentional. There is, indeed, no obvious reason why non-commercial activities would be caught under Article 3(11) but not the other activities caught by the AIA.⁴⁸⁵ However, the lack of explicit clarification on this point is problematic,

⁴⁷⁸ Case C-230/14 *Weltimmo* EU:C:2015:639, para 41.

⁴⁷⁹ *Ibid.*

⁴⁸⁰ This criterion is of course not determinative in relation to languages shared by different Member States.

⁴⁸¹ Blue Guide 2022 (n 186) point 2.2.

⁴⁸² Articles 2(1) and 2(8) AIA.

⁴⁸³ Blue Guide 2022 (n 186) point 2.6. The notion of the end user ought to be translated into the deployer in the context of the AIA.

⁴⁸⁴ Blue Guide 2022 (n 186) point 2.2.

⁴⁸⁵ It is also worth noting that a commercial activity is explicitly required under Article 4(9) of the Product Liability Directive (n 445).

as it leaves open the question of what precisely constitutes ‘putting into service’ and, more broadly, where the boundaries of the research exemption lie.⁴⁸⁶

- 2.191** While the notion of putting a product into service is also used in other EU harmonizing instruments, it is generally used as the first use of a product, not the supply of an AI system for first use. To illustrate, Article 3(13) of the MR defines the concept as ‘the first use, for its intended purpose, in the Union, of machinery or related product.’⁴⁸⁷ Under the AI Act, ‘putting into service’ is an act of the provider, not the deployer.⁴⁸⁸ The supply of an AI system for first use to a deployer, for instance, occurs when an AI system developed outside the Union by a **third-country provider** is placed on the Union market for the first time—whether it is offered in return for payment or free of charge—in one or more Member States.⁴⁸⁹ In turn, the supply of an AI system for own use, for instance, occurs when a public authority develops a scoring system in-house and deploys it to predict the risk of fraud of household allowance beneficiaries, thereby putting it into service.⁴⁹⁰
- 2.192** A central question in relation to this concept is whether the supply of SaaS can be framed as the *supply* of an AI system.⁴⁹¹ If the interpretation excludes SaaS-based provision from the notion of supply, then AI systems or GPAIM delivered exclusively via SaaS would, in principle, fall outside the traditional categories of provider, importer, or distributor—given that such models do not involve placing or making the system available on the market. Consequently, only the users of the system would be subject to regulation, classified solely as deployers.⁴⁹² Yet, such an interpretation would defeat the rationale behind the legislation as many AI systems and GPAIMs are supplied as software and the co-legislators explicitly intended to cover not only products but also service within the scope of the regulation.⁴⁹³
- 2.193** Putting into service takes place at the moment of first use within the EU for the **purposes for which it was intended**.⁴⁹⁴ This last requirement ought to be interpreted in line with the notion of intended purpose, defined in Article 3(12), which is introduced just below.
- xii. Article 3(12): Intended purpose
- 2.194** Article 3(12) defines the intended purpose of an AI system as the use for which it is intended by the provider, including the specific context and conditions of use, as specified in the information supplied by the provider in the instructions for use⁴⁹⁵, promotional or sales materials and statements, and in the technical documentation. The intended purpose of an AI system is a **central interpretative concept** of the obligations set out in Chapter III for high-risk AI systems.⁴⁹⁶

⁴⁸⁶ See further Article 2(10) AIA.

⁴⁸⁷ Article 3(13) of the Machinery Regulation (n 213).

⁴⁸⁸ Articles 2(1)(a) and (b) As well as Articles 3(3) AIA. It should be noted, however, that this act may also lead to a requalification of the provider into a deployer when they subsequently use the product under their authority in accordance with Article 3(4) AIA.

⁴⁸⁹ Commission Guidelines on prohibited artificial intelligence practices (n 166) 4.

⁴⁹⁰ Ibid 5.

⁴⁹¹ See generally Jennifer Cobbe and Jatinder Singh, ‘Artificial Intelligence as a Service: Legal Responsibilities, Liabilities, and Policy Challenges’ (2021) 42 Computer Law & Security Review 105573.

⁴⁹² Article 6(2) MDR and Blue Guide 2022 (n 186) point 2.2. could be read to support such an interpretation.

⁴⁹³ Recital 8 AIA states that ‘rules regarding the placing on the market, the putting into service and the use of certain AI systems should be laid down, thus ensuring the smooth functioning of the internal market and allowing those systems to benefit for the principle of the free movement of goods and services [...]’.

⁴⁹⁴ Blue Guide 2022 (n 186) point 2.6.

⁴⁹⁵ See further Article 3(15) and Article 13(3) AIA.

⁴⁹⁶ See Article 8(1) as well as Article 9(2)(a) and (b) and Article 10(2) AIA.

The notion of ‘use’ is not legislatively defined. According to the Commission Guidelines on the interpretation of the prohibited practices in Article 5, this notion ought to be interpreted ‘in a broad manner to cover the use or deployment of the system at any moment of its lifecycle after having been placed on the market or put into service. This may also cover the integration of the AI system in the services and processes of the person(s) making use of the AI system, including as part of more complex systems, processes or infrastructure.’⁴⁹⁷ Therefore, the intended purpose of the AI system aligns with its intended use as defined by the provider. This also echoes the definition in the Blue Guide according to which the concept refers to the intended purpose of the product as defined by the manufacturer under conditions which can be reasonably foreseen. Usually, this is the end use of the product.’⁴⁹⁸

The intended purpose is **explicitly or implicitly defined by the provider**. The specific context and conditions of use, as specified in the information supplied by the provider in the instructions for use, promotional or sales materials and statements, as well as in the technical documentation need to be considered to determine what the intended purpose is.⁴⁹⁹ These definitions of context and conditions of use are also operated by the provider, underlining its centrality in defining the intended purpose of an AI system. Indeed, the provider needs to formulate the intended purpose of a high-risk AI system in its instructions for use.⁵⁰⁰

By requiring consideration of a broad range of factors, the regulation ensures that a provider cannot narrowly define the intended purpose in the instructions for use while simultaneously promoting a broader application to potential deployers. These objective criteria **safe-guard against a purely subjective interpretation** of ‘intention’, grounding it instead in the purpose explicitly defined and communicated by the provider. Yet, if the deployer changes the intended purpose of a high-risk AI system to a purpose not initially envisaged by the provider and this amounts to a **substantial modification**⁵⁰¹, then the deployer becomes the provider.⁵⁰² If the deployer changes the intended purpose with the result of the AI system qualifying as high-risk, the deployer themselves becomes the provider.⁵⁰³

By defining the intended purpose, providers can effectively manage the risks inherent to AI systems. Indeed, the idea behind the concept is that providers are ‘required to match a level of protection for the users of the products or of other public interests which corresponds to the use that the manufacturer prescribes for the product in the product information.’⁵⁰⁴

As the entities responsible for the development of AI systems, **providers** are best positioned to identify and evaluate potential risks. It is therefore logical that they bear primary responsibility for defining the boundaries of appropriate use. For instance, a provider may restrict the deployment of a HRAIS in certain sectors or delineate specific contexts in which its use would be unsuitable. A salient example might involve a cancer detection algorithm trained predominantly on data from a particular demographic, prompting the provider to caution against its application to broader or unrepresented populations.

⁴⁹⁷ Commission Guidelines on prohibited artificial intelligence practices (n 166) 5.

⁴⁹⁸ Blue Guide 2022 (n 186) point 2.2.

⁴⁹⁹ Article 3(12) and (15) and Article 13(3) AIA.

⁵⁰⁰ Article 11(1) and Point 1(a) of Annex IV AIA.

⁵⁰¹ For a definition, refer to Article 3(23) AIA.

⁵⁰² Article 25(1)(b) AIA.

⁵⁰³ Article 25(1)(c) AIA.

⁵⁰⁴ Blue Guide 2022 (n 186) 25.

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2.200 Other provisions in the regulation use the expression of ‘**intended to be used**’.⁵⁰⁵ Considering that this concept also refers to the intention of the provider, there is reason to believe that this formulation can be interpreted as a synonym of the notion of intended purpose. In these contexts, the notion of intended use reflects that deployers determine the use of an AI system, yet providers need to assess whether an AI system qualifies as high-risk prior to its placing on the market or putting into service.⁵⁰⁶ As a consequence, the provider may not know to what ends a deployer will use an AI system and, accordingly, they need to speculate about its intended use through—it can be argued—its own definition of the intended purpose, which in turn influences in which way the deployer can use the system.⁵⁰⁷

xiii. Article 3(13): Reasonably foreseeable misuse

2.201 Article 3(13) defines the reasonably foreseeable misuse as the use of an AI system in a way that is not in accordance with its intended purpose, but which may result from reasonably foreseeable human behaviour or interaction with other systems, including other AI systems.

2.202 While the intended purpose is defined by the provider, a reasonably foreseeable misuse is **caused by the deployer or end user**. The notion is of relevance in the context of the risk management system under Article 9, transparency obligations under Article 13, as well as human oversight under Article 14.

2.203 The concept of reasonably foreseeable misuse **originates in EU product safety law**, echoing a general recognition that it is foreseeable by reasonableness standards that products are sometimes used beyond their intended purpose.⁵⁰⁸ For example, while a slide is intended for sliding down, children naturally use it to run up as well. Similarly, the AI Act recognizes that AI systems can be used beyond their intended purpose in a manner that is, however, reasonably foreseeable and hence does not amount to a substantial modification, as defined by Article 3(23). In its non-binding guidance on the Machinery Directive, the Commission states that while the product manufacturer cannot be expected to take account of ‘all possible misuse of the machinery’, they must account for ‘certain kinds of misuse, whether intentional or unintentional, are predictable on the basis of experience of past use of the same type of machinery or of similar machinery, accident investigations and knowledge about human behaviour’.⁵⁰⁹ This criterion is, of course, much harder to assess in relation to AI, as experience with many of these systems is limited or lacking.⁵¹⁰ The Commission guidance on the Machinery Directive provides the following examples of reasonable foreseeable misuse: loss of control of the machine by the operator; reflex behaviour of a person in case of malfunction, incident or failure during the use of the machine; behaviour resulting from lack of concentration or carelessness; behaviour resulting from taking the line of least resistance in carrying out a task; behaviour resulting from pressures to keep machinery running in all circumstances; the behaviour of certain persons, such as children.⁵¹¹

⁵⁰⁵ See, inter alia, Articles 6(1)(a); Article 7(1)(a); Article 10(3) and (4); Article 43(1)(d) AIA.

⁵⁰⁶ See further Articles 3(3) and (4) AIA.

⁵⁰⁷ See analysis of Article 7(2) AIA.

⁵⁰⁸ See, e.g. Recital 11 TSD or Annex I, Chapter I (3)(c) MDR.

⁵⁰⁹ Commission, ‘Guide to Application of the Machinery Directive 2006/42/EC’ (2.3. edn, April 2024) <<https://ec.europa.eu/docsroom/documents/60145>> accessed 1 July 2025.

⁵¹⁰ It might, for instance, be argued, that few people foresaw a few years ago that some humans could feel romantically attached to an AI system. See further Jerlyn QH Ho and others, ‘Potential and Pitfalls of Romantic Artificial Intelligence (AI) Companions: A Systematic Review’ (2025) 19 *Computers in Human Behavior Reports* 100715.

⁵¹¹ Commission Guide to Application of the Machinery Directive (n 509) 157.

While the concept of reasonably foreseeable misuse extends beyond the intended purpose explicitly defined by the provider, the misuse must still fall within the realm of **what can be anticipated**.⁵¹² This includes scenarios arising from reasonably foreseeable human behaviour or interactions with other systems, including other AI systems.

First, a reasonably foreseeable misuse can result from **reasonably foreseeable human behaviour**. Providers must therefore anticipate how users might be naturally incentivized to interact with a system in ways not explicitly envisioned in the definition of its intended purpose. This necessitates evaluating the technical capabilities of an AI system, as users may explore functionalities that are technically possible, even if not explicitly covered by the intended purpose, as defined in the instructions for use.⁵¹³ Indeed, when a capability is both technically feasible and easily detectable, it is reasonable to anticipate that users may take advantage of it. Regarding the reasonableness criterion, the Blue Guide explains that ‘manufacturers have to look beyond what they consider the intended use of a product and place themselves in the position of the average user of a particular product and envisage in what way they would reasonably consider to use the product’.⁵¹⁴

Second, a reasonably foreseeable misuse can result from a reasonably foreseeable **interaction with other systems**, including other AI systems. The Blue Guide also explains that regarding products intended for outdoor use, **climate change** must also be accounted for as providers ought to ‘consider how projections about the changing climate in the EU will affect the safety and performance of the product during use’.⁵¹⁵ This is particularly relevant for AI systems that have been trained on historical climate data. It also raises the question of whether the notion of ‘systems’ ought to be interpreted broadly, to account also for, *inter alia*, interactions with climate systems, or whether it was intended to refer to IT systems only. In any event, Article 3(13) makes explicit that the reasonably **foreseeable interaction with AI systems** needs to be accounted for. In this respect, reasonable foreseeableness is a particularly intriguing notion. Indeed, the more these systems learn and evolve, the more difficult it will become to assess what is reasonably foreseeable. Whereas the intended purpose is defined by the provider, a reasonably foreseeable misuse is caused by the deployer or end user. Yet, providers can become deployers where they themselves use an AI system under their authority. It is questionable whether in such circumstances a reasonably foreseeable misuse can really occur.

xiv. Article 3(14): Safety component

Article 3(14) defines the safety component as a component of a product or of an AI system which fulfils a safety function for that product or AI system, or the failure or malfunctioning of which endangers the health and safety of persons or property.

Recital 55 provides a distinct definition for safety components used in **critical infrastructure**. It provides that ‘safety components of critical infrastructure, including critical digital infrastructure, are systems used to directly protect the physical integrity of critical infrastructure or the health and safety of persons and property but which are not necessary in order for the system to function’.⁵¹⁶ Two key aspects stand out in this definition. First, it

⁵¹² See further Article 3(12) AIA.

⁵¹³ Article 13(2) AIA and Article 13(3)(b)(i) AIA.

⁵¹⁴ Blue Guide 2022 (n 186) 29.

⁵¹⁵ Ibid 26.

⁵¹⁶ Recital 55 AIA.

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acknowledges risks to property, which are not typically within the scope of the AIA, as the regulation primarily protects against risks to human health, safety, and fundamental rights. Second, it clarifies that a safety component does not need to be essential for the system to function, meaning a HRAIS must be capable of operating without it for the component to qualify as a safety component. Recital 55 provides the examples of systems for monitoring water pressure or fire alarm controlling systems in cloud computing centres.⁵¹⁷ It is, however, worth recalling that recitals are not strictly legally binding.⁵¹⁸ Recital 47 emphasizes the importance of recognizing safety components within a risk-based legal framework, given the specific risks AI systems pose to health and safety when functioning as safety components in products.⁵¹⁹

- 2.209** The definition of the safety component is crucial as it determines whether the obligations under Chapter III apply to an AI system. Indeed, in accordance with Article 6(1), an AI system qualifies as a high-risk AI system subject to the obligations in Chapter III when it is intended to be used as a safety component of a product, or when the AI system itself is a product covered by the Union harmonization legislation listed in Annex I, and where that product is required to undergo a third-party conformity assessment prior to its placing on the market or putting into service, pursuant to the Union harmonization legislation listed in Annex I.⁵²⁰ Furthermore, Point 2 of Annex III qualifies as HRAIS used in critical infrastructure that are intended to be used as safety components in the management and operation of critical digital infrastructure, road traffic, or in the supply of water, gas, heating, or electricity.⁵²¹ Articles 102–109 make changes to existing product legislation concerned with safety components.⁵²²
- 2.210** Given its significance within the broader framework of the AI Act, the definition of the safety component is **surprisingly vague**. Contrast Article 3(14) with Article 3(3) of the MR, which defines the safety component as ‘physical or digital component, including software, of a product within the scope of this Regulation, which is designed or intended to fulfil a safety function and which is independently placed on the market, the failure or malfunction of which endanger the safety of persons, but which is not necessary in order for that product to function or for which normal components may be substituted in order for that product to function.’⁵²³ Article 3(14) identifies two distinct scenarios where a component of a product or AI system qualifies as a safety component.
- 2.211** First, Article 3(14) provides that a component qualifies as a safety component where it **fulfils a safety function** for that product or AI system. Crucially, the notion of the safety function is not defined, in contrast to what we find in other areas of EU product safety law. Indeed, Article 3(4) MR defines the safety function as ‘a function that serves to fulfil a protective measure designed to eliminate, or, if that is not possible, to reduce, a risk, which, if it fails, could result in an increase of that risk.’⁵²⁴

⁵¹⁷ Ibid.

⁵¹⁸ On the legal status of recitals in EU law, refer to the corresponding section in the Introduction.

⁵¹⁹ Recital 47 AIA. See also Recital 55 AIA.

⁵²⁰ Article 6(1) AIA.

⁵²¹ Point 2 of Annex III and Article 6(2) AIA.

⁵²² Articles 102–109 AIA. See also Recital 49 AIA.

⁵²³ Article 3(3) of the Machinery Regulation (n 213).

⁵²⁴ Article 3(4) of the Machinery Regulation (n 213).

In any event, the safety component can be part of an AI system, just as a HRAIS can itself be a safety component of a product covered by EU harmonizing legislation.⁵²⁵ Regarding HRAIS used in critical infrastructure, Recital 55 provides that ‘components intended to be used solely for cybersecurity purposes should not qualify as safety components’.⁵²⁶

Article 3(14) specifies that there are also safety components, the failure or malfunctioning of which can endanger the **health and safety of persons or property**. The rationale behind this inclusion contrasts with the general objectives of the AIA, which focus solely on mitigating risks to human health, safety, and fundamental rights. The decision to specifically incorporate property was presumably motivated by the fact that the right to property, which is protected under EU law was deemed to be the only relevant fundamental right in this context.⁵²⁷

Considering the **circularity and redundancy of this definition**, the following test has been suggested to identify a safety component: a component qualifies as a safety component if it is either responsible for ensuring safety or if its failure would endanger safety. Alternatively, if the hypothetical removal of the component compromises safety, then it is to be regarded as a safety component.⁵²⁸

xv. Article 3(15): Instructions for use

Article 3(15) defines the instructions for use as the information provided by the provider to inform the deployer of, in particular, an AI system’s intended purpose and proper use.

A fundamental premise of the regulation is that providers possess greater knowledge of their AI systems compared to deployers.⁵²⁹ Thus, whether providers develop an AI system or have it developed, they determine its key characteristics, which they have more control over compared to any actor. If a distributor, importer, deployer, or other third party exercises decisive control over a HRAIS—such as by making a substantial modification to it—they themselves assume the role of a provider.⁵³⁰ Instructions for use should be understood within this context, as they serve as a **means for providers to convey essential information to deployers** regarding the HRAIS in question.

This notion is important as it **shapes many legal obligations** foreseen for HRAIS. For example, deployers are obliged to adopt appropriate technical and organizational measures to ensure that HRAIS are used only in accordance with the instructions for use.⁵³¹ Importers can only place on the market a HRAIS provided that the system is accompanied by instructions for use.⁵³² They must also keep a copy of the instructions for use for 10 years.⁵³³ Distributors can also only make available a HRAIS in the internal market after verification of the presence of instructions for use.⁵³⁴ Where a fundamental rights impact assessment (‘FRIA’) ought to be carried out by virtue of Article 27, deployers ought to perform

⁵²⁵ Article 25(3) AIA.

⁵²⁶ Recital 55 AIA.

⁵²⁷ Article 17 CFR.

⁵²⁸ Henrik Nolte (PhD Thesis, University of Tübingen forthcoming).

⁵²⁹ Article 3(3) AIA states that the provider is the entity that ‘develops an AI system or a general-purpose AI model or that has an AI system or a general-purpose AI model developed and places it on the market or puts the AI system into service under its own name or trademark, whether for payment or free of charge’.

⁵³⁰ Article 25(1) AIA.

⁵³¹ Article 26(1) AIA.

⁵³² Article 23(1)(c) AIA.

⁵³³ Article 23(5) AIA.

⁵³⁴ Article 24(1) AIA.

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a description of implementation of the human oversight measures, as defined in the instructions for use.⁵³⁵ Where an AI system is tested in real world conditions in a regulatory sandbox, providers must make available instructions for use to deployers and prospective deployers.⁵³⁶

2.218 Article 3(15) insists that the instructions for use must, in any event, contain **information** regarding the intended purpose of the AI system, as well as its proper use.⁵³⁷ Yet, Article 13(3) provides a list of other information that must always be included in the instructions for use of a high-risk AI system such as, for instance, the identity and contact details of the provider and potentially its authorized representative⁵³⁸ or the human oversight measures adopted pursuant to Article 14.⁵³⁹ It is hence not enough for providers to focus on the two elements mentioned by this definition; rather, they must adhere to the much longer list in Article 13(3), as detailed by Recital 72.⁵⁴⁰ Pursuant to Article 13(2), instructions for use must take the form of an ‘appropriate digital format or otherwise that include concise, complete, correct and clear information that is relevant, accessible and comprehensible to deployers’.⁵⁴¹

xvi. Article 3(16): Recall of an AI system

2.219 Article 3(16) defines the recall of an AI system as any measure aiming to achieve the return to the provider or taking out of service or disabling the use of an AI system made available to deployers. The recall is an **enforcement measure** under the regulation. In case of formal non-compliance, the competent market surveillance authority can order a recall of a HRAIS.⁵⁴² Similarly, in case of AI systems presenting a risk, market surveillance authorities may also order recalls.⁵⁴³ Recalls also form part of the enforcement structure around GPAIM as the AI Office is empowered to request providers to recall a model.⁵⁴⁴

2.220 This concept can also be found in other domains of EU **product safety law**, such as the MR⁵⁴⁵ or the Toy Safety Directive (‘TSD’).⁵⁴⁶ In all instances, those having made available a product must seek to obtain it back from end users.

2.221 The recall can be ‘**any measure**’, meaning that the regulation does not prescribe through which means a recall is to be operated. In line with existing definitions, Article 3(16) also only formulates an **obligation of means**, not an obligation of ends as measures only need to be aimed at achieving the return to the provider or taking out of service/disabling the use of an AI system rather than actually achieving that goal. There appears to be an implicit assumption that the concept of recall applies to products, yet it remains unclear how a recall mechanism would function in the context of a service.

⁵³⁵ Article 27(1)(e) AIA.

⁵³⁶ Article 60(4)(h) AIA.

⁵³⁷ Article 3(15) AIA.

⁵³⁸ Article 13(3)(a) AIA.

⁵³⁹ Article 13(3)(d) AIA.

⁵⁴⁰ Article 13(3) AIA.

⁵⁴¹ Article 13(2) AIA.

⁵⁴² Article 83(2) AIA.

⁵⁴³ Article 79(2) AIA.

⁵⁴⁴ Article 93(1)(c) AIA.

⁵⁴⁵ Article 3(32) of the Machinery Regulation (n 213) defines recall as ‘any measure aimed at achieving the return of a product within the scope of this Regulation that has already been made available to a user’.

⁵⁴⁶ Article 3(13) TSD defines recall as ‘any measure aimed at achieving the return of a toy that has already been made available to the end user’.

xvii. Article 3(17): Withdrawal of an AI system

Article 3(17) defines the withdrawal of an AI system as any measure aiming to prevent an AI system in the supply chain being made available on the market. Withdrawal is an **enforcement measure** under the regulation. In case of formal non-compliance, the competent market surveillance authority can order a withdrawal of a HRAIS.⁵⁴⁷ Similarly, in case of AI systems presenting a risk, market surveillance authorities may also order the withdrawal of the system.⁵⁴⁸ Withdrawals also form part of the enforcement structure around general-purpose AI models as the AI Office is empowered to request providers to withdraw a model.⁵⁴⁹

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xviii. Article 3(18): Performance of an AI system

Article 3(18) defines the performance of an AI system as the ability of an AI system to achieve its intended purpose. The performance of an AI system is a **guiding interpretational principle** in the AI Act. Providers have a number of obligations pertaining to performance, such as to test HRAIS to ensure that they perform consistently for their intended purpose⁵⁵⁰ or to note limitations to performance in the instructions for use.⁵⁵¹ The technical documentation of a HRAIS must include a detailed description of its performance⁵⁵² and detailed information regarding the monitoring, functioning and control of the AI system, in respect of its capabilities and limitations in performance. This must include—an assessment of performance in relation to its intended purpose; the foreseeable unintended outcomes and sources of risks to health and safety, fundamental rights and discrimination in view of the intended purpose of the AI system; and the human oversight measures needed, including the technical measures put in place to facilitate the interpretation of the outputs of AI systems by the deployers and, as appropriate, specifications on input data. This formulation not only details what information must be included with respect to the technical documentation governed by Article 11(1), but also implies that accuracy is an important aspect of the performance of a HRAIS. Intriguingly, the objectives governed by Article 15 in addition to accuracy, namely robustness and cybersecurity, are not mentioned.⁵⁵³

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The performance of an AI system ought to be determined with the help of **performance metrics**. The expected level of performance metrics should be declared in the instructions for use in a clear and easily understandable way, free of misunderstandings or misleading statements so that deployers can transparently assess the performance of the HRAIS.⁵⁵⁴ In collaboration with relevant stakeholders and organizations, including metrology and benchmarking authorities, the Commission has to actively promote, where appropriate, the development of benchmarks and measurement methodologies for AI systems.⁵⁵⁵

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⁵⁴⁷ Article 83(2) AIA.

⁵⁴⁸ Article 79(2) AIA.

⁵⁴⁹ Article 93(1)(c) AIA.

⁵⁵⁰ Article 9(6) AIA.

⁵⁵¹ Article 13(3)(b) AIA. As per Recital 72 AIA, 'Those would cover information on possible known and foreseeable circumstances related to the use of the high-risk AI system, including deployer action that may influence system behaviour and performance, under which the AI system can lead to risks to health, safety, and fundamental rights, on the changes that have been pre-determined and assessed for conformity by the provider and on the relevant human oversight measures, including the measures to facilitate the interpretation of the outputs of the AI system by the deployers.'

⁵⁵² Point 2(f) of Annex IV AIA.

⁵⁵³ Contrast this with Recital 74 AIA, which provides that 'high-risk AI systems should perform consistently throughout their lifecycle and meet an appropriate level of accuracy, robustness and cybersecurity, in light of their intended purpose and in accordance with the generally acknowledged state of the art.'

⁵⁵⁴ Recital 74 AIA.

⁵⁵⁵ Ibid.

2.225 The **intended purpose** of an AI system is the use for which it is explicitly and implicitly intended by the provider, including the specific context and conditions of use, as specified in the information supplied by the provider in the instructions for use, promotional or sales materials and statements, as well as in the technical documentation.⁵⁵⁶ This concept has been examined in detail above.

2.226 The instructions for use must contain **information** about the performance of an AI system as well as about any known or foreseeable circumstances, related to the use of the HRAIS in accordance with its intended purpose or under conditions of reasonably foreseeable misuse, which may lead to risks to the health and safety or fundamental rights.⁵⁵⁷ In addition, providers must note in the instructions for use the performance of the HRAIS ‘regarding specific persons or groups of persons on which the system is intended to be used’ where this is deemed appropriate.⁵⁵⁸

2.227 The **post-market monitoring** for HRAIS must actively and systematically collect, document and analyse relevant data (provided by deployers or collected by other sources) on the performance of a HRAIS throughout its lifetime in order to evaluate the continuous compliance of the system with the essential requirements applicable to high-risk AI systems.⁵⁵⁹ Changes to the performance of an AI system can be a ground necessitating a new conformity assessment.⁵⁶⁰

xix. Article 3(19): Notifying authority

2.228 Article 3(19) defines a notifying authority as the national authority responsible for setting up and carrying out the necessary procedures for the assessment, designation, and notification of conformity assessment bodies and for their monitoring.

2.229 Notifying authorities are one of the ‘**national competent authorities**’ alongside market surveillance authorities.⁵⁶¹ Their status and role is governed by Article 28, which provides, inter alia, that each Member State has to designate or establish at least one notifying authority responsible for setting up and carrying out the necessary procedures for the assessment, designation and notification of conformity assessment bodies, and for their monitoring.⁵⁶² Notifying authorities must be set up so that no conflict of interest arises with conformity assessment bodies.⁵⁶³

xx. Article 3(20): Conformity assessment

2.230 Article 3(20) defines a conformity assessment as the process of demonstrating whether the essential requirements applicable to high-risk AI system have been fulfilled.⁵⁶⁴

⁵⁵⁶ Article 3(12) AIA.

⁵⁵⁷ Article 13(3)(b)(iii) AIA.

⁵⁵⁸ Article 13(3)(b)(v) AIA.

⁵⁵⁹ Article 72(2) AIA. This obligation does not cover sensitive operational data of deployers which are law enforcement authorities. Section 2 of Chapter III AIA outlines the essential requirements applicable to providers of high-risk AI systems.

⁵⁶⁰ Article 3(23) and Article 43(4) AIA.

⁵⁶¹ Article 3(48) AIA.

⁵⁶² Member States may designate a national accreditation body governed by European Accreditation Regulation (n 122).

⁵⁶³ Article 28(3) AIA.

⁵⁶⁴ Section 2 of Chapter III AIA outlines the essential requirements applicable to providers of high-risk AI systems.

The conformity assessment is a **familiar figure in EU product safety law**, where it refers to the ‘process carried out by the manufacturer of demonstrating whether specified requirements relating to a product have been fulfilled’.⁵⁶⁵ In the context of the AIA specifically, it refers to the process by which the provider demonstrates whether the essential requirements applicable to HRAIS, as set out in Section 2 of Chapter III, have been met.⁵⁶⁶ This Section outlines the essential requirements high-risk AI systems must meet before they can be placed on the market or put into service. It is one of the techno-correctionist instruments in the AIA, aimed at rectifying technological issues of high-risk AI systems to enable their use in the economy.⁵⁶⁷

The conformity assessment must occur prior to a high-risk AI system being placed on the market or put into service to **avert potential risks** to health, safety and fundamental rights. The regulation outlines **various forms of conformity assessment**, which are applied based on the risk categorization of the underlying AI system. For most high-risk AI systems, conformity assessments can be carried out internally by the provider.⁵⁶⁸ In doing so, they must follow the requirements set out in Annex VI, which requires, *inter alia*, that the established quality management system follows the requirements of Article 17.⁵⁶⁹ For biometric systems that qualify as high-risk AI systems by virtue of point 1 of Annex III, the alternative conformity assessment procedures foreseen under Article 43(1) apply.⁵⁷⁰ If a product regulated by EU harmonization legislation listed in Section A of Annex I qualifies as a high-risk AI system or contains a high-risk AI system, it may undergo conformity assessments governed by relevant legal frameworks.⁵⁷¹ However, the specific requirements for high-risk AI system under the AI Act must also be duly considered in this context.⁵⁷²

xxi. Article 3(21): Conformity assessment body

Article 3(21) defines a conformity assessment body as a body that performs third-party conformity assessment activities, including testing, certification and inspection.

Conformity assessment bodies are relevant in the context of the **conformity assessment** of high-risk AI systems with related essential requirements.⁵⁷³ A positive conformity assessment will result in the issuance of the CE marking.⁵⁷⁴

Conformity assessment bodies are **usually notified bodies**, as defined in Article 3(22), which is examined just below. They could also be another authority, particularly in respect of Article 43(1)(a), in case providers want to outsource this obligation to specialized third parties. Article 74 foresees two exceptions to this general rule. Article 74(8) requires that, where a HRAIS is used for law enforcement, border management, or justice and democracy purposes, Member States have to designate as market surveillance authorities either the competent data protection supervisory authorities or any other authority designated

⁵⁶⁵ Blue Guide 2022 (n 186) 70.

⁵⁶⁶ Article 43(1) AIA.

⁵⁶⁷ Jessica Eaglin, ‘When Critical Race Theory Enters the Law & Technology Frame’ (2021) *Michigan Journal of Race & Law* 151, 155–157.

⁵⁶⁸ Providers can carry out conformity assessment procedures internally by AIS that qualify as HRAIS by virtue of Article 6(2) AIA.

⁵⁶⁹ Point 2 of Annex VI AIA.

⁵⁷⁰ Article 43(1) AIA.

⁵⁷¹ Article 43(3) AIA.

⁵⁷² *Ibid.*

⁵⁷³ These essential requirements are outlined in Section 2 of Chapter III AIA.

⁵⁷⁴ Article 3(24) AIA.

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pursuant to the same conditions laid down in Articles 41 to 44 LED.⁵⁷⁵ Pursuant to Article 74(9), the European Data Protection Supervisor acts as the market surveillance authority of Union institutions, bodies, offices or agencies except in relation to the Court of Justice of the European Union acting in its judicial capacity.⁵⁷⁶

xxii. Article 3(22): Notified body

2.236 Article 3(22) defines a notified body as a conformity assessment body notified in accordance with the AIA and other relevant Union harmonization legislation.

2.237 This notion needs to be interpreted in line with the legislative definition in Article 3(21), which was examined just above. The **notification** occurs through the notifying authority, as defined by Article 3(19), which also acts as a surveillance authority for notified bodies.⁵⁷⁷ Article 29 governs the modalities of notification whereas Article 31(2) sets out the requirements related to notified bodies, such as that they must satisfy the organizational, quality management, resources and process requirements that are necessary to fulfil their tasks.⁵⁷⁸ Importantly, notified bodies have to be independent from providers of HRAIS as well as any other operators that have an economic interest in the HRAIS to be assessed.⁵⁷⁹ The Commission will assign single identification numbers to each notified body⁵⁸⁰ and make publicly available a list of all notified bodies certified under the AI Act.⁵⁸¹ Article 36 governs the circumstances where changes to notifications can occur, whereas Article 37 empowers the Commission to investigate cases where there is reason to doubt the competence or continued fulfilment of responsibilities under the regulation of a notified body.⁵⁸²

xxiii. Article 3(23): Substantial modification

2.238 Article 3(23) defines a substantial modification as a change to an AI system, after it has been placed on the market or put into service, which is not foreseen or planned in the initial conformity assessment carried out by the provider, and as a result of which, the compliance of the AI system with related essential requirements is affected or results in a modification to the intended purpose for which the AI system has been assessed.⁵⁸³ The notion of substantial modification is relevant in respect of the legal regime applicable to HRAIS. It is a commonly established notion for products regulated by EU harmonizing legislation.⁵⁸⁴ Logging capability of HRAIS must enable the recording of events that may result in a substantial modification.⁵⁸⁵

2.239 Substantial modifications refer to **changes that are made to the AI system after it has been placed on the market or put into service**, actions that are, as per the definition in Article 3(3), necessarily carried out by the provider.⁵⁸⁶ Article 3(9) defines **placing on the market**

⁵⁷⁵ LED.

⁵⁷⁶ See also Article 3(48) AIA.

⁵⁷⁷ Article 28(1) AIA.

⁵⁷⁸ Article 31(2) AIA.

⁵⁷⁹ Article 31(4) AIA.

⁵⁸⁰ Article 35(1) AIA.

⁵⁸¹ Article 35(2) AIA.

⁵⁸² Article 37(1) AIA.

⁵⁸³ See also Recital 128 AIA. Section 2 of Chapter III AIA outlines the essential requirements applicable to high-risk AI systems.

⁵⁸⁴ Recital 128 AIA.

⁵⁸⁵ Article 12(2)(a) AIA.

⁵⁸⁶ Article 3(3) AIA.

the first making available of an AI system or a general-purpose AI model on the Union market.⁵⁸⁷ Article 3(11) defines **putting into service** as the supply of an AI system for first use directly to the deployer or for own use in the Union for its intended purpose.⁵⁸⁸ Article 3(23) does not provide a definition of ‘change’. Recital 128 clarifies that this can, for instance, be a change of operating system or software architecture.⁵⁸⁹

Only changes that are **not foreseen or planned in the initial conformity assessment** that was carried out by the provider in accordance with Article 43 qualify as substantial modifications.⁵⁹⁰ A change of operating system or software architecture that is anticipated in the initial conformity assessment would hence not qualify as a substantial modification.⁵⁹¹

Article 43(4) determines that high-risk AI system that have already been subject to a conformity assessment have to undergo a **new conformity assessment** in the event of a substantial modification (regardless of whether the modified system is intended to be further distributed or continues to be used by the current deployer).⁵⁹² As the Blue Guide explains, in such circumstances the product is essentially considered ‘a new product’, which implies that ‘the compliance of the modified product with the applicable essential requirements has to be reassessed and the person carrying out the substantial modification has to fulfil the same requirements as an original manufacturer, for example preparation of the technical documentation, drawing up a EU declaration of conformity and affixing the CE marking on the product.’⁵⁹³

Pursuant to Article 3(1), AI systems may exhibit **adaptiveness after deployment**, meaning that they ‘learn’ and hence change over time.⁵⁹⁴ This raises the question of when learning should be considered an unforeseen or unplanned change. Article 43(4) clarifies that for HRAIS that continue to learn after being placed on the market or put into service, as envisaged by the definition in Article 3(1), changes to the system and its performance that were predetermined by the provider at the moment of the initial conformity assessment and are part of the technical documentation do not constitute a substantial modification.⁵⁹⁵ An example of such learning would be where AI systems are ‘automatically adapting how functions are carried out.’⁵⁹⁶ The placement of this important caveat raises the question of whether it is applicable only in the context of Article 43(4) or whether it also precludes the shifting liability governed by Article 25(1)(b). The location of this limitation in this specific provision and not in Article 3(23) or also in Article 25(1)(b) might lead to the conclusion that it is only applicable in the context of the conformity assessment. Logically, however, one would expect this limitation to apply holistically, a conclusion which also seems supported by the wording of Recital 128.⁵⁹⁷ It would also be in line with the Blue Guide’s stipulations

⁵⁸⁷ Article 3(9) AIA.

⁵⁸⁸ Article 3(11) AIA.

⁵⁸⁹ Recital 128 AIA.

⁵⁹⁰ Articles 3(23) and 43 AIA.

⁵⁹¹ Recital 128 AIA.

⁵⁹² Article 43(4) AIA.

⁵⁹³ Blue Guide 2022 (n 186) 17.

⁵⁹⁴ Article 3(1) AIA.

⁵⁹⁵ Article 43(4) AIA.

⁵⁹⁶ Recital 128 AIA.

⁵⁹⁷ Recital 128 AIA provides that ‘changes occurring to the algorithm and the performance of AI systems which continue to ‘learn’ after being placed on the market or put into service, namely automatically adapting how functions are carried out, should not constitute a substantial modification, provided that those changes have been predetermined by the provider and assessed at the moment of the conformity assessment.’

on product updates, according to which modifications that entail products ‘perform better due to technical progress’ mean the resulting product ‘is not to be considered as new’.⁵⁹⁸ In relation to software specifically, the Blue Guide notes that ‘a product should be considered as substantially modified by a software change where: (i) the software update modifies the original intended functions, type or performance of the product and this was not foreseen in the initial risk assessment; (ii) the nature of the hazard has changed or the level of risk has increased because of the software update; and (iii) the product is made available (or put into service where this is covered by the specific Union harmonisation legislation)’.⁵⁹⁹ The exact contours of the meaning and implications of this concept will become clearer with time as Article 96(1)(c) empowers the Commission to develop guidelines on the practical implementation of ‘the provisions related to substantial modification’.⁶⁰⁰

2.243 Article 3(23) qualifies as substantial modifications those changes that are not only unanticipated, but which furthermore (i) affect the compliance of the HRAIS with related essential requirements⁶⁰¹ or (ii) which results in a modification to the intended purpose for which the HRAIS was assessed.⁶⁰²

2.244 A substantial modification can have **major implications for the distribution of obligations alongside the AI value chain**. Pursuant to Article 25(1)(b), the distributor, importer, deployer, or other third-party is considered a provider if they make a substantial modification to a HRAIS that has already been placed on the market or put into service in such a way that it remains a HRAIS.⁶⁰³ As a result, they are subject to the provider’s obligations under Article 16.⁶⁰⁴ A substantial modification to a HRAIS can also be considered to end its ‘lifecycle’, a notion that is relevant, for instance, under Article 9.⁶⁰⁵

2.245 There could also be scenarios where **corrective measure** adopted under Article 20(1) gives rise to a substantial modification of a HRAIS, for instance, when the adoption of cybersecurity measures has detrimental effects on the transparency of such systems or alters their architecture.⁶⁰⁶

xxiv. Article 3(24): CE marking

2.246 Article 3(24) defines a CE marking as a marking by which a provider indicates that an AI system is in conformity with related essential requirements and other applicable Union harmonization legislation providing for its affixing.⁶⁰⁷ The notion is hence relevant in the context of the regime for high-risk AI systems governed by Chapter III.

2.247 The acronym ‘CE’ stands for *Conformité Européenne*, meaning ‘European Conformity’. The CE marking is **central to the EU internal market** as it needs to be fixed on many products for them to be admitted to the internal market. Indeed, it is so ubiquitous that it is ‘deeply

⁵⁹⁸ Blue Guide 2022 (n 186) 18. Note, also, that Recital 128 AIA explicitly references the ‘commonly established notion of substantial modification for products’.

⁵⁹⁹ Blue Guide 2022 (n 186) 18–19.

⁶⁰⁰ Article 96(1)(c) AIA.

⁶⁰¹ The essential requirements for high-risk AI systems are outlined in Section 2 of Chapter III AIA.

⁶⁰² The notion of the intended purpose is defined in Article 3(12) AIA.

⁶⁰³ Article 25(1)(b) AIA. Article 6 AIA outlines the classification rules for high-risk AI systems.

⁶⁰⁴ Article 25(1)(b) AIA.

⁶⁰⁵ See Jonas Schuett, ‘Risk Management in the Artificial Intelligence Act’ (2024) 15 *European Journal of Risk Regulation* 367.

⁶⁰⁶ See further Article 15(1) and Recital 128 AIA.

⁶⁰⁷ The essential requirements for high-risk AI systems are outlined in Section 2 of Chapter III AIA.

embedded within our collective visual memories.⁶⁰⁸ It serves several important functions as it is 'equally a source of consumer confidence in a product's regulatory compliance and a mechanism for a kind of regional 'nation branding', a logo 'in the promotion of [the EC's] interests in the global marketplace' and a visual reminder of the far reaching influence of Brussels and its rule makers.⁶⁰⁹ Most importantly, marks of conformity like the CE dually function as vectors for the circulation of goods within and among markets on the one hand and as non-tariff barriers restricting market access on the other.'⁶¹⁰

The **provider** bears responsibility for the affixing of the CE marking.⁶¹¹ Pursuant to Article 48(1), the general principles of CE marking, set out in Article 30 of the European Accreditation Regulation are applicable to CE markings under the AI Act.⁶¹² These foresee, inter alia, that only the manufacturer/provider or their representative may affix the CE marking⁶¹³ to products for which this is foreseen by EU harmonizing legislation,⁶¹⁴ or that the manufacturer/provider takes responsibility for the conformity of the product with all applicable requirements by affixing the marking.⁶¹⁵ Article 48 outlines additional requirements fashioned for the high-risk AI systems it applies to. These include that digital CE markings can be used for high-risk AI systems provided digitally⁶¹⁶ or that where a high-risk AI system is subject to other EU law which also provides for the affixing of CE markings, the marking shall indicate that the high-risk AI system also complies with the regulation.⁶¹⁷ Article 83 foresees that where a CE marking has been affixed in violation of Article 48, or has not been affixed at all, this results in formal non-compliance. Consequently, the national supervisory authority has to require the provider of the high-risk AI systems to put an end to the non-compliance within a given period.⁶¹⁸ If the non-compliance persists, the national market surveillance authority has to take appropriate and proportionate measures to restrict or prohibit the high-risk AI system being made available on the market or to ensure that it is recalled or withdrawn from the market without delay.⁶¹⁹

Importers and distributors of HRAIS also have duties in relation to the CE marking. Importers must ensure that the HRAIS is in conformity with the regulation by verifying that the system bears the CE marking.⁶²⁰ Distributors must verify that the HRAIS bears the required CE marking before making it available on the market.⁶²¹

xxv. Article 3(25): Post-market monitoring system

Article 3(25) defines a post-market monitoring system as all **activities** carried out by providers of AI systems to collect and review experience gained from the use of AI systems they place on the market or put into service for the purpose of identifying any need to

⁶⁰⁸ Grace Ballor, 'The Origins of CE Marking: Standards, Business, and the European Market in the 1980s–1990s' (Harvard Business School Working Paper 2021) No 21-142, 1 <<https://www.hbs.edu/faculty/Pages/item.aspx?num=60460>> accessed 16 January 2025.

⁶⁰⁹ Ibid.

⁶¹⁰ Ibid.

⁶¹¹ Article 16(h) AIA.

⁶¹² European Accreditation Regulation (n 122).

⁶¹³ Article 30(1) of the European Accreditation Regulation (n 122).

⁶¹⁴ Article 30(2) of the European Accreditation Regulation (n 122).

⁶¹⁵ Article 30(3) of the European Accreditation Regulation (n 122).

⁶¹⁶ Article 48(1) AIA.

⁶¹⁷ Article 48(5) AIA.

⁶¹⁸ Article 83(1)(a) and (b) AIA.

⁶¹⁹ Article 83(2) AIA.

⁶²⁰ Article 23(1)(c) AIA.

⁶²¹ Article 24(1) AIA.

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immediately apply any necessary corrective or preventive actions. This notion matters in the context of the post-market monitoring obligations falling upon providers of high-risk AI systems by virtue of Article 72.

- 2.251** This system forms part of the **risk management system**⁶²² as well as the **quality management system** of a HRAIS.⁶²³ It is governed specifically by Article 72, which sets out the requirements for a post-market monitoring system. Under this regime, providers have an obligation to establish and document a post-market monitoring system in a manner that is proportionate to the ‘nature of the AI technologies’ and the risks of the system.⁶²⁴ The system is an active and systematic collection, documentation and analysis of relevant data on the performance of the AI system (meaning its ability to achieve its intended purpose) provided by the deployer or collected through other means.⁶²⁵ The system must be based on a **post-market monitoring plan**, a concept that, unlike a post-monitoring system, is not independently defined by Article 3. This plan needs to be included in the technical documentation under Annex IV.⁶²⁶ Article 72(3) obliges the Commission to adopt an implementing act establishing a template for the post-market monitoring plan and the list of elements to be included by 2 February 2026, pursuant to the examination procedure under Article 98(2) AIA.

xxvi. Article 3(26): Market surveillance authority

- 2.252** Article 3(26) defines a market surveillance authority as the national authority carrying out the activities and taking the measures pursuant to the MSR.

- 2.253** The market surveillance authority is a **well-known figure in EU product safety law**. Under the AI Act, each Member State has to establish or designate at least one MSA.⁶²⁷ Market surveillance authorities have to cooperate and exchange in the context of one of the two standing sub-groups to be established under the Artificial Intelligence Board.⁶²⁸ The Scientific Panel of Independent Experts has to support the work of MSAs at their request.⁶²⁹ The Union Safeguard Procedure moreover foresees that MSAs or the Commission can raise objections against the measures adopted by other MSAs.⁶³⁰

xxvii. Article 3(27): Harmonized standard

- 2.254** Article 3(27) defines a harmonized standard as set out in Article 2(1)(c) of the European Standardisation Regulation, which in turn defines the harmonized standard as ‘a European standard adopted on the basis of a request made by the Commission for the application of Union harmonisation legislation.’⁶³¹

- 2.255** Harmonized standards are **at the core of EU product safety law** and under the AI Act ‘standardisation should play a key role to provide technical solutions to providers to ensure

⁶²² See Article 9(2)(c) AIA.

⁶²³ See further Article 9(2)(c) and Article 17(1)(h) AIA.

⁶²⁴ Article 72(1) AIA.

⁶²⁵ Article 72(2) AIA.

⁶²⁶ Article 72(3) AIA.

⁶²⁷ For an overview of these developments, see ‘Responsible Authorities for the Enforcement of the AI Act on National Level’ (*gdpr.blog.hu*, 2 October 2024) <https://gdpr.blog.hu/2024/10/02/responsible_authorities_for_the_enforcement_of_the_ai_act_on_national_level> accessed 24 December 2024.

⁶²⁸ Article 65(6) AIA.

⁶²⁹ Article 68(3)(b) AIA.

⁶³⁰ Article 81(1) AIA.

⁶³¹ Article 2(1)(c) SR.

compliance.⁶³² Harmonized standards are the technical rules which put flesh on the bones of more general legislative texts. Standards are documents that describe ‘the best way of doing something. It could be about making a product, managing a process, delivering a service or supplying materials—standards cover a huge range of activities.’⁶³³ Under the NLF, EU secondary legislation outlines essential requirements that ‘define the results to be attained, or the hazards to be dealt with, [...] [without] specify[ing] the technical solutions for doing so.’⁶³⁴ Standards are designed to clarify essential requirements by implementing them through technical guidelines, which operationalize the legal requirements for practical implementation.⁶³⁵

Article 40(2) obliges the Commission to request standards for all essential requirements applicable to **high-risk AI systems**.⁶³⁶ The Commission has already issued a standardization request during the legislative process, and these standards are expected to be in place by the time the corresponding legal obligations take effect.⁶³⁷ Whether this objective can be achieved remains to be seen. Compliance with the resulting harmonized standards lead⁶³⁸ to a presumption of conformity. Article 40(2) also empowers the Commission to request harmonized standards in relation to the requirements arising for **GPAIM**.⁶³⁹ These standards had not been requested at the time of writing, as efforts to form codes of practice for these models were still underway.⁶⁴⁰

xxviii. Article 3(28): Common specification

Article 3(28) defines a common specification as a set of technical specifications as defined in Article 2(4) of the European Standardisation Regulation, providing means to comply with certain requirements established under the AIA. Article 2(4) of the European Standardisation Regulation (‘SR’) defines the common specification as a ‘document that prescribes technical requirements to be fulfilled by a product, process, service or system’ and which lays down information regarding various aspects, such as the characteristics required of a product, product methods and processes, characteristics of services.

Common specifications provide a **means to comply** with certain requirements established under the regulation. The Commission can adopt common specifications in relation to the essential requirements applicable to HRAIS and the obligations the regulation imposes on providers of general-purpose AI models and general-purpose AI models with systemic risk under certain circumstances due to which harmonized standards are unavailable, such as when a standardization request is refused or standards are delayed.⁶⁴¹ Given that common specifications become relevant when there is a delay in the development of harmonized

⁶³² Recital 121 AIA.

⁶³³ ‘Standards’ (ISO) <<https://www.iso.org/standards.html>> accessed 14 April 2025.

⁶³⁴ Blue Guide 2022 (n 186) 48.

⁶³⁵ Commission, ‘Explanatory Memorandum to the Proposal for Regulation of the European Parliament and of the Council Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts’ COM (2021) 206 final, sections 2.1 and 2.3 <<https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52021PC0206>> accessed 14 March 2025.

⁶³⁶ Article 40(2) AIA. Section 2 of Chapter III AIA outlines the essential requirements applicable to high-risk AI systems.

⁶³⁷ Commission, ‘Draft Standardisation Request to the European Standardisation Organisations in Support of Safe and Trustworthy Artificial Intelligence’ (2022) <<https://ec.europa.eu/docsroom/documents/52376>> accessed 6 January 2025.

⁶³⁸ Article 40(1) AIA.

⁶³⁹ Article 40(2).

⁶⁴⁰ See further Article 56 AIA.

⁶⁴¹ Article 41(1) AIA. See also Section 2 of Chapter III and Sections 2 and 3 of Chapter V AIA.

standards or when the delivered standards are inadequate, they are likely to, if at all, be significant in the early stages of the regulation if the expected harmonized standards are not delivered on time. To date, no action has been taken by the Commission to create common specifications. Should the Commission be satisfied with the harmonized standards that are produced following its request, common specifications may never emerge in relation to the AI Act. Indeed, Recital 121 underlines that the common specification is only an ‘exceptional fall back solution.’⁶⁴²

xxix. Article 3(29): Training data

2.259 Article 3(29) defines training data as data used for training an AI system through fitting its learnable parameters.

2.260 Training data is used in **machine learning** in order to optimize trainable parameters in the training algorithm.⁶⁴³ Training data is used differently depending on whether a machine learning algorithm is based on supervised, unsupervised or reinforcement learning.⁶⁴⁴ In supervised machine learning, the model is ‘told’ during the optimization process what the training data it is confronted with represents.⁶⁴⁵ This is not the case in unsupervised learning, where the data used is unlabelled and the model is trained to identify similarities, parallels or differences in data.⁶⁴⁶ Reinforcement learning does not rely on pre-existing datasets but rather gathers data from simulations or games and the algorithm determines the rules based on continuous feedback on the actions it takes during training.⁶⁴⁷

2.261 The AI Act creates various legal obligations aimed at ensuring the quality of training data used in **high-risk AI systems**. The data and data governance obligations in Article 10 apply to high-risk AI systems ‘involving the training of AI models.’⁶⁴⁸ Moreover, high-risk AI systems ought to be resilient against cyberattacks, including attacks that try to manipulate training data or pre-trained components used in training (so-called data poisoning attacks).⁶⁴⁹ Pursuant to Article 74(12), MSAs have a conditional right to access training data sets used for the development of high-risk AI systems.⁶⁵⁰

2.262 Furthermore, the regulation also creates an obligation for providers of **general-purpose AI models** to provide ‘information on the data used for training, testing and validation.’⁶⁵¹ In addition, providers of general-purpose AI models are required to draw up and make publicly available a sufficiently detailed summary about the ‘content used for training of the

⁶⁴² Recital 121 AIA.

⁶⁴³ Josef Drexler and others, ‘Technical Aspects of Artificial Intelligence: An Understanding from an Intellectual Property Law Perspective’ (*Social Science Research Network*, 8 October 2019) <<https://papers.ssrn.com/abstract=3465577>> accessed 21 March 2025.

⁶⁴⁴ For an analysis of training data under EU law, see Philipp Hacker, ‘A Legal Framework for AI Training Data—from First Principles to the Artificial Intelligence Act’ (2021) 13 *Law, Innovation and Technology* 257.

⁶⁴⁵ *Ibid* 7.

⁶⁴⁶ *Ibid* 8.

⁶⁴⁷ *Ibid*.

⁶⁴⁸ Article 10(2)–(5) AIA define quality criteria that apply to training data, which are defined by Articles 3(30)–(32) AIA. Pursuant to Article 10(3) AIA, training data sets shall be relevant, sufficiently representative, and to the best extent possible, free of errors and complete in view of the intended purpose. Article 13(3)(b)(vi) AIA foresees that the instructions for use of a HRAIS ought to include, where appropriate, ‘specifications for the input data, or any other relevant information in terms of the training, validation and testing data sets used, taking into account the intended purpose of the high-risk AI system.’

⁶⁴⁹ Article 15(5) AIA.

⁶⁵⁰ Article 74(12) AIA.

⁶⁵¹ Article 53(1)(a) and Point 2(c) of Annex XI AIA.

general-purpose AI model'.⁶⁵² While this diverging formulation is regrettable from the perspective of legislative quality and certainty, there also seem to be no reasons why this formulation should not be used as a synonym for, respectively, training, testing and validation data. The interpretation of the 'content used for training of the general-purpose AI model' should in any event be clarified by the template the AI Office ought to create in relation to this obligation.⁶⁵³

xxx. Article 3(30): Validation data

Article 3(30) defines validation data as data used for providing an evaluation of the trained AI system and for tuning its non-learnable parameters and its learning process in order to, *inter alia*, prevent underfitting or overfitting.

The AIA creates various legal obligations aimed at ensuring the quality of validation data used in **high-risk AI systems**. The data and data governance obligations in Article 10 apply to high-risk AI systems 'involving the training of AI models'.⁶⁵⁴ According to Article 15(5), high-risk AI systems ought to be resilient against cyberattacks, including attacks that try to manipulate training data or pre-trained components used in training (so-called data poisoning attacks).⁶⁵⁵ Pursuant to Article 74(12), market surveillance authorities have a conditional right to access training, validation and testing data sets used for the development of high-risk AI systems.⁶⁵⁶

Furthermore, the regulation also creates an obligation for providers of **general-purpose AI models** to provide 'information on the data used for training, testing and validation'.⁶⁵⁷ While this diverging formulation is regrettable from the perspective of legislative quality and certainty, there also seem to be no reasons why this formulation should not be used as a synonym for, respectively, training, testing and validation data.

xxxi. Article 3(31): Validation data set

Article 3(31) defines the validation data set as a separate data set or part of the training data set, either as a fixed or variable split. The added value of this definition in addition to Article 3(30) is unclear, especially considering there is no separate definition for training or testing data sets.

xxxii. Article 3(32): Testing data

Article 3(32) defines testing data as data used for providing an independent evaluation of the AI system in order to confirm the expected performance of that system before its placing on the market or putting into service.

The AI Act creates various legal obligations aimed at ensuring the quality of testing data used in **high-risk AI systems**. The data and data governance obligations in Article 10 apply

⁶⁵² Article 53(1)(d) AIA.

⁶⁵³ *Ibid.*

⁶⁵⁴ Article 10(2)–(5) AIA define quality criteria that apply to validation data, which are defined by Articles 3(30)–(32) AIA. Pursuant to Article 10(3) AIA, validation data sets shall be relevant, sufficiently representative, and to the best extent possible, free of errors and complete in view of the intended purpose. Article 13(3)(b)(vi) foresees that the instructions for use of a HRAIS ought to include, where appropriate, 'specifications for the input data, or any other relevant information in terms of the training, validation and testing data sets used, taking into account the intended purpose of the high-risk AI system.'

⁶⁵⁵ Article 15(5) AIA.

⁶⁵⁶ Article 74(12) AIA.

⁶⁵⁷ Article 53(1)(a) and Point 2(c) of Annex XI AIA.

to high-risk AI systems ‘involving the training of AI models’.⁶⁵⁸ According to Article 15(5), high-risk AI systems ought to be resilient against cyberattacks, including attacks that try to manipulate training data or pre-trained components used in training (so-called data poisoning attacks).⁶⁵⁹ Pursuant to Article 74(12), market surveillance authorities have a conditional right to access testing data sets used for the development of high-risk AI systems.⁶⁶⁰

- 2.269** Furthermore, the regulation also creates an obligation for providers of **general-purpose AI models** to provide ‘information on the data used for training, testing and validation’.⁶⁶¹ While this diverging formulation is regrettable from the perspective of legislative quality and certainty, there also seem to be no reasons why this formulation should not be used as a synonym for, respectively, training, testing and validation data.

xxxiii. Article 3(33): Input data

- 2.270** Article 3(33) defines input data as data provided to or directly acquired by an AI system on the basis of which the system produces an output.

- 2.271** Unlike training, testing and validation data, input data is not used during the development of an AI system but rather while it is in use.⁶⁶² Input data can take many forms, including data that is subject to specific legal protections, such as copyright law or data protection law. Indeed, where input data is personal data, data protection law applies.⁶⁶³

- 2.272** The AI Act creates specific legal obligations applicable to input data. For instance, **high-risk AI systems** that continue to learn after being placed on the market or put into service must be developed to avoid biased outputs influencing future inputs,⁶⁶⁴ and remote biometric identification systems must have certain logging capabilities regarding input data.⁶⁶⁵ Where appropriate, providers must also include in the instructions for use specifications for input data, taking into account the intended purpose of the high-risk AI system.⁶⁶⁶

xxxiv. Article 3(34): Biometric data

- 2.273** Article 3(34) defines biometric data as personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, such as facial images or dactyloscopic data.

- 2.274** The notion of biometric data determines the scope of related qualified prohibitions under Article 5⁶⁶⁷, whether a system qualifies as a high-risk AI system by virtue of Article 6(2) read in conjunction with Points 1(a) and (b) of Annex III, and the obligations for deployers of post-remote biometric identification systems under Article 26(10).⁶⁶⁸ Indeed, remote

⁶⁵⁸ Article 10(2)–(5) AI define quality criteria that apply to testing data, which are defined by Articles 3(30)–(32) AIA. Pursuant to Article 10(3) AIA, testing data sets shall be relevant, sufficiently representative, and to the best extent possible, free of errors and complete in view of the intended purpose. Article 13(3)(b)(vi) AIA foresees that the instructions for use of a HRAIS ought to include, where appropriate, ‘specifications for the input data, or any other relevant information in terms of the training, validation and testing data sets used, taking into account the intended purpose of the high-risk AI system’.

⁶⁵⁹ Article 15(5) AIA.

⁶⁶⁰ Article 74(12) AIA.

⁶⁶¹ Article 53(1)(a) and Point 2(c) of Annex XI AIA.

⁶⁶² Input data can also be used in training to simulate the later use of an AIS.

⁶⁶³ See further Article 3(50) AIA and Article 4(1) GDPR.

⁶⁶⁴ Article 15(4) AIA.

⁶⁶⁵ Article 12(3) AIA.

⁶⁶⁶ Article 13(3)(b)(vi) AIA.

⁶⁶⁷ See Article 5(1)(g) and (h) as well as Article 5(2)–(7) AIA.

⁶⁶⁸ Article 26(10) AIA.

biometric identification systems are high-risk AI systems by virtue of Point 1(a) of Annex III. In addition to the obligations in Chapter III, which apply to any remote biometric identification system that is not prohibited by Article 5(1)(h), Articles 5(1)(h) and 5(2)–(7) create specific deployer obligations in respect of ‘real-time’ remote biometric identification systems, while Article 26(1) foresees deployer obligations in respect of post-remote biometric identification systems.

Biometric data is subject to **specific protections** as it is deemed particularly sensitive considering that it ‘can allow for the authentication, identification or categorisation of natural persons and for the recognition of emotions of natural persons.’⁶⁶⁹ The latter point is controversial though, as many have pointed out that emotion recognition technologies are mere forms of pseudoscience.⁶⁷⁰

Biometric data is a form of **personal data** that results from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, such as facial images or dactyloscopic data. Pursuant to Recital 14, this notion has to be interpreted in light of the definition of biometric data in Article 4(14) GDPR.⁶⁷¹ In EU data protection law, biometric data, indeed, forms a subcategory of special category personal data.⁶⁷² There is, however, an important distinction between the definitions of biometric data in the AIA and the GDPR as Article 4(14) GDPR foresees that biometric data ‘allow[s] or confirm[s] the unique identification of that natural person.’⁶⁷³ While this formulation was initially also included in the dAIA, it was later dropped due to criticisms that this requirement makes the definition too narrow and fails to keep pace with technological developments.⁶⁷⁴ This definition would in fact only have caught AI systems that allow for the unique identification of natural persons, not however ‘emotion recognition system based on pulse frequency, body temperature and non-unique facial expressions (such as smiling, raising of the brow or yawning) or non-unique voice signals (such as volume or trembling).’⁶⁷⁵ Recital 14, indeed, recognizes that biometric data can allow for the ‘authentication, identification or categorisation of natural persons and for the recognition of emotions of natural persons.’⁶⁷⁶

The definition of biometric data in Article 3(34) forms the **basis for subsequent definitions**, namely the definition of biometric identification in Article 3(35), biometric verification in Article 3(36), emotion recognition system in Article 3(39), biometric categorization system in Article 3(40), remote biometric identification system in Article 3(41), real-time remote

⁶⁶⁹ Recital 14 AIA.

⁶⁷⁰ Jay Stanley, ‘Experts Say “Emotion Recognition” Lacks Scientific Foundation’ (American Civil Liberties Union, 18 July 2019) <<https://www.aclu.org/news/privacy-technology/experts-say-emotion-recognition-lacks-scientific>> accessed 27 June 2025. For a general analysis, see Elisabeth Steindl, ‘Does the European Data Protection Framework Adequately Protect Our Emotions? Emotion Tech in Light of the Draft AI Act and Its Interplay with the GDPR’ (2022) 8 European Data Protection Law Review 311.

⁶⁷¹ Article 4(14) GDPR reads as follows: ‘biometric data means personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, which allow or confirm the unique identification of that natural person, such as facial images or dactyloscopic data.’ The same definition can also be found in Article 3(18) EUDPR and in Article 3(13) LED.

⁶⁷² See also Article 3(37) AIA.

⁶⁷³ Article 4(14) GDPR.

⁶⁷⁴ Four Columns AI Act (n 1) line 160. Christiane Wendehorst, *The Proposal for an Artificial Intelligence Act COM(2021) 206 from a Consumer Policy Perspective: Part I: The Qualification of AI Systems and Practices as Prohibited or High-Risk* (Bundesministerium für Soziales, Gesundheit, Pflege und Konsumentenschutz 2021) 1, 94.

⁶⁷⁵ Ibid.

⁶⁷⁶ Recital 14 AIA.

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biometric identification system in Article 3(42), post-remote biometric identification system in Article 3(43).

xxv. Article 3(35): Biometric identification

- 2.278** Article 3(35) defines biometric identification as the automated recognition of physical, physiological, behavioural, or psychological human features for the purpose of establishing the identity of a natural person by comparing biometric data of that individual to biometric data of individuals stored in a database.⁶⁷⁷
- 2.279** The notion of biometric identification defines the following: the scope of high-risk AI systems that qualify as such by virtue of Article 6(2) read in conjunction with Point 1(a) of Annex III; the qualified prohibition in practices in Articles 5(1)(h) and Article 5(2)–(7) (which applies only to ‘real-time’ remote biometric identification systems); as well as the deployer obligations in Article 26(10) (which apply only to post-remote biometric identification systems). Indeed, remote biometric identification systems are high-risk AI systems by virtue of Point 1(a) of Annex III. In addition to the obligations in Chapter III, which apply to any remote biometric identification system not prohibited by Article 5(1)(h), Articles 5(1)(h) and 5(2)–(7) create specific deployer obligations in respect of ‘real-time’ remote biometric identification systems, while Article 26(1) establishes deployer obligations in respect of post-remote biometric identification systems.
- 2.280** Biometric identification constitutes a **specific form of processing biometric data** to establish the identity of a natural person. Remote biometric identification systems qualify as high-risk AI systems and are subject to related legal obligations arising under the regulation.⁶⁷⁸
- 2.281** The notion of biometric identification **forms the basis for the definitions** of remote biometric identification system in Article 3(41), real-time remote biometric identification system in Article 3(42), post-remote biometric identification system in Article 3(43).
- 2.282** Only the **automated** recognition of human features falls within the scope of the AI Act. The notion of automation is not, however, defined in the regulation, which particularly raises the question of whether the processing needs to be solely automated or, if not, what degree of human involvement is permissible.⁶⁷⁹
- 2.283** The definition of biometric identification in **Recital 15** is narrower and broader than the legislative formulation provided in Article 3(35) as it provides that biometric identification ‘should be defined as the automated recognition of physical, physiological and behavioural human features such as the face, eye movement, body shape, voice, prosody, gait, posture, heart rate, blood pressure, odour, keystrokes characteristics, for the purpose of establishing an individual’s identity by comparing biometric data of that individual to stored biometric data of individuals in a reference database, irrespective of whether the individual has given its consent or not.’⁶⁸⁰ First, the definition in Recital 15 is narrower as it omits reference to psychological traits. Given the normative superiority of the legislative text, this omission should not, however, undermine the interpretation of Article 3(35).⁶⁸¹ Second, the definition in Recital 15 is also broader as it (i) provides examples of physical, physiological, and

⁶⁷⁷ See also Recital 15 AIA.

⁶⁷⁸ Point 1 (a) of Annex III AIA.

⁶⁷⁹ By analogy, see Article 22(1) GDPR.

⁶⁸⁰ Recital 15 AIA.

⁶⁸¹ On the legal status of recitals in EU law, see further related analysis in the Introduction.

behavioural human features that biometric identification systems automatically recognize; (ii) clarifies that, for the purposes of the regulation, an individual's consent to the processing of such personal data is not a determining factor. Recital 15 also clarifies that the definition of biometric identification excludes AI systems 'intended to be used for biometric verification, which includes authentication, whose sole purpose is to confirm that a specific natural person is the person he or she claims to be and to confirm the identity of a natural person for the sole purpose of having access to a service, unlocking a device or having security access to premises'.⁶⁸²

xxxvi. Article 3(36): Biometric verification

Article 3(36) defines biometric verification as the automated, one-to-one verification, including authentication, of the identity of natural persons by comparing their biometric data to previously provided biometric data.⁶⁸³

Recital 15 defines **authentication** as a process 'whose sole purpose is to confirm that a specific natural person is the person he or she claims to be and to confirm the identity of a natural person for the sole purpose of having access to a service, unlocking a device or having security access to premises'.⁶⁸⁴ The same limitation is also found in Point 1 (a) of Annex III, which excludes biometric verification systems—'the sole purpose of which is to confirm that a specific natural person is the person he or she claims to be'—from the classification of high-risk AI systems in the area of biometrics.⁶⁸⁵ This means that if a system has functions beyond authentication, it falls outside the scope of this provision.

Although biometric verification is closely related to biometric identification, it is considered less risky to fundamental rights due to its more limited scope of identifying a specific natural person.⁶⁸⁶

xxxvii. Article 3(37): Special categories of personal data

Article 3(37) qualifies as special categories of personal data the categories of personal data referred to in Article 9(1) GDPR, Article 10 LED and Article 10(1) EUDPR.⁶⁸⁷

Pursuant to these provisions, 'processing of personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation shall be prohibited'.⁶⁸⁸ Where such personal data is processed, **data protection law** requires an additional justification, such as the data subject's explicit consent⁶⁸⁹ or the data having been manifestly made public by the data subject,⁶⁹⁰ that applies in addition to

⁶⁸² Recital 15 AIA.

⁶⁸³ Ibid.

⁶⁸⁴ Ibid.

⁶⁸⁵ Point 1 (a) of Annex III AIA.

⁶⁸⁶ Remote biometric identification systems are high-risk AI systems by virtue of Point 1(a) of Annex III AIA. In addition, to the obligations in Chapter III AIA, which apply to any remote biometric identification system that is not prohibited by Articles 5(1)(h) and Articles 5(2)–(7) AIA create specific deployer obligations in respect of 'real-time remote biometric identification systems', whereas Article 26(1) foresees deployer obligations in respect of 'post-remote biometric identification systems'.

⁶⁸⁷ GDPR; LED; EUDPR.

⁶⁸⁸ Article 9(1) GDPR.

⁶⁸⁹ Article 9(2)(a) GDPR.

⁶⁹⁰ Article 9(2)(e) GDPR.

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the lawful grounds of processing in Article 6 GDPR. The notion can also be found in other instruments of EU digital law, such as Article 26(3) DSA.⁶⁹¹ In the context of data protection law, **sensitive data** is sometimes also used as a synonym for special category data. This echoes that such data is considered worthy of additional legal protections compared to ordinary personal data as it relates to personal characteristics that are deemed particularly sensitive. The notion must be interpreted expansively to include personal data which only indirectly relates to sensitive personal data.⁶⁹²

- 2.289** The notion of special categories of personal data **plays different roles** in the context of the AIA. For instance, when the Commission assesses whether to adopt delegated acts under Article 97 to amend Annex III, it shall consider whether special categories of personal data are used.⁶⁹³ Article 10(5) empowers providers to process special categories of personal data where this is strictly necessary for the purposes of ensuring bias detection and correction in relation to HRAIS.⁶⁹⁴

xxxviii. Article 3(38): Sensitive operational data

- 2.290** Article 3(38) defines sensitive operational data as operational data related to activities of prevention, detection, investigation or prosecution of criminal offences, the disclosure of which could jeopardize the integrity of criminal proceedings.

- 2.291** Due to its **importance in the criminal domain**, sensitive operational data is deemed particularly important and worthy of protection. For this reason, the regulation at times exempts such data from the general obligations of transparency and disclosure. For example, sensitive operational data does not need to be included in the information provided to the relevant market surveillance authority and the national data protection authority in respect of real time remote biometric identification systems used in publicly accessible spaces for law enforcement purposes.⁶⁹⁵ Further exemptions can be found in Articles 26(5) and 46(3), 59(1)(j), 72(2) and 78(3).

- 2.292** This notion should not be confused with that of **special categories of personal data**.⁶⁹⁶ In fact, where overlaps are possible, sensitive operational data is not necessarily personal data, for instance, when it concerns a general tactic of crime prevention.

xxxix. Article 3(39): Emotion recognition system

- 2.293** Article 3(39) defines an emotion recognition system as an AI system for the purpose of identifying or inferring emotions or intentions of natural persons based on their biometric data.⁶⁹⁷

- 2.294** **Biometric data** is defined by Article 3(34), which was examined just above. Recital 18 lists examples of emotions that can be inferred from the emotion recognition system: happiness, sadness, anger, surprise, disgust, embarrassment, excitement, shame, contempt, satisfaction and amusement.⁶⁹⁸ In contrast, it does not provide examples of intentions, which could be

⁶⁹¹ Article 26(3) DSA reads as follows: 'Providers of online platforms shall not present advertisements to recipients of the service based on profiling as defined in Article 4, point (4), of Regulation (EU) 2016/679 using special categories of personal data referred to in Article 9(1) of Regulation (EU) 2016/679'.

⁶⁹² Case C-184/20 *Vyriausioji tarnybinės etikos komisija* EU:C:2022:601.

⁶⁹³ Article 7(2)(c) AIA.

⁶⁹⁴ Article 10(5) AIA.

⁶⁹⁵ Article 5(4) AIA.

⁶⁹⁶ See further Article 3(37) AIA.

⁶⁹⁷ See also Recital 18 AIA.

⁶⁹⁸ *Ibid.*

interpreted to include indicators that forecast certain physical movements and potentially also other acts. An emotion recognition system in any event does not include 'physical states, such as pain or fatigue, including, for example, systems used in detecting the state of fatigue of professional pilots or drivers for the purpose of preventing accidents.'⁶⁹⁹ The distinction between emotional and physical states will be difficult to draw in practice considering that emotional states are often expressed through physical changes. Take, for instance, the example of a lowering or raising of voice, which will often be the result of a changed emotional state. Emotion recognition systems also do not include 'the mere detection of readily apparent expressions, gestures or movements, unless they are used for identifying or inferring emotions,' such as basic facial expressions like a frown or a smile, or gestures like the movement of hands, arms or head, or characteristics of a person's voice like a raised voice or whispering.⁷⁰⁰

Emotion recognition systems have faced widespread criticism for their unreliability, with many dismissing them as **pseudoscience**.⁷⁰¹ The regulation regulates them as high-risk AI systems.⁷⁰² Furthermore, in addition to the provider-centric obligations arising under Chapter III, deployers of emotion recognition systems are also subject to special transparency obligations as they must inform the natural persons exposed thereto of the operation of the system.⁷⁰³

xl. Article 3(40): Biometric categorization system

Article 3(40) defines biometric categorization system as an AI system for the purpose of assigning natural persons to specific categories based on their biometric data, unless it is ancillary to another commercial service and strictly necessary for objective technical reasons.⁷⁰⁴

The notion of the biometric categorization system defines the scope of Article 5(1)(g), which formulates a qualified prohibition regarding the placing on the market, the putting into service for this specific purpose, or the use of biometric categorization systems that categorize individual natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation.⁷⁰⁵ The notion is crucial to determine whether an AI system qualifies as high-risk by virtue of Article 6(2) read in conjunction with Point 1(c) of Annex III.

The related definition of biometric data can be found under Article 3(34), which was already examined above.⁷⁰⁶ The AIA embraces a **wide definition** of the specific categories people can be assigned to on the basis of their biometric data, such as 'sex, age, hair colour, eye colour, tattoos, behavioural or personality traits, language, religion, membership of a national minority, sexual or political orientation'.⁷⁰⁷ This wide definition risks being overinclusive as many systems operate on the basis of biometric categorization, such as those used for medical diagnosis.

⁶⁹⁹ Ibid.

⁷⁰⁰ Ibid.

⁷⁰¹ Stanley (n 670). For a general analysis, see Steindl (n 670).

⁷⁰² Article 6(2) and Point 1(c) of Annex III AIA.

⁷⁰³ Article 50 (3) AIA.

⁷⁰⁴ See also Recital 16 AIA.

⁷⁰⁵ See also Recital 30 AIA.

⁷⁰⁶ Article 3(34) AIA.

⁷⁰⁷ Recital 16 AIA.

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2.299 The definition in Article 3(40) underlines that biometric categorization systems do not include systems that are **ancillary to another commercial service and strictly necessary for objective technical reasons**. Pursuant to Recital 16, this means that ‘the feature cannot, for objective technical reasons, be used without the principal service, and the integration of that feature or functionality is not a means to circumvent the applicability of the rules of this Regulation.’⁷⁰⁸ The recital provides examples, such as filters categorizing facial or body features used on online marketplaces could constitute such an ancillary feature as they can be used only in relation to the principal service which consists in selling a product by allowing the consumer to preview the display of the product to make a purchase decision.⁷⁰⁹ Another example given is that of filters used on online social network services which categorize facial or body features to allow users to add or modify pictures or videos.⁷¹⁰ This limitation is perplexing for two reasons. First, it is unclear why it is limited to ‘commercial services’ only. There does not seem to be any objective reason for such limitation. Second, this exclusion is antithetical to the broader framework for high-risk AI systems, pursuant to which a broader exemption is available under Article 6(3) for systems that would otherwise qualify as high-risk AI systems by virtue of Article 6(2) but which do ‘not pose a significant risk of harm to the health, safety or fundamental rights of natural persons, including by not materially influencing the outcome of decision making.’⁷¹¹

xli. Article 3(41): Remote biometric identification system

2.300 Article 3(41) defines a remote biometric identification system (‘RBIS’) as an AI system for the purpose of identifying natural persons, without their active involvement, typically at a distance through the comparison of a person’s biometric data with the biometric data contained in a reference database.⁷¹²

2.301 The regulation distinguishes between ‘real-time’ and ‘post’ remote biometric identification systems. The use of ‘real-time’ remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement is subject to a qualified prohibition under Chapter II.⁷¹³ Where real-time remote biometric identification systems fall under the exemptions foreseen in relation to this prohibition, they still qualify as high-risk AI systems and must meet related requirements.⁷¹⁴ In contrast, post remote biometric identification systems are not prohibited but qualify as HRAIS in respect of which special duties arise for the deployer.⁷¹⁵ The definition announced in Article 3(41) serves as the basis for the definitions of ‘real-time remote biometric identification system’ in Article 4(42) and of ‘post-remote biometric identification system’ in Article 4(42).

2.302 The AI Act embraces a **functional definition** of an RBIS which does not focus on the system’s technical features but rather its purpose, namely the identification of natural persons without their active involvement. While Article 3(41) notes that such identification ‘typically’ occurs at a distance, it does not necessarily have to, thereby contradicting the ‘remote’ aspect in the notion of remote biometric identification systems. Indeed, Recital 17 clarifies that a RBIS is

⁷⁰⁸ Ibid.

⁷⁰⁹ Ibid.

⁷¹⁰ Ibid.

⁷¹¹ Article 6(3) AIA.

⁷¹² See also Recital 41 AIA.

⁷¹³ Articles 5(1)(h) and 5(2) AIA.

⁷¹⁴ Point 1(a) of Annex III and Article 6(2) AIA.

⁷¹⁵ See Point 1(a) of Annex III and Article 26(10) AIA.

defined through its function ‘irrespective of the particular technology, processes or types of biometric data used.’⁷¹⁶

RBIS are typically used to perceive multiple persons or their behaviour simultaneously in order to **facilitate the identification of natural persons without their active involvement**.⁷¹⁷ RBIS hence do not include biometric verification systems that merely serve to confirm the identity of one particular person, such as for authentication purposes.⁷¹⁸ Recital 17 specifies that that exclusion is justified ‘by the fact that such systems are likely to have a minor impact on fundamental rights of natural persons compared to the remote biometric identification systems which may be used for the processing of the biometric data of a large number of persons without their active involvement.’⁷¹⁹

xlii. Article 3(42): Real-time remote biometric identification system

Article 3(42) defines a real-time remote biometric identification system (‘RTRBIS’) as a remote biometric identification system, whereby the capturing of biometric data, the comparison and the identification all occur without a significant delay, comprising not only instant identification, but also limited short delays in order to avoid circumvention.⁷²⁰

The AI Act distinguishes between ‘real-time’ and ‘post’ remote biometric identification systems. The use of ‘real-time’ remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement is subject to a qualified prohibition under Chapter II.⁷²¹ Where real-time remote biometric identification systems fall under the exemptions foreseen in relation to this prohibition, they still qualify as HRAIS and must meet related requirements.⁷²² In contrast, post remote biometric identification systems are not prohibited but qualify as HRAIS in respect of which special duties arise for the deployer.⁷²³

The notion of **remote biometric identification systems** needs to be interpreted in line with Article 3(41), which defines a remote biometric identification system as an AI system for the purpose of identifying natural persons, without their active involvement, typically at a distance through the comparison of a person’s biometric data with the biometric data contained in a reference database.⁷²⁴ This definition was examined in further detail above.

The key concept in the definition outlined in Article 3(42) is **identification without significant delay**, allowing for only minimal short delays to prevent circumvention. This distinction is crucial, as it determines which AI practices fall under the qualified prohibition in Article 5(1)(h) and which AI systems will instead be subject to the obligations for high-risk AI systems outlined in Chapter II. Recital 17 provides further clarification, namely that real-time systems ‘involve the use of “live” or “near-live” material, such as video footage, generated by a camera or other device with similar functionality.’⁷²⁵

⁷¹⁶ Recital 17 AIA.

⁷¹⁷ Ibid.

⁷¹⁸ Ibid.

⁷¹⁹ Ibid.

⁷²⁰ See also Recital 17 AIA.

⁷²¹ Articles 5(1)(h) and 5(2) AIA.

⁷²² Point 1(a) of Annex III and Article 6(2) AIA.

⁷²³ See Point 1(a) of Annex III and Article 26(10) AIA.

⁷²⁴ See also Recital 41 AIA.

⁷²⁵ See also Recital 17 AIA.

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xl.iii. Article 3(43): Post remote biometric identification system

- 2.308** Article 3(43) defines a post remote biometric identification systems as a remote biometric identification system other than a RTRBIS.⁷²⁶
- 2.309** The regulation distinguishes between ‘real-time’ and ‘post’ remote biometric identification systems. The use of ‘real-time’ remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement is subject to a qualified prohibition under Chapter II.⁷²⁷ Where real-time remote biometric identification systems fall under the exemptions foreseen in relation to this prohibition, they still qualify as high-risk AI systems and must meet related requirements.⁷²⁸ In contrast, post remote biometric identification systems are not prohibited but qualify as HRAIS in respect of which special duties arise for the deployer.⁷²⁹
- 2.310** The notion of remote biometric identification systems needs to be interpreted in line with Article 3(41), which defines a remote biometric identification system as an AI system for the purpose of identifying natural persons, without their active involvement, typically at a distance through the comparison of a person’s biometric data with the biometric data contained in a reference database.⁷³⁰ This definition was examined in further detail above.
- 2.311** Article 3(43) embraces an **exclusionary definition** of the post remote biometric identification system as any remote biometric identification system that is not a real-time remote biometric identification system. The notion of a real-time remote biometric identification system needs to be interpreted in line with Article 3(42), which defines real-time remote biometric identification systems as a remote biometric identification system, whereby the capturing of biometric data, the comparison and the identification all occur without a significant delay, comprising not only instant identification, but also limited short delays in order to avoid circumvention.⁷³¹
- 2.312** In post remote biometric identification systems, the biometric data has already been captured and the comparison and identification occur only after a ‘**significant delay**’.⁷³² They involve material, such as pictures or video footage generated by closed circuit television cameras or private devices, which has been generated before the use of the system in respect of the natural persons concerned.⁷³³ A post remote biometric identification system is a HRAIS by virtue of Point 1(a) of Annex III.
- xliv. Article 3(44): Publicly accessible space
- 2.313** Article 3(44) defines a publicly accessible space as any publicly or privately owned physical place accessible to an undetermined number of natural persons, regardless of whether certain conditions for access may apply, and regardless of potential capacity restrictions.⁷³⁴

⁷²⁶ Ibid.

⁷²⁷ Articles 5(1)(h) and Article 5(2) AIA.

⁷²⁸ Point 1(a) of Annex III and Article 6(2) AIA.

⁷²⁹ See Point 1(a) of Annex III and Article 26(10) AIA.

⁷³⁰ See also Recital 41 AIA.

⁷³¹ See also Recital 17 AIA.

⁷³² Ibid.

⁷³³ Ibid.

⁷³⁴ See also Recital 19 AIA.

This definition matters in the context of Chapter II as **Article 5(1)(h)** formulates a prohibition of the use of real-time remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement.⁷³⁵

A publicly accessible space can be used for **any activity**, such as ‘for commerce, for example, shops, restaurants, cafés; for services, for example, banks, professional activities, hospitality; for sport, for example, swimming pools, gyms, stadiums; for transport, for example, bus, metro and railway stations, airports, means of transport; for entertainment, for example, cinemas, theatres, museums, concert and conference halls; or for leisure or otherwise, for example, public roads and squares, parks, forests, playgrounds.’⁷³⁶ A space should also be classified as being publicly accessible if, regardless of potential capacity or security restrictions, **access is subject to certain predetermined conditions** which can be fulfilled by an undetermined number of persons, such as the purchase of a ticket or transport pass (e.g. public transportation), prior registration, or meeting certain age requirements (e.g. playgrounds reserved for little children).⁷³⁷

A space should not be considered publicly accessible when ‘access is limited to specific and defined natural persons through either Union or national law directly related to public safety or security or through the clear manifestation of will by the person having the relevant authority over the space.’⁷³⁸ **Company and factory premises**, as well as offices and workplaces that are intended to be accessed only by relevant employees and service providers are not publicly accessible spaces.⁷³⁹ Recital 19 usefully clarifies that the **factual possibility of access** alone, ‘such as an unlocked door or an open gate in a fence’ does not imply that the space is publicly accessible when there are indications or circumstances suggesting the contrary, like signs prohibiting or restricting access.⁷⁴⁰

Recital 19 explicitly qualifies ‘**prisons and border control**’ as spaces that are not publicly accessible.⁷⁴¹ The exclusion of ‘border control’ may be surprising as within the European Union, where most Member States participate in the Schengen Agreement⁷⁴², most border areas are in fact publicly accessible. The notion of ‘border control’ hence ought to be interpreted restrictively to not include a broader area at a border but only the premises where controls are carried out, in line with the formulations in other linguistic versions of the AIA.⁷⁴³ In practice this may be difficult to delineate with precision, such as where a highway runs along a border and police occasionally carries out controls on the sidelines. According to the Guidelines on Article 5, a border crossing point is not a publicly accessible space, while the street leading to the border crossing point or a forest in the vicinity normally is.⁷⁴⁴ The Guidelines also specify that where a space has a dual function, such as an airport, where the common areas are publicly accessible but the border control area is not, then this space

⁷³⁵ See also Articles 5(3)–(7) AIA.

⁷³⁶ Recital 19 AIA.

⁷³⁷ Ibid.

⁷³⁸ Ibid.

⁷³⁹ Ibid.

⁷⁴⁰ Ibid.

⁷⁴¹ Ibid.

⁷⁴² The Schengen *acquis* – Agreement between the Governments of the States of the Benelux Economic Union, the Federal Republic of Germany and the French Republic on the gradual abolition of checks at their common borders [2000] OJ L239/13.

⁷⁴³ The French text refers to ‘le contrôle aux frontières’ whereas the German version speaks of ‘Grenzkontrollbereiche’.

⁷⁴⁴ Commission Guidelines on prohibited artificial intelligence practices (n 166) 102.

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is excluded from Article 5(1)(h).⁷⁴⁵ The conceptual grounding for this rule, and in particular the difference with the street example provided above, is unclear.

- 2.318** The regulation acknowledges that some areas ‘may comprise **both publicly accessible and non-publicly accessible spaces**, such as the hallway of a private residential building necessary to access a doctor’s office or an airport.’⁷⁴⁶ Importantly, **online spaces** are not covered, ‘as they are not physical spaces.’⁷⁴⁷

xliv. Article 3(45): Law enforcement authority

- 2.319** Article 3(45) classifies as law enforcement authority: (a) any public authority competent for the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security; or (b) any other body or entity entrusted by Member State law to exercise public authority and public powers for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security.

- 2.320** The concept of law enforcement in essence includes **two different sets of authorities**. Their definition is closely connected to that of ‘law enforcement’ in Article 3(46), meaning that it would have been more logical to invert the order of these two definitions. The definition is also aligned with Article 2(2)(d) GDPR, and Article 4(7)(a) LED. Yet, there remains considerable uncertainty regarding which authorities are precisely covered by these definitions.⁷⁴⁸ In any event, tax authorities and safety directorates do not qualify as law enforcement authorities in EU law.⁷⁴⁹

- 2.321** Law enforcement authorities are **defined functionally** to include any of the activities listed in Article 3(45), irrespective of whether these activities are carried out by a public authority or whether these are bodies or entities entrusted with these activities by Member State law. This is in line with the functional definition of the national security exemption in Article 2(3).

- 2.322** The regulation on the one hand recognizes that the use of AI systems by law enforcement authorities can pose **particularly impactful risks to fundamental rights**, which is why these systems qualify as high-risk AI systems.⁷⁵⁰ On the other hand, the regulation provides exemptions for law enforcement authorities from certain obligations applicable to high-risk AI systems, acknowledging that **not all transparency requirements can be fulfilled in the context of law enforcement**.⁷⁵¹

xlvi. Article 3(46): Law enforcement

- 2.323** Article 3(46) defines law enforcement as activities carried out by law enforcement authorities or on their behalf for the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including safeguarding against and preventing threats to public security.

⁷⁴⁵ Ibid.

⁷⁴⁶ Recital 19 AIA.

⁷⁴⁷ Ibid.

⁷⁴⁸ Luca Tosoni and Lee A Bygrave, ‘Definitions’ in Eleni Kosta and Franziska Boehm (eds), *The EU Law Enforcement Directive (LED)* (1st edn, OUP 2024).

⁷⁴⁹ Case C-175/20 *Valsts ienēmumu dienests* EU:C:2022:124 and Case C-439/19 *Latvijas Republikas Saeima* EU:C:2021:504.

⁷⁵⁰ See further Article 6(2) and Point 6 of Annex III AIA.

⁷⁵¹ See, e.g. Article 78(3) AIA.

This definition is in line with that of ‘law enforcement authority’ in Article 3(45). The definition is also aligned with Article 2(2)(d) GDPR, meaning that the CJEU may build on its case law under this provision for future interpretations of Article 3(46).⁷⁵² In practice, the distinction between law enforcement, as defined by Article 3(46), and national security, which benefits from the exemption in Article 2(3), will be challenging to delineate, for instance, in relation to terrorist activity or large-scale organized crime.⁷⁵³

xlvi. Article 3(47): AI Office

Article 3(47) defines the AI Office as the Commission’s function of contributing to the implementation, monitoring and supervision of AI systems and general-purpose AI models, and AI governance, provided for in Commission Decision of 24 January 2024.⁷⁵⁴ The provision further states that references to the AI Office in the AI Act are to be construed as references to the Commission.⁷⁵⁵

The AI Office is an **important institutional actor** concerning the implementation of the regulation. Commission Decision of 24 January 2024 broadly describes the role of the AI Office as enabling the laying of the foundations for a single governance system for AI in the EU through a structure overseeing the advancements in artificial intelligence models, including as regards general-purpose AI models, the interaction with the scientific community, and which shall play a key role in investigations and testing, enforcement, and have a global vocation.⁷⁵⁶ The AI Office is furthermore exclusively competent to supervise and enforce the obligations falling under providers of general-purpose AI models, as well as general-purpose AI models with systemic risk.⁷⁵⁷ Article 75(1) provides that the AI Office ‘shall have all the powers of a market surveillance authority’ in the context of its monitoring and supervisory duties in relation to general-purpose AI models.⁷⁵⁸ This forms an important distinction with the other legal sub regimes created by the AI Act, which are largely overseen by national authorities in the Member States.⁷⁵⁹

The AI Office **forms part of the European Commission’s** Directorate-General for Communication Networks, Content and Technologies (‘DG Connect’), the administrative structure of the European Commission in charge of digital technologies.⁷⁶⁰ Although the AIO is a sub-unit of the Commission, the AI Act refers to them independently. For instance, Article 57 creates different sets of obligations for the AI Office in relation to AI regulatory sandboxes as well as separate tasks for the Commission.⁷⁶¹ While obligations directed specifically to the AI Office ought to be carried out by that unit, the broader reference to the Commission leaves open which part of the Commission ought to carry out related obligations, and whether this may also be the AI Office. This will be a crucial institutional question moving forward.

⁷⁵² Article 3(46) AIA.

⁷⁵³ Articles 3(46) and 2(3) AIA respectively.

⁷⁵⁴ Commission Decision of 24 January 2024 Establishing the Artificial Intelligence Office [2024] OJ C1459. The website of the AI Office is at <https://digital-strategy.ec.europa.eu/en/policies/ai-office>.

⁷⁵⁵ This can be confusing as Article 88(1) AIA states that ‘the Commission shall entrust the implementation of these tasks to the AI Office, without prejudice to the powers of organisation of the Commission’.

⁷⁵⁶ Recital 5 of Commission Decision C/2024/1459 (n 756).

⁷⁵⁷ Article 88(1) AIA. These obligations are laid down in Chapter V AIA.

⁷⁵⁸ Article 75(1) AIA.

⁷⁵⁹ For an overview of these legal sub-regimes, see Article 1(2) AIA.

⁷⁶⁰ Article 1 of Commission Decision C/2024/1459 (n 754).

⁷⁶¹ See Article 57(17) AIA.

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- 2.328** The **organizational structure** of the AI Office consists of five units and two advisors, reflecting its mandate. These units include the ‘Excellence in AI and Robotics’ unit, the ‘Regulation and Compliance’ unit, the ‘AI Safety’ unit, the ‘AI Innovation and Policy Coordination’ unit, and the ‘AI for Societal Good’ unit. In addition, the office includes the Lead Scientific Advisor and the Advisor for International Affairs.⁷⁶² The AIO is expected to employ over 140 staff, some of which will come from other parts of the European Commission and some of which will be seconded national experts.⁷⁶³ The work of the AIO will be supported by the European Artificial Intelligence Board⁷⁶⁴, the Advisory Forum,⁷⁶⁵ as well as the scientific panel of independent experts.⁷⁶⁶ Member States moreover have a duty to facilitate the tasks entrusted to the AI Office.⁷⁶⁷
- 2.329** The **responsibilities** of the AI Office are manifold. Article 64(1) provides that the European Commission shall develop expertise and capabilities in the field of AI through the AI Office.⁷⁶⁸ The AI Office also coordinates the operation of the European Artificial Intelligence Board. It provides the secretariat for the Board, convenes its meetings upon request of the Chair, and prepares the agenda in accordance with the tasks of the Board and its rules of procedure.⁷⁶⁹ While the AI Office has a right to attend the meetings of the Board, it does not have voting rights.⁷⁷⁰ Article 25(4) further empowers the AI Office to develop and recommend voluntary model terms for contracts between providers of high-risk AI systems and third parties.⁷⁷¹ The AI Office is, moreover, obliged to develop a template for a questionnaire, including through an automated tool, to facilitate deployers in complying with the obligation to carry out a fundamental rights impact assessment.⁷⁷² Article 50(7) furthermore obliges the AIO to encourage and facilitate the drawing up of codes of practice to facilitate the effective implementation of the transparency obligations regarding artificially generated or manipulated content.⁷⁷³
- 2.330** Regarding **general-purpose AI models**, the AIO is empowered to exclusively oversee the supervision and enforcement of the legal obligations imposed on providers and general-purpose AI models and general-purpose AI models with systemic risk.⁷⁷⁴ It also plays an important role with respect to the codes of practice for general-purpose AI models.⁷⁷⁵ It will encourage and facilitate the drawing up of such codes⁷⁷⁶ and monitor compliance by those that sign up to the codes of practice,⁷⁷⁷ as well as encourage the review and adaptation of these codes.⁷⁷⁸

⁷⁶² ‘European AI Office’ (*European Commission*) <<https://digital-strategy.ec.europa.eu/en/policies/ai-office>> accessed 28 May 2025.

⁷⁶³ ‘Why work at the EU AI Office?’ (*EU Artificial Intelligence Act*, 7 June 2024) <<https://artificialintelligence.act.eu/de/why-work-at-the-eu-ai-office/>> accessed 16 December 2024. December 2024.

⁷⁶⁴ Article 65 AIA.

⁷⁶⁵ Article 67 AIA.

⁷⁶⁶ Article 68 AIA.

⁷⁶⁷ Article 64(2) AIA.

⁷⁶⁸ Article 64(1) AIA.

⁷⁶⁹ Article 65(8) AIA.

⁷⁷⁰ Article 65(2) AIA.

⁷⁷¹ Article 25(4) AIA.

⁷⁷² Article 27(5) AIA.

⁷⁷³ Article 50(7) AIA.

⁷⁷⁴ Article 88(1) AIA.

⁷⁷⁵ Article 56 AIA.

⁷⁷⁶ Articles 56(1)–(4) AIA.

⁷⁷⁷ Article 56(5) and (6) AIA.

⁷⁷⁸ Article 56(8) AIA.

The AI Office also tracks developments in relation to **national regulatory sandboxes**⁷⁷⁹ and is responsible for facilitating the compliance of SMEs and start-ups with the AI Act.⁷⁸⁰ It provides coordination support for joint investigations by national market surveillance authorities⁷⁸¹ and implements the Commission's enforcement actions under Article 88.⁷⁸² Finally, the AI Office coordinates the anticipated codes of conduct for AI systems not subject to legal obligations under the regulation.⁷⁸³ Commission Decision C/2024/1459 outlines more detailed tasks for the AI Office compared to the regulation, and, for example, highlights the AI Office's role in assisting the Commission on all matters related to standardization within the framework of the AI Act.⁷⁸⁴

Regarding **legislative history**, the AIO originated from a parliamentary proposal to replace the AI Board initially envisioned in the draft regulation. The Parliament's original concept for the AIO was an independent EU body with legal personality, featuring a management board, secretariat, and an advisory forum. The final AI Act incorporates both an AI Office and an AI Board.⁷⁸⁵ Crucially, the AI Office in the final text is significantly different from the Parliament's initial vision—it is not an independent EU body, but rather operates as a component of the European Commission.⁷⁸⁶ However, Article 112(5) foreshadows that the AIO could in the future be upgraded into an independent EU body.

xlvi. Article 3(48): National competent authority

Article 3(48) defines the national competent authority as a notifying authority or a market surveillance authority. Regarding AI systems put into service or used by Union institutions, agencies, offices, and bodies, references to national competent authorities or market surveillance authorities in the AI Act must be construed as references to the European Data Protection Supervisor.

Articles 3(19) and 3(26) provide the relevant definitions of notifying authority and market surveillance authority respectively and their commentary should be referred to for interpretations of these notions. Articles 70(9) and 74(9) govern the status and role of the EDPS when acting as a competent authority.⁷⁸⁷ The EDPS is not competent in relation to the Court of Justice of the European Union acting in its judicial capacity.⁷⁸⁸ Although not mentioned in Article 3(48), it must be noted that Article 75(1) provides that the AI Office 'shall have all the powers of a market surveillance authority' in the context of its monitoring and supervisory duties in relation to general-purpose AI models.⁷⁸⁹

xlix. Article 3(49): Serious incident

Article 3(49) defines a serious incident as an incident or malfunctioning of an AI system that directly or indirectly leads to any of the following: (a) the death of a person, or serious harm to a person's health; (b) a serious and irreversible disruption of the management or

⁷⁷⁹ Articles 57(11), (15) and (16) AIA.

⁷⁸⁰ Article 62(3) AIA.

⁷⁸¹ Article 74(9) AIA.

⁷⁸² Article 88(1) AIA.

⁷⁸³ Article 95 AIA.

⁷⁸⁴ Article 3(2)(d) of Commission Decision C/2024/1459 (n 754).

⁷⁸⁵ See further Article 3(65) AIA.

⁷⁸⁶ Ibid.

⁷⁸⁷ Articles 70(9) and 74(9) AIA.

⁷⁸⁸ Article 74(9) AIA.

⁷⁸⁹ Article 75(1) AIA.

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operation of critical infrastructure; (c) the infringement of obligations under Union law intended to protect fundamental rights; (d) serious harm to property or the environment.

2.336 Similar concepts exist in other instances of product safety law, such as that of ‘serious adverse event’ in Article 2(58) MDR.⁷⁹⁰ The concept is of relevance in the context of high-risk AI systems as well as general-purpose AI models.⁷⁹¹ Regarding **high-risk AI systems**, providers have to report serious incidents related to high-risk AI systems to the market surveillance authorities of the Member States where that incident occurred.⁷⁹² In the context of **general-purpose AI models with systemic risk**, the regulation foresees an obligation for providers to monitor, document, and report relevant information about serious incidents and possible corrective measures to the AI Office without undue delay and, where appropriate, to national competent authorities.⁷⁹³

2.337 Article 3(4) defines a serious incident as an **incident or malfunctioning** that has one of the four consequences listed by the provision. While the notion of incident or malfunctioning is not legislatively defined, the seriousness of these occurrences is given if they fall within one of the four alternative scenarios listed by the provision. The link between the incident or malfunctioning can be direct or indirect, but in any event a causal link between the occurrence and the consequence is required.

2.338 First, an incident or malfunction of an AI system that directly or indirectly leads to the **death of a person or serious harm to their health** constitutes a serious incident.⁷⁹⁴ The formulation of this provision presupposes a causal link between the incident or malfunction and the death or serious harm to health. In this context, we can, for instance, imagine malfunctions of medical devices that cause these events.⁷⁹⁵ While death is a binary condition that is easily identifiable, the concept of serious harm to an individual’s health is far more subjective and context-dependent. Notably, the regulation provides no clear guidance on whether this includes harm to mental health or is limited solely to physical health, or whether the harm needs to be permanent or not.

2.339 Second, an incident or malfunction of an AI system that causes a **serious and irreversible disruption of the management or operation of critical infrastructure** constitutes a serious incident under the AI Act. Article 3(62) defines the notion of critical infrastructure by referencing Article 2(4) of the Directive on the Resilience of Critical Entities. This provision defines the critical infrastructure as ‘an asset, a facility, equipment, a network or a system, or a part of an asset, a facility, equipment, a network or a system, which is necessary for the provision of an essential service.’⁷⁹⁶ Article 2(5) of that Directive defines an essential service as ‘a service which is crucial for the maintenance of vital societal functions, economic activities,

⁷⁹⁰ Article 2(58) MDR reads as follows: ‘serious adverse event means any adverse event that led to any of the following: (a) death, (b) serious deterioration in the health of the subject, that resulted in any of the following: (i) life-threatening illness or injury, (ii) permanent impairment of a body structure or a body function, (iii) hospitalisation or prolongation of patient hospitalisation, (iv) medical or surgical intervention to prevent life-threatening illness or injury or permanent impairment to a body structure or a body function, (v) chronic disease, (c) foetal distress, foetal death or a congenital physical or mental impairment or birth defect’.

⁷⁹¹ See further Chapters III and V AIA.

⁷⁹² Article 73(1) AIA.

⁷⁹³ Article 55(1)(c) and Recital 115 AIA.

⁷⁹⁴ Article 73 AIA.

⁷⁹⁵ Which, in that case, would be covered by Article 73 AIA as well as Article 87 MDR.

⁷⁹⁶ Article 2(4) of the Directive on the Resilience of Critical Entities (Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities and repealing Council Directive 2008/114/EC [2022] OJ L333/164).

public health and safety, or the environment’.⁷⁹⁷ In 2023, the Commission adopted delegated legislation providing a non-exhaustive list of essential services.⁷⁹⁸

What remains ambiguous is the interpretation of the requirement that disruptions to the management or operation of critical infrastructure be both ‘serious’ and ‘irreversible’. While the concept of ‘seriousness’ can generally be assessed through a contextual, common-sense lens, the notion of ‘irreversibility’ presents greater interpretive complexity. For example, consider an AI system used as a safety component in electricity grid operations that triggers a system failure, resulting in a week-long blackout. Although the technical issue may eventually be resolved, the broader consequences—such as halted industrial production or unrecoverable economic losses—may be irreversible in practice. Thus, ‘irreversibility’ should not be interpreted too narrowly, as doing so would risk excluding the very types of high-impact incidents the provision is intended to address.

Third, an incident or malfunction of an AI system that leads to the **infringement of obligations under EU law intended to protect fundamental rights** amounts to a serious incident. This is a vague and imprecise formulation for a potentially far-reaching concept, which could potentially cover any failure by a data controller operating a high-risk AI system related to the obligations arising under the GDPR. Consider, for instance, a data controller who operates an AI system to determine access to vocational training, but the system fails to correctly interpret the data subject’s rights under the GDPR. Although data protection laws provide mechanisms to remedy such failures, it is questionable whether these should truly be classified as a ‘serious incident’ under the high-risk AI system framework. As such, it seems appropriate to apply a *de minimis* threshold in relation to this clause, which is not, however, foreseen by the AI Act.

Fourth, an incident or malfunction of an AIS that leads to **serious harm to property or the environment**. Like the preceding scenario, this one also lacks clarity, as the definition of what constitutes serious harm to property or the environment is open to varying interpretations. This is particularly problematic considering that the AIA does not explicitly address risks to property or the environment but is rather focused on risks to human health, safety and fundamental rights. The right to personal property is, indeed, an EU fundamental right, but it is unclear why an infringement of this right is singled out as opposed to all other fundamental rights protected by the Charter. Furthermore, the Charter recognizes environmental protection as a principle and not a right, making this reference even more puzzling.⁷⁹⁹

1. Article 3(50): Personal data

Article 3(50) defines personal data by referencing Article 4(1) GDPR, which defines personal data as ‘any information relating to an identified or identifiable natural person (“data subject”); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person.’⁸⁰⁰

⁷⁹⁷ Article 2(5) of the Directive on the Resilience of Critical Entities (n 797).

⁷⁹⁸ Commission Delegated Regulation (EU) 2023/2450 of 25 July 2023 supplementing Directive (EU) 2022/2557 of the European Parliament and of the Council by establishing a list of essential services [2023] OJ L2450.

⁷⁹⁹ See further Articles 17 and 37 CFR.

⁸⁰⁰ Article 4 (1) GDPR.

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2.344 EU data protection law embraces a **broad definition of personal data**, according to which a wide range of data qualifies as personal data.⁸⁰¹ In the context of artificial intelligence, the broad scope of the concept of personal data carries several significant implications.⁸⁰² First, much of the data used to train AI models qualifies as personal data.⁸⁰³ This implies that data controllers who use personal data to train AI models need to comply with data protection law requirements in relation to such data. Second, the output of an AI system is oftentimes personal data, such as a medical diagnosis generated in relation to a specific natural person. In such instances, data protection law needs to be respected in relation to such outputs.⁸⁰⁴ Third, an AI model can also qualify as personal data.⁸⁰⁵

2.345 In the context of the AI Act, the notion of personal data serves more specific purposes. First, the notion forms an **element of other definitions**, such as that of biometric data, and indirectly also of the definitions linked to that concept, such as that of biometric identification.⁸⁰⁶ Second, the regulation also creates a few **legal obligations** in relation to personal data used in the context of AI systems. For example, Article 10(2)(b) and (5) create specific data governance obligations in relation to personal data, as does Article 59 regarding the further processing of personal data in AI regulatory sandboxes, and Article 60(4) in relation to personal data tested in real-world conditions. Third, references to personal data in the AI Act serve to **recall that data protection law applies** in its entirety when an AI system falls within the scope of EU law, and that accordingly, the dynamic of *lex posterior derogat legi priori* does not play out between the various data protection law instruments⁸⁰⁷ and the Artificial Intelligence Act.⁸⁰⁸

li. Article 3(51): Non-personal data

2.346 Article 3(51) defines non-personal data as data other than personal data as defined in Article 4(1) GDPR. This definition is in line with data found in other EU secondary legislation, such as Article 2(4) DGA.⁸⁰⁹

lii. Article 3(52): Profiling

2.347 Article 3(52) defines profiling by referencing Article 4(4) GDPR, according to which profiling refers to ‘any form of automated processing of personal data consisting of the use of personal data to evaluate certain personal aspects relating to a natural person, in particular to analyse or predict aspects concerning that natural person’s performance at work, economic situation, health, personal preferences, interests, reliability, behaviour, location or movements.’⁸¹⁰

⁸⁰¹ Finck and Pallas (n 250) and Purtova (n 253).

⁸⁰² Henrik Nolte, Michèle Finck and Kristof Meding, ‘Machine Learners Should Acknowledge the Legal Implications of Large Language Models as Personal Data’ (*arXiv*, 3 March 2025) <<http://arxiv.org/abs/2503.01630>> accessed 4 March 2025.

⁸⁰³ Finck and Pallas (n 250); Lauren Leffer, ‘Your Personal Information Is Probably Being Used to Train Generative AI Models’ (*Scientific American*, 19 October 2023) <<https://www.scientificamerican.com/article/your-personal-information-is-probably-being-used-to-train-generative-ai-models/>> accessed 23 December 2024.

⁸⁰⁴ Christopher Kuner and others, ‘Machine Learning with Personal Data: Is Data Protection Law Smart Enough to Meet the Challenge?’ (2017) 7 *International Data Privacy Law* 1.

⁸⁰⁵ Veale, Binns and Edwards (n 256); Purtova and Leenes (n 255); Nolte, Finck and Meding (n 802).

⁸⁰⁶ See Articles 3(34) and (36) AIA respectively.

⁸⁰⁷ GDPR; LED; EUDPR.

⁸⁰⁸ See, instead of many, Recital 10 AIA.

⁸⁰⁹ It is worth noting that Article 3(1) of the Free Flow of Non-Personal Data Regulation (n 20) considered any data that is not personal data to be ‘data’ rather than ‘non-personal data.’

⁸¹⁰ Article 4(4) GDPR.

Article 5(1)(d) relies on the notion of profiling as it prohibits placing on the market, putting into service for this specific purpose, or using an AI system for making risk assessments of natural persons in order to assess or predict the risk of a natural person committing a criminal offence, based solely on profiling or the assessment of personality traits and characteristics.⁸¹¹ Similarly, **Article 6(3)** determines that its derogation from the risk classification in Article 6(2) is not available in the context of systems that profiles natural persons.⁸¹²

liii. Article 3(53): Real-world testing plan

Article 3(53) defines a real-world testing plan as a document that describes the objectives, methodology, geographical, population and temporal scope, monitoring, organization and conduct of testing in real-world conditions. This definition matters in the context of the testing of high-risk AI systems in real-world conditions outside of regulatory sandboxes as a real-world testing plan must be drawn up to manage such testing.⁸¹³

This legislative definition remains **vague**, as it fails to specify the various elements that must be included in the documentation. Its precise contours are expected to emerge over time, particularly as Article 60(1) mandates the Commission to adopt implementing acts that will define the detailed requirements for real-world testing plans.⁸¹⁴

The testing of AI systems in real-world conditions outside of regulatory sandboxes is governed by **Article 60**.⁸¹⁵ This mechanism allows providers or prospective providers of high-risk AI systems that qualify as such by virtue of the combined reading of Article 6(2) and Annex II prior to placing them on the market or putting them into service.⁸¹⁶ Such testing is subject to a number of conditions, including that providers, as well as prospective providers, draw up a real-world testing plan, which has to be submitted to the market surveillance authority in the Member State in which the testing is to be conducted.⁸¹⁷

liv. Article 3(54): Sandbox plan

Article 3(54) defines the sandbox plan as a document agreed between the participating provider and the competent authority describing the objectives, conditions, timeframe, methodology, and requirements for the activities carried out within the sandbox.

This definition is notably **vague** and suffers from a lack of legal certainty, particularly with respect to the interpretation of terms, such as ‘conditions’ and ‘requirements’. Greater clarity is expected to emerge over time, as Article 58(1)(b) empowers the Commission to adopt implementing acts that establish common principles for sandbox plans, thereby aiming to prevent regulatory fragmentation across Member States.⁸¹⁸

The AI Act obliges Member States to ensure that their competent authorities establish at least one **AI regulatory sandbox**, which has to be operational by 2 August 2026.⁸¹⁹ The regulation imposes a number of requirements on sandboxes and providers, as well as prospective

⁸¹¹ Article 5(1)(d) AIA. See also Recital 42 AIA.

⁸¹² Articles 6(2) and (3) AIA.

⁸¹³ Article 60(1) AIA.

⁸¹⁴ These implementing acts ought to be adopted in accordance with the examination procedure referred to in Article 98(2) AIA.

⁸¹⁵ Article 60 AIA.

⁸¹⁶ Articles 60(1) and (2) AIA.

⁸¹⁷ Article 60(4)(a) AIA.

⁸¹⁸ Article 58(1)(b) AIA.

⁸¹⁹ Article 57(1) AIA.

providers participating in them, including the need for a specific sandbox plan to be agreed between the (prospective) provider and the competent authority.⁸²⁰ Provided that the prospective providers observe the specific plan and requirements related to their participation in the regulatory sandbox and respect in good faith the guidance given by the national competent authority, no administrative fines can be imposed on them under the AIA.⁸²¹ These sandboxes are one of the instruments foreseen by Chapter VI in order to promote the innovation-enhancing objective of the AI Act.⁸²²

lv. Article 3(55): AI regulatory sandbox

2.355 Article 3(55) defines the AI regulatory sandbox as a controlled framework set up by a competent authority which offers providers or prospective providers of AI systems the possibility to develop, train, validate and test an innovative AI system, where appropriate, in real-world conditions, pursuant to a sandbox plan and for a limited time under regulatory supervision.⁸²³

2.356 It appears that the notion of the competent authority should be interpreted in line with that of ‘**national competent authority**’ in Article 3(48), which refers both to national competent authorities and also clarifies that this includes the EDPS.⁸²⁴ Pursuant to Article 57(3), the EDPS is competent for the establishment of AI regulatory sandboxes or Union institutions, bodies, offices and agencies.⁸²⁵ Given that the EDPS explicitly qualifies as a national competent authority by virtue of Article 3(48), it would have been preferable to also use this notion in Article 3(55).

2.357 **Regulatory sandboxes** were first debuted in the financial sector and provide a controlled testing environment for select innovative projects, offering flexibility in compliance with applicable norms, tailored compliance guidance, and adjusted enforcement measures.⁸²⁶ The term originates from software development, where a ‘sandbox’ refers to a testing environment that isolates untested code changes and experimental features from the live production environment or main code repository. Thus, sandboxes serve as innovation-promoting tools under the AI Act.

2.358 While the regulation defines the notion of the AI system, it does not provide an interpretation of what an ‘**innovative AI system**’ is. It has been suggested that innovative AI systems are those that address existing market gaps, introduce new functionalities, or offer distinct value propositions.⁸²⁷

⁸²⁰ Article 60(5) AIA.

⁸²¹ Article 60(12) AIA.

⁸²² See further Article 1(1) AIA.

⁸²³ See also Article 57(5) AIA.

⁸²⁴ Article 3(48) AIA defines the national competent authority as follows: ‘A notifying authority or a market surveillance authority; as regards AI systems put into service or used by Union institutions, agencies, offices and bodies, references to national competent authorities or market surveillance authorities in this Regulation shall be construed as references to the European Data Protection Supervisor’.

⁸²⁵ Article 57(3) AIA.

⁸²⁶ For an overview of regulatory sandboxes in the financial sector, see Jayoung James Goo and Joo-Yeun Heo, ‘The Impact of the Regulatory Sandbox on the Fintech Industry, with a Discussion in the Relation between Regulatory Sandboxes and Open Innovation’ (2020) 6 *Journal of Open Innovation: Technology, Market, and Complexity* 43; Hilary J Allen, ‘Regulatory Sandboxes’ (2019) 87 *George Washington Law Review* 579; Lauren A. Fahy, ‘Fostering Regulator–Innovator Collaboration at the Frontline: A Case Study of the UK’s Regulatory Sandbox for Fintech’ (2022) 44 *Law & Policy* 162.

⁸²⁷ Fabio Seferi, ‘A Comparative Analysis of Regulatory Sandboxes From Selected Use Cases: Insights From Recurring Operational Practices’ in Filippo Bagni and Fabio Seferi (eds), *Regulatory sandboxes for AI and Cybersecurity. Questions and answers for stakeholders* (Cybersecurity National Lab 2025) 150.

Article 3(54) defines the **sandbox plan** as a document agreed between the participating provider and the competent authority describing the objectives, conditions, timeframe, methodology, and requirements for the activities carried out within the sandbox.⁸²⁸ This notion has been examined just above. The rules listed by Article 57 in relation to sandboxes will be complemented by more detailed arrangements that the Commission will adopt in the form of implementing legislation.⁸²⁹

Ivi. Article 3(56): AI literacy

Article 3(56) defines AI literacy as skills, knowledge and understanding that allow providers, deployers and affected persons, considering their respective rights and obligations in the context of the AIA, to make an informed deployment of AI systems, as well as to gain awareness about the opportunities and risks of AI and possible harm it can cause.

AI literacy is governed by **Article 4** and should hence have come earlier in the list of definitions as opposed to in between the definitions related to Chapter VI. Article 4 formulates an obligation of means for providers and deployers to ensure ‘a sufficient level of AI literacy of their staff and other persons dealing with the operation and use of AI systems on their behalf’.⁸³⁰ Article 95(2)(c) also mentions AI literacy in the context of codes of conduct for voluntary application of specific requirements and foresees that these codes of conduct may serve to promote AI literacy.⁸³¹

The definition of AI literacy in Article 3(56) is **vague**, as it relies on several undefined concepts, such as ‘skills, knowledge and understanding’ and ‘opportunities, risks, and harms of AI’. However, given that the related obligation in Article 4 is highly context-specific, formulating a more precise definition would have been challenging. In any case, Article 4 is unlikely to have significant legal consequences, making the lack of legal certainty in Article 3(56) less problematic.

Ivii. Article 3(57): Testing in real-world conditions

Article 3(57) defines testing in real world conditions (‘TRWC’) as the temporary testing of an AI system for its intended purpose in real-world conditions outside a laboratory or otherwise simulated environment, with a view to gather reliable and robust data and to assess and verify the conformity of the AI system with the requirements of the regulation. Testing in real-world conditions does not qualify as placing the AI system on the market or putting it into service within the meaning of the AI Act, provided that all the conditions laid down in Articles 57 or 60 are fulfilled. TRWC is one of the instruments foreseen by Chapter VI to promote the **innovation-enhancing** objective of the AI Act.⁸³²

Such testing is always based on a **real-world testing plan**, as defined by Article 3(53). The main addition of the definition in Article 3(57) is the clarification that TRWC does not amount to placing an AI system on the market or putting it into service, a clarification that would have made more sense under Articles 3(9) and (11), which provide definitions of these important concepts.⁸³³

⁸²⁸ Article 3(54) AIA.

⁸²⁹ Article 58 AIA.

⁸³⁰ Article 4 AIA.

⁸³¹ Article 95(2)(c) AIA.

⁸³² See further Article 1(1) AIA.

⁸³³ See further Articles 3(9) and (11) AIA.

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lviii. Article 3(58): Subject

- 2.365** Article 3(58) defines the subject for the purpose of real-world testing as a natural person who participates in TRWC. Subjects are mentioned in Articles 60 and 61, which govern TRWC, and might also be addressed by the detailed elements of testing of high-risk AI systems that the Commission will adopt by means of implementing acts pursuant to Article 60(1).⁸³⁴

lix. Article 3(59): Informed consent

- 2.366** Article 3(59) defines informed consent as a subject's freely given, specific, unambiguous and voluntary expression of his or her willingness to participate in a particular testing in real-world conditions, after having been informed of all aspects of the testing that are relevant to the subject's decision to participate.
- 2.367** Subjects must provide informed consent prior to their **participation in TRWC**.⁸³⁵ Such consent can be revoked without any resulting detriment or without the need to provide any justification.⁸³⁶
- 2.368** The notion of informed consent is **reminiscent of that found in other EU law instruments** such as Article 2(55) MDR.⁸³⁷ Existing case law and literature on these will likely inform future interpretations of informed consent under the AI Act. Consent is also an important notion in the GDPR for it can serve as one of various possible legal bases for the processing of personal data.⁸³⁸ The informed consent of subjects to participate in real-world testing is, however, distinct from, and without prejudice to, consent of data subjects for the processing of their personal data.⁸³⁹

lx. Article 3(60): Deepfake

- 2.369** Article 3(60) defines a deepfake as an AI-generated or manipulated image, audio or video content that resembles existing persons, objects, places, entities, or events, and would falsely appear to a person to be authentic or truthful. Given that deepfakes are referenced solely in Article 50(4), it may have been more appropriate to define the term directly within that provision. The inclusion of a separate definition in Recital 134, which appears to diverge in substance, risks introducing unnecessary complexity and reducing the overall clarity of the legislative text.
- 2.370** This definition matters as pursuant to **Article 50(4)**, deployers of AI systems constituting deepfakes must disclose that the content has been artificially generated or manipulated unless the use has been authorized by law and is used to combat crime. This definition is in line with that provided in Article 35(1)(k) of the Digital Services Act (which does not, however, use the term 'deepfake').⁸⁴⁰ This is, however, a very broad definition, which is problematic

⁸³⁴ Articles 60 and 61 AIA.

⁸³⁵ Articles 60(4) and 61(1) AIA.

⁸³⁶ Article 60(5) AIA.

⁸³⁷ Article 2(55) MDR reads as follows: 'informed consent means a subject's free and voluntary expression of his or her willingness to participate in a particular clinical investigation, after having been informed of all aspects of the clinical investigation that are relevant to the subject's decision to participate or, in the case of minors and of incapacitated subjects, an authorisation or agreement from their legally designated representative to include them in the clinical investigation'.

⁸³⁸ Article 6(1)(a) GDPR.

⁸³⁹ Recital 141 AIA.

⁸⁴⁰ This provision reads as follows refers to an item or information: 'a generated or manipulated image, audio or video that appreciably resembles existing persons, objects, places or other entities or events and falsely appears to a person to be authentic or truthful is distinguishable through prominent markings when presented on their

considering that the distinction between real and synthetic images is, even from a computer vision perspective, far from trivial.⁸⁴¹ Indeed, research has shown that even standard photo editing functions are potentially caught by Article 50(4).⁸⁴² It is also worth pointing out that Recital 134 refers to ‘content that appreciably resembles an existing person’, which implies a higher threshold of similarity to a natural person compared to the legislative definition.

Deepfakes are **images, audio, or video content that have been generated or altered using AI**. While the concepts of images, audio, and videos are straightforward, questions may arise regarding what precisely qualifies as AI, a notion that is not defined by the regulation⁸⁴³, and what degree of manipulation is required for content to be considered AI-altered.⁸⁴⁴

Article 3(60) defines a deepfake as an AI-generated or manipulated content that ‘**resembles existing persons, objects, places, entities or events**’. In contrast, Recital 134 requires that such content ‘appreciably resembles’ these elements and forms yet another example of lacking coherence in the AI Act.⁸⁴⁵

A key question regarding the final part of this legislative definition is what standard should be used to determine whether content falsely appears **authentic or truthful to a person**. Given the wide variations in AI literacy and general education, what one person can easily recognize as AI-generated or manipulated content may be nearly impossible for another to identify.⁸⁴⁶

lxi. Article 3(61): Widespread infringement

Article 3(61) defines widespread infringement as any act or omission contrary to Union law protecting the interest of individuals, who: (a) has harmed or is likely to harm the collective interests of individuals residing in at least two Member States other than the Member State in which: (i) the act or omission originated or took place; (ii) the provider concerned, or, where applicable, its authorized representative is located or established; or (iii) the deployer is established, when the infringement is committed by the deployer; (b) has caused, causes or is likely to cause harm to the collective interests of individuals and has common features, including the same unlawful practice or the same interest being infringed, and is occurring concurrently, committed by the same operator, in at least three Member States. This definition relies on many other definitions in Chapter 1, such as that of the serious incident in Article 3(49); that of the provider, authorized representative, deployer, or operator.⁸⁴⁷

This definition is important for the reporting of serious incidents of HRAIS.⁸⁴⁸ Indeed, where an incident is considered to be widespread or a serious incident has occurred, providers of HRAIS must report the infringement to the market surveillance authorities of the

online interfaces, and, in addition, providing an easy to use functionality which enables recipients of the service to indicate such information.’

⁸⁴¹ Kristof Meding and Christoph Sorge, ‘What Constitutes a Deep Fake? The Blurry Line between Legitimate Processing and Manipulation under the EU AI Act’ (*arXiv*, 4 February 2025) <<http://arxiv.org/abs/2412.09961>> accessed 24 April 2025.

⁸⁴² *Ibid.*

⁸⁴³ Article 3(1) AIA defines AI systems.

⁸⁴⁴ See further Meding and Sorge (n 841).

⁸⁴⁵ On the legal status of recitals in EU law, see further related analysis in the Introduction.

⁸⁴⁶ Laura Gozzi, ‘AI Brad Pitt Dupes French Woman out of €830,000’ *BBC* (15 January 2025) <<https://www.bbc.com/news/articles/ckgnz8rw1xgo>> accessed 19 February 2025.

⁸⁴⁷ Articles 3(3), (4), (5) and (8) AIA.

⁸⁴⁸ Article 73(3) AIA.

Member States in which it occurred no later than two days after the provider or, where applicable, the deployer, became aware of it.⁸⁴⁹

lxii. Article 3(62): Critical infrastructure

2.376 Article 3(62) defines the notion of critical infrastructure by referencing Article 2(4) of the Directive on the Resilience of Critical Entities, which reads as follows: ‘an asset, a facility, equipment, a network or a system, or a part of an asset, a facility, equipment, a network or a system, which is necessary for the provision of an essential service.’⁸⁵⁰ Article 2(5) of that directive defines an essential service as ‘a service which is crucial for the maintenance of vital societal functions, economic activities, public health and safety, or the environment.’⁸⁵¹ In 2023, the Commission adopted delegated legislation providing a non-exhaustive list of essential services.⁸⁵²

2.377 In the context of the AI Act, AI systems used in the area of critical infrastructure can legally qualify as HRAIS.⁸⁵³

lxiii. Article 3(63): General-purpose AI model

2.378 Article 3(63) defines the general-purpose AI model as an AI model, including where such an AI model is trained with a large amount of data using self-supervision at scale, that displays significant generality and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market, and that can be integrated into a variety of downstream systems or applications, but except those that are used for research, development or prototyping activities before they are placed on the market.⁸⁵⁴

2.379 This definition matters in the context of Chapter V as well as Section 5 of Chapter IX and Article 101. It also forms the basis of the definition of a general-purpose AI system in Article 3(66).

2.380 The AI Act generally regulates AI systems, as defined in Article 3(1), rather than models, with the notable exception of GPAIM. Regarding **legislative history**, the draft regulation did not anticipate the regulation of models. However, during the legislative process, the release of ChatGPT by OpenAI sparked widespread attention and raised concerns about the risks of generative AI, prompting discussions on the need for targeted regulation.⁸⁵⁵ This led to the inclusion of specific obligations addressed to general-purpose AI models, which are listed in Chapter V. General-purpose AI models deemed to pose systemic risk are subject to additional legal obligations.⁸⁵⁶ Article 3(63) determines the scope of these obligations by defining general-purpose AI models. During the trilogue negotiations, there were proposals to distinguish between general-purpose AI models and foundation models as separate categories.⁸⁵⁷ However, in the final text, this distinction was removed, and the term ‘foundation models’ was not recognized as a distinct category.⁸⁵⁸ Recital 97 provides that ‘the

⁸⁴⁹ Ibid.

⁸⁵⁰ Article 2(4) of the Directive on the Resilience of Critical Entities (n 796).

⁸⁵¹ Article 2(5) of the Directive on the Resilience of Critical Entities (n 797).

⁸⁵² Commission Delegated Regulation (EU) 2023/2450 (n 798).

⁸⁵³ Article 6(2) and Point 2 of Annex III AIA.

⁸⁵⁴ See also Recitals 97–99 AIA.

⁸⁵⁵ Claire Boine, ‘L’IA Générale et La Proposition de Règlement de La Commission Européenne’ (2022) 2 DALLOZ IP/IT 79.

⁸⁵⁶ Article 55 AIA.

⁸⁵⁷ On the concept of foundation models, see further Rishi Bommasani and others, ‘On the Opportunities and Risks of Foundation Models’ (*arXiv*, 2021) <<https://arxiv.org/abs/2108.07258>> accessed 2 May 2025.

⁸⁵⁸ Gstrein, Haleem, and Zwitter (n 304).

notion of general-purpose AI models should be clearly defined and set apart from the notion of AI systems to enable legal certainty.⁸⁵⁹

General-purpose AI models are **models**, not systems. The notion of ‘model’ is not, however, legislatively defined.⁸⁶⁰ In the literature, it has been defined as a ‘physical, mathematical or otherwise logical representation of a system, entity, phenomenon, process or data.’⁸⁶¹ Recital 97 states that ‘although AI models are essential components of AI systems, they do not constitute AI systems on their own. AI models require the addition of further components, such as for example a user interface, to become AI systems. AI models are typically integrated into and form part of AI systems.’⁸⁶² The category includes, but is not limited to, models that are trained with a large amount of data using self-supervision at scale. This aspect was likely included in the legislative definition because it reflects the characteristics of most GPAIM currently in use. However, it appears to serve merely as an illustrative example rather than a strict criterion, ensuring that future models can also fall within the legislative scope even if they do not share these specific characteristics.⁸⁶³

First, the model must **display significant generality, demonstrating broad applicability across various domains**. A definition of significant generality is provided in Recital 98, according to which ‘the generality of a model could, inter alia, also be determined by a number of parameters, models with at least a billion of parameters and trained with a large amount of data using self-supervision at scale should be considered to display significant generality and to competently perform a wide range of distinctive tasks.’⁸⁶⁴

Second, the model should be **capable of competently performing a wide range of distinct tasks**, regardless of how it is placed on the market. GPAIM can be deemed capable of competently performing a wide range of distinct tasks because they are trained on vast, diverse datasets and designed with architectures that enable transferability and adaptability across multiple domains. This means that they often lay the foundation for other AI systems.⁸⁶⁵ Large generative AI models ought to be considered as one form of a general-purpose AI model as they allow for flexible generation of content (such as text, audio, images or videos) that can readily accommodate a wide range of distinctive tasks.⁸⁶⁶

Third, the model must be **adaptable**, with the ability to integrate into diverse downstream systems or applications. Recital 97 adds that general-purpose AI models are typically trained on large amounts of data and that they can be placed on the market through different means, including through libraries, application programming interfaces (APIs), as direct download, or as physical copy. These models may moreover be further modified or fine-tuned into new models.⁸⁶⁷

⁸⁵⁹ Recital 97 AIA.

⁸⁶⁰ The notion of the model is also not defined in other provisions that mention the concept such as Recital 110 AIA, which moreover talks about the capabilities and reach of the model.

⁸⁶¹ ISO/IEC 22989:2022 (n 333).

⁸⁶² Recital 97 AIA.

⁸⁶³ Ibid.

⁸⁶⁴ Recital 98 AIA.

⁸⁶⁵ It is for this reason that these models are sometimes referred to as ‘foundation models’. See further Jeanina Matias, ‘What Is a Foundation Model? An Explainer for Non-Experts’ (*Stanford HAI*, 10 May 2023) <<https://hai.stanford.edu/news/what-foundation-model-explainer-non-experts>> accessed 3 July 2025.

⁸⁶⁶ Recital 99 AIA.

⁸⁶⁷ Recital 97 AIA.

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2.385 Article 3(63) also excludes from the concept of GPAIM ‘AI models that are used for **research, development or prototyping activities** before they are placed on the market.’⁸⁶⁸ Not only is it conceptually curious to embed an exemption within a definition, but the description of covered activities is vague and again raises the question of whether Article 2(8) meaningfully supplements Article 3(3). Article 53 adds to this confusion as it imposes obligations not only for training but also for pre-training, suggesting a broader scope than the exemptions might imply.⁸⁶⁹ Recital 109 further excludes from the scope of the AIA ‘persons that develop or use models for scientific research purposes,’ yet notably omits the qualifier ‘sole purpose’ found in Article 2(6), which is, however, found in Recital 97.⁸⁷⁰ This omission is significant, as GPAIM rarely serve a single purpose, and most models can plausibly be used, at least in part, for research.

2.386 Interestingly, a second exemption has been tucked away into **Recital 97** and provides that ‘the obligations laid down for models should in any case not apply when an own model is used for purely internal processes that are not essential for providing a product or a service to third parties and the rights of natural persons are not affected.’⁸⁷¹ This limitation is broadly phrased, yet it might be evident anyway, as the ‘use’ of a GPAIM, in the context of Chapter V, never comes within the scope of the AIA, since these obligations are only targeted at providers that develop or have developed and placed on the market a GPAIM.

lxiv. Article 3(64): High-impact capabilities

2.387 Article 3(64) defines high-impact capabilities as those capabilities that match or exceed the capabilities recorded in the most advanced general-purpose AI models.⁸⁷²

2.388 This notion only appears in **Article 51** on the classification of general-purpose AI models with systemic risk, raising the question of why it was necessary to define it in Article 3 instead of Article 51.⁸⁷³ Pursuant to the latter provision, GPAIM are deemed to have systemic risk, and are hence subject to the related legal regime in Article 55. This applies when GPAIM, inter alia, have high impact capabilities evaluated on the basis of appropriate technical tools and methodologies, including indicators and benchmarks,⁸⁷⁴ and the cumulative amount of computation used for their training measured in floating point operations is greater than $10^{(25)}$.⁸⁷⁵ It has been argued that by exclusively referencing the most advanced models at any given moment, there is a risk of excluding models that were previously considered cutting-edge but have since been superseded.⁸⁷⁶

lxv. Article 3(65): Systemic risk

2.389 Article 3(65) defines systemic risk as a risk that is specific to the high-impact capabilities of general-purpose AI models, having a significant impact on the Union market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety,

⁸⁶⁸ Article 3(63) AIA.

⁸⁶⁹ See further Article 53(1)(a) and Recital 107 AIA.

⁸⁷⁰ On the legal status of recitals in EU law, see further related analysis in the Introduction.

⁸⁷¹ Recital 97 AIA.

⁸⁷² See also Recital 110 AIA.

⁸⁷³ Article 51 AIA.

⁸⁷⁴ Article 51(1)(a) AIA.

⁸⁷⁵ Article 51(2) AIA.

⁸⁷⁶ Philipp Hacker and Matthias Holweg, ‘The Regulation of Fine-Tuning: Federated Compliance for Modified General-Purpose AI Models’ (*Social Science Research Network*, 11 June 2025) <<https://papers.ssrn.com/abstract=5289125>> accessed 8 July 2025.

public security, fundamental rights, or the society as a whole, that can be propagated at scale across the value chain.⁸⁷⁷

This definition relies on the preceding definition of the GPAIM in Article 3(63) as well as that of high-impact capabilities in Article 3(64).⁸⁷⁸ According to Recital 110, **systemic risks** include, but are not limited to, ‘any actual or reasonably foreseeable negative effects in relation to major accidents, disruptions of critical sectors and serious consequences to public health and safety; any actual or reasonably foreseeable negative effects on democratic processes, public and economic security; the dissemination of illegal, false, or discriminatory content.’⁸⁷⁹ Systemic risks increase with model capabilities and model reach, meaning that this is not a static concept but rather one that can dynamically evolve throughout the lifecycle of the model.⁸⁸⁰ Systemic risks are influenced by factors such as conditions of misuse, model reliability, model fairness and model security, the level of autonomy of the model, its access to tools, novel or combined modalities, release and distribution strategies, the potential to remove guardrails and other factors.⁸⁸¹ Recital 110 makes explicit reference to systemic risks that can arise from potential intentional misuse or unintended control issues relating to alignment with human intent, including chemical, biological, radiological, and nuclear risks.⁸⁸² Such reference has also been made in the US Executive Order on Artificial Intelligence.⁸⁸³

These models are classified in accordance with the **classification rules set out in Article 51**, pursuant to which a general-purpose AI model is of systemic risk when it either has high-impact capabilities evaluated on the basis of appropriate technical tools and methodologies, including indicators and benchmarks,⁸⁸⁴ or is defined as such on the basis of a Commission decision having regard to the criteria in Annex XIII.⁸⁸⁵ The Commission is empowered to adopt delegated acts in accordance with Article 97 to amend these criteria as well as to supplement benchmarks and indicators in light of evolving technological developments, such as algorithmic improvements or increased hardware efficiency, when necessary, for these thresholds to reflect the state of the art.⁸⁸⁶

lxvi. Article 3(66): General-purpose AI system

Article 3(66) defines a general-purpose AI system (‘GPAIS’) as an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems.⁸⁸⁷

⁸⁷⁷ See further Recital 110 AIA.

⁸⁷⁸ See Articles 3(63) and (64) AIA.

⁸⁷⁹ Recital 110 AIA.

⁸⁸⁰ Ibid.

⁸⁸¹ Ibid.

⁸⁸² On these risks, see further Janet Egan and Eric Rosenbach, ‘Biosecurity in the Age of AI: What’s the Risk?’ (*The Belfer Center*, 6 November 2023) <<https://www.belfercenter.org/publication/biosecurity-age-ai-whats-risk>> accessed 2 May 2025.

⁸⁸³ Executive Order No 14110 (n 371) Section 3(k) reads as follows: ‘[...] and that exhibits, or could be easily modified to exhibit, high levels of performance at tasks that pose a serious risk to security, national economic security, national public health or safety, or any combination of those matters, such as by: (i) substantially lowering the barrier of entry for non-experts to design, synthesize, acquire, or use chemical, biological, radiological, or nuclear (CBRN) weapons; (ii) enabling powerful offensive cyber operations through automated vulnerability discovery and exploitation against a wide range of potential targets of cyber attacks.’

⁸⁸⁴ Article 51(1)(a) AIA.

⁸⁸⁵ Article 51(1) AIA.

⁸⁸⁶ Article 51(3) AIA.

⁸⁸⁷ See further Recital 100 AIA.

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- 2.393** This definition relies on the definition of a GPAIM in Article 3(63), which was examined above. The **definition has two components**: first, a GPAIS is an AI system that is based on a GPAIM, and second, has the capability to serve a variety of purposes, both for direct use as well as for integration with other AI system.
- 2.394** A GPAIM can form the **basis of a GPAIS**. A GPAIS is ‘an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems.’⁸⁸⁸ A GPAIS can be **used directly or be integrated into other AI systems**.⁸⁸⁹ An example of a GPAIS are the most commonly used large-language models, which integrate general-purpose AI models into systems by adding an interface that maintain the capacity to serve a variety of purposes and can be used directly as well as be integrated into other systems.⁸⁹⁰ It is worth noting that a GPAIS can also be further **integrated into a HRAIS**.⁸⁹¹
- 2.395** The notion is mainly relevant in the context of **Article 75**, which regulates mutual assistance, market surveillance and control of general-purpose AI systems. Article 75(1) empowers the AI Office with the competences of a market surveillance authority to carry out monitoring and supervision tasks in relation to general-purpose AI systems.
- lxvii. **Article 3(67): Floating-point operation**
- 2.396** Article 3(67) defines a floating-point operation as any mathematical operation or assignment involving floating-point numbers, which are a subset of the real numbers typically represented on computers by an integer of fixed precision scaled by an integer exponent of a fixed base.⁸⁹²
- 2.397** This definition provides clarification to the definition of **high-impact capabilities** under Article 3(64) and Article 51, according to which general-purpose AI models are deemed to have systemic-risk, and are hence subject to the related legal regime in Article 55. This applies when GPAIM, inter alia, have high impact capabilities evaluated on the basis of appropriate technical tools and methodologies, including indicators and benchmarks,⁸⁹³ and the cumulative amount of computation used for their training measured in floating point operations is greater than $10(^{25})$.⁸⁹⁴
- lxviii. **Article 3(68): Downstream provider**
- 2.398** Article 3(68) defines a downstream provider as a provider of an AI system, including a general-purpose AI system, which integrates an AI model, regardless of whether the AI model is provided by themselves and vertically integrated, or provided by another entity based on contractual relations.
- 2.399** The reference to vertical integration underlines that a provider using their own model in an AI system or general-purpose AI system is also considered a downstream provider. This provision acknowledges the complexity of AI value chains, noting that providers of AI systems

⁸⁸⁸ Article 3(66) AIA.

⁸⁸⁹ Recital 97 AIA.

⁸⁹⁰ Recital 100 AIA. Recital 97 AIA explicitly confirms that the addition of a user interface transforms a GPAIM into a GPAIS.

⁸⁹¹ Article 3(1) of Commission Decision C/2024/1459 (n 754).

⁸⁹² See further Recital 100 AIA.

⁸⁹³ Article 51(1)(a) AIA.

⁸⁹⁴ Article 51(2) AIA.

often depend on models developed by third parties. In accordance with Article 53(1)(b), providers of general-purpose AI models ought ‘to draw up, keep up-to-date and make available information and documentation to providers of AI systems who intend to integrate the general-purpose AI model into their AI systems.’⁸⁹⁵ This obligation is intended to benefit downstream providers, although it strangely does not use this terminology despite the definition in Article 3(68).⁸⁹⁶ In contrast, downstream providers are explicitly referred to in Article 89, which covers monitoring actions and provides a right to lodge a duly reasoned complaint alleging an infringement of the AI Act.⁸⁹⁷

V. Article 4: AI Literacy

Providers and deployers of AI systems shall take measures to ensure, to their best extent, a sufficient level of AI literacy of their staff and other persons dealing with the operation and use of AI systems on their behalf, taking into account their technical knowledge, experience, education and training and the context the AI systems are to be used in, and considering the persons or groups of persons on whom the AI systems are to be used.

A. Overview

Article 4 aims to foster AI literacy by equipping providers, deployers, and affected persons with the conceptual tools needed to make informed decisions regarding AI systems. More broadly, it is intended to enable all actors across the AI value chain to acquire the **insights necessary for effective compliance** with the AI Act.⁸⁹⁸ The provision imposes obligations on both providers and deployers and relates uniformly to all AI systems within the scope of the AI Act, regardless of their risk classification. The AI literacy requirement extends to personnel employed by these entities, as well as to individuals utilizing the AI system on their behalf—including collaborators and third parties, such as an external contractor or service provider.⁸⁹⁹ It is worth noting that AI literacy requirements may eventually extend beyond the AI systems that fall within the scope of the regulation. The voluntary application of specific requirements, including by deployers, is envisaged to include the objective of ‘promoting AI literacy, in particular that of persons dealing with the development, operation and use of AI.’⁹⁰⁰

The objective of Article 4 is to **systematically build AI competencies** to enable better use of AI throughout the European Union. The provision hence acknowledges the complexity and opacity of AI technology.⁹⁰¹ This goal further aligns with the broader purpose of the AIA, which is to harness the benefits of AI for the internal market while addressing its risks to health, safety, and fundamental rights.⁹⁰² The co-legislators associated high hopes with

⁸⁹⁵ Article 53(1)(b) AIA.

⁸⁹⁶ Article 3(68) AIA.

⁸⁹⁷ Article 89(2) AIA.

⁸⁹⁸ Recital 20 AIA.

⁸⁹⁹ According to the non-legally binding FAQs on Article 4, these are ‘persons broadly under the organisational remit’ of providers and deployers. See further ‘AI Literacy - Questions & Answers’ (*European Commission*) <<https://digital-strategy.ec.europa.eu/en/faqs/ai-literacy-questions-answers>> accessed 3 July 2025.

⁹⁰⁰ Article 95(2)(c) AIA.

⁹⁰¹ On this, see further Introduction.

⁹⁰² Article 1(1) AIA.

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AI literacy, namely that it will ‘contribute to improving working conditions and ultimately sustain the consolidation, and innovation path of trustworthy AI in the Union.’⁹⁰³ This may be a bit too euphoric, however, considering the limited scope and strength of the obligation, which are examined further below.

- 2.402** AI literacy is **not a new concept** in digital regulation. For instance, Article 20 of the Council of Europe Framework Convention on AI also stresses the importance of promoting digital literacy.⁹⁰⁴ Similarly, the OECD Recommendations on AI stress the importance of what is referred to as ‘AI knowledge.’⁹⁰⁵ Notwithstanding, the draft regulation initially lacked any provisions on AI literacy.⁹⁰⁶ The European Parliament’s mandate proposed introducing two distinct AI literacy obligations: one requiring the EU and Member States to ‘promote measures for the development of a sufficient level of AI literacy’ and another that served as the precursor to the obligation now established in Article 4.⁹⁰⁷ Notably, the Parliament’s mandate did not include the phrase ‘to their best extent’, meaning that the original AI literacy obligation was formulated as an obligation of results. In contrast, the final text only establishes an obligation of means, thereby softening its legal impact.⁹⁰⁸
- 2.403** Like all provisions in Chapters I and II, Article 4 became applicable on 2 February 2025.⁹⁰⁹ In practice, however, Member States had until 2 August 2025 to designate the market surveillance authorities responsible for enforcing this legal obligation, meaning that any enforcement action prior to that date was extremely unlikely.⁹¹⁰

B. Analysis

- 2.404** Article 3(56) provides a **definition** of AI literacy as ‘skills, knowledge and understanding that allow providers, deployers and affected persons, taking into account their respective rights and obligations in the context of this Regulation, to make an informed deployment of AI systems, as well as to gain awareness about the opportunities and risks of AI and possible harm it can cause.’⁹¹¹ Interestingly, this refers to providers, deployers, and affected persons as the beneficiaries of AI literacy, whereas Article 4 frames AI literacy as an obligation resting on providers and deployers towards their staff and ‘other persons’, which according to Article 3(56) ought to be interpreted as including affected persons.⁹¹² According to Recital 20, affected persons benefit also of ‘the knowledge necessary to understand how decisions taken with the assistance of AI will have an impact on them.’⁹¹³ Such knowledge serves as an

⁹⁰³ Recital 20 AIA.

⁹⁰⁴ Article 20 of Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law (opened for signature 5 September 2024, not yet in force) CETS No 225.

⁹⁰⁵ OECD Recommendation of the Council on Artificial Intelligence (n 280).

⁹⁰⁶ For an overview of the legislative history of Article 4 AIA, see Manuela Paolini E Silva, Aurelia Tamar-Larrieux and Odile Ammann, ‘AI Literacy Under the AI Act: Tracing the Evolution of a Weakened Norm’ (6 June 2025) <https://osf.io/th7c8_v2> accessed 3 July 2025.

⁹⁰⁷ Four Columns AI Act (n 1) line 177m.

⁹⁰⁸ Ibid line 177n.

⁹⁰⁹ Article 113(a) AIA.

⁹¹⁰ Article 74 and Article 113 AIA.

⁹¹¹ Article 3(56) AIA.

⁹¹² Ibid. For a detailed analysis of the status of affected persons under Article 4 AIA see further Tommaso Fia, ‘Article 4 AI Act’ in Gianclaudio Malgieri and others, *The EU Artificial Intelligence Act: A Thematic Commentary* (Hart forthcoming).

⁹¹³ Recital 20 AIA.

enabling factor for the effective exercise of the rights granted to affected persons under the AI Act, including the right to lodge a complaint with a market surveillance authority and the right to an explanation of individual decision-making.⁹¹⁴

Article 4 tasks the **AI Board** with supporting the Commission to promote AI literacy tools as well as public awareness in relation to the use of AI systems.⁹¹⁵ The AI Office qua EU Commission⁹¹⁶ has gathered some of the ongoing practices among the pledgers of the AI Pact in a living repository that provides examples on ongoing AI literacy practices.⁹¹⁷ The repository collects various AI literacy practices as they have been self-declared by the respective industry actors, without any evaluation or endorsement of these practices, which should hence not be understood as examples of compliance with Article 4.⁹¹⁸ Article 4 also assigns to the Commission and Member States the responsibility of promoting AI literacy by collaborating with stakeholders to develop codes of conduct for the responsible development, operation, and use of AI.⁹¹⁹ It should also be noted that the AI Office published an FAQ on AI literacy on its website in May 2025.⁹²⁰

Article 4 applies to both **providers and deployers**, as well as to their personnel and other individuals operating or using AI systems on their behalf. According to the non-legally binding FAQs on Article 4, these are ‘persons broadly under the organisational remit’ of providers and deployers.⁹²¹ Notably, its broad scope extends to persons outside the immediate organisation, over whom providers and deployers may have limited control or knowledge. However, it is important to emphasize that the obligation applies exclusively to AI systems and not to general-purpose AI models. While this is consistent with the broader structure of the regulation—which does not impose obligations on deployers of general-purpose AI models—the exclusion of these models from the scope of the AI literacy requirement is questionable, particularly given that such literacy may be equally, if not more, critical in relation to general-purpose AI models.⁹²² Recital 91 specifically engages with the AI literacy obligations of deployers, which must ensure that those assigned to ‘implement the instructions for use and human oversight’ have the necessary competence, including ‘an adequate level of AI literacy, training and authority to properly fulfil those tasks.’⁹²³ This needs to be read together with Article 26(2), according to which deployers have to ‘assign human oversight to natural persons who have the necessary competence, training and authority, as well as the necessary support’. The regulation also refers to AI in other instances, such as Article 60(4) (j), which specifies that effective oversight of TRWC can only occur through persons that are suitably qualified in the relevant field and have the necessary capacity, training and authority to perform their tasks.

⁹¹⁴ Articles 85 and 86 AIA.

⁹¹⁵ Recital 20 AIA. According to Article 66(f) AIA, one of the tasks of the Board is to ‘support the Commission in promoting AI literacy, public awareness and understanding of the benefits, risks, safeguards and rights and obligations in relation to the use of AI systems’.

⁹¹⁶ Pursuant to Article 3(47), the AI Office refers to the Commission’s function of contributing to the implementation, monitoring and supervision of AI systems and general-purpose AI models, and AI governance.

⁹¹⁷ See further ‘Living Repository to Foster Learning and Exchange on AI Literacy’ (*European Commission*, 4 February 2025) <<https://digital-strategy.ec.europa.eu/en/library/living-repository-foster-learning-and-exchange-ai-literacy>> accessed 20 February 2025.

⁹¹⁸ Recital 20 AIA.

⁹¹⁹ *Ibid.*

⁹²⁰ Commission AI Literacy - Questions & Answers (n 899).

⁹²¹ *Ibid.*

⁹²² Chapter V AIA indeed only imposes obligations on providers of general-purpose AI models.

⁹²³ Recital 21 AIA. See also Articles 14 and 26 AIA.

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- 2.407** According to Recital 20, AI literacy is a **highly contextually obligation**. In some circumstances, it can ‘include understanding the correct application of technical elements during the AI system’s development phase, the measures to be applied during its use, the suitable ways in which to interpret the AI system’s output, and, in the case of affected persons, the knowledge necessary to understand how decisions taken with the assistance of AI will have an impact on them.’⁹²⁴ The Commission’s FAQ on AI literacy encourages organizations to consider their role (being providers or deployers of AI systems) as well as the risks associated to the AI systems they provide and/or deploy.⁹²⁵
- 2.408** This highlights that Article 4 is a contextual requirement that various actors must consider within the broader framework of their compliance efforts under different legal sub-regimes and provisions of the AIA. Indeed, Article 4 applies universally, regardless of risk level. However, its context-dependent nature suggests that AI literacy requirements will vary depending on the specific risk category and the risks involved. Indeed, there is a question whether Article 4 will be **enforced independently** of other obligations in the AI Act at all. The Irish data protection authority has for instance stated that this is unlikely to be the case.⁹²⁶ Yet, it has also been reported that the European Commission has urged all Member States to introduce a ‘fallback’ clause to address potential violations comprehensively, including the inclusion of specific sanctions for infringements relating to AI literacy, fundamental rights impact assessments, and the right to an explanation of individual decision-making.⁹²⁷
- 2.409** In any event, Article 4 is formulated as a **qualified obligation of means** that obliges providers and deployers of AI systems to ‘take measures’ to ensure—only to their best extent—a sufficient level of AI literacy, a notion that is certainly context-dependent and flexible in its interpretation.⁹²⁸
- 2.410** The future **enforcement of Article 4** is hard to predict. Indeed, the regulation does not establish specific fines for non-compliance with the AI literacy obligation. Nevertheless, such non-compliance may be considered in the enforcement of other substantive provisions of the regulation.⁹²⁹ Yet, the regulation also obliges Member States to lay down national rules on penalties and other enforcement measures, which may also include warnings and non-monetary measures, applicable to any infringement of the AI Act by operators.⁹³⁰ These could include specific penalties for non-compliance with Article 4.⁹³¹ The **guidelines** the Commission is expected to adopt on this matter can be expected to shed further light on this matter.⁹³²

⁹²⁴ Ibid.

⁹²⁵ Commission AI Literacy - Questions & Answers (n 899).

⁹²⁶ Luca Bertuzzi, ‘EU’s AI Literacy Requirement Unlikely to Be Enforced on Its Own, Irish DPA Says’ (*MLex*, 14 May 2025) <<https://www.mlex.com/mlex/articles/2340227/eu-s-ai-literacy-requirement-unlikely-to-be-enforced-on-its-own-irish-dpa-says>> accessed 3 July 2025.

⁹²⁷ Ibid.

⁹²⁸ Article 4 AIA.

⁹²⁹ See also Article 85 AIA. Its broad formulation enables affected persons to signal to market surveillance authorities that there may have been a breach of Article 4. Market surveillance authorities however do not have to individually follow up on any such signal.

⁹³⁰ Article 99(1) AIA.

⁹³¹ It has indeed been reported that the EU Commission is pushing Member States in that direction: Bertuzzi (n 926).

⁹³² Article 96 and 99(1) AIA.